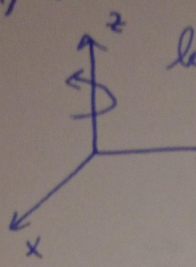
CURS 1

MECANICA CLASICĂ

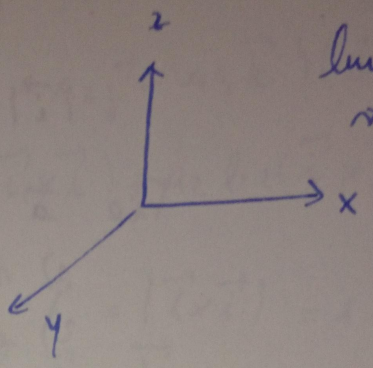
Coordonate (Sisteme de coordonate)

1) Coordonate carteziene



lunghiul drept

sistem drept



lunghiul stâng

Regula Burghiului

x rotit peste y pe drumul cel mai scurt dă sensul pozitiv al axei Oz.

$\mathbb{R}$: ⟶ axa reală

$\mathbb{E}^3$ =

3D : $\mathbb{R} \times \mathbb{R} \times \mathbb{R} = \mathbb{R}^3$; euclidian tridimensional

Spațiul fizic → Mec. clasică $(\mathbb{E}^3, E_+)$ $t > 0$

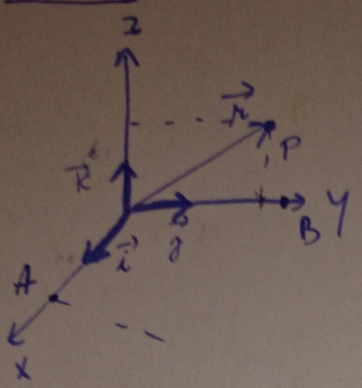
Mec. rel. $(\mathbb{E}^3 \times E \equiv \mathbb{E}^4)$

d = distanța euclidiană

$d^2 = (x-x_0)^2 + (y-y_0)^2 + (z-z_0)^2$, $O(x_0, y_0, z_0)$

MR : $x^2 + y^2 + z^2 - c^2 t^2$, $x_0 = 0, y_0 = 0, z_0 = 0$

Vectori



$\vec{r} = \overrightarrow{OP} = x\vec{i} + y\vec{j} + z\vec{k}$

$\vec{r_1} = x_1\vec{i} + y_1\vec{j} + z_1\vec{k}$

$\vec{i}, \vec{j}, \vec{k}$ vectori

$|\vec{i}| = |\vec{j}| = |\vec{k}| = 1$

Fie $\vec{a}$ vector

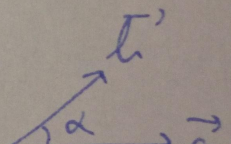
1) modul
2) direcție
3) sens

Operații cu vectori

Produsul a 2 vectori { scalar $(\cdot)$; vectorial $(\times)$

Fie $\vec{a}$, $\vec{b}$ 2 vectori

Atunci $\vec{a} \cdot \vec{b} = \vec{b} \cdot \vec{a} = a\,b\cos\alpha$



$a = |\vec{a}|$, $b = |\vec{b}|$

$\vec{a} \cdot \vec{b} = \vec{b} \cdot \vec{a}$ - comutativ

$\vec{a} = a_1\vec{i} + a_2\vec{j} + a_3\vec{k}$

$\vec{b} = b_1\vec{i} + b_2\vec{j} + b_3\vec{k}$

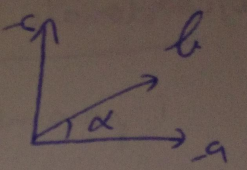
$\vec{a} \cdot \vec{b} = a_1 b_1 + a_2 b_2 + a_3 b_3$

$\vec{i} \cdot \vec{i} = |\vec{i}|\,|\vec{i}|\cos 0 = 1$

$\vec{i}^2 = \vec{j}^2 = \vec{k}^2 = 1$

$\vec{i} \cdot \vec{j} = \vec{j} \cdot \vec{k} =$

Fie $\vec{a}$, $\vec{b}$ vectori



(prod. ved) $\vec{a} \times \vec{b} = \vec{c}$

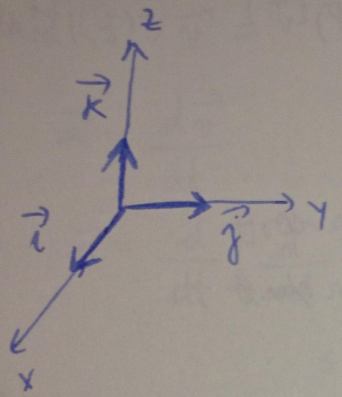
$|\vec{c}| = |\vec{a} \times \vec{b}| = ab \sin\alpha$

$c = ab \sin\alpha$

necomutativ: $\vec{a} \times \vec{b} = -(\vec{b} \times \vec{a})$

$\vec{a} \times \vec{b} = (a_1\vec{i} + a_2\vec{j} + a_3\vec{k})(b_1\vec{i} + b_2\vec{j} + b_3\vec{k})$

$= a_1 b_1 (\vec{i} \times \vec{i}) + a_2 b_2 (\vec{j} \times \vec{j}) + a_3 b_3 (\vec{k} \times \vec{k})$



$\vec{i} \times \vec{i} = |\vec{i} \times \vec{i}| = 1 \cdot 1 \sin 0 = 0$

$\vec{i} \times \vec{j} = \vec{k}$

$\vec{j} \times \vec{k} = \vec{i}$

$\vec{k} \times \vec{i} = \vec{j}$

$\vec{j} \times \vec{i} = -\vec{k}$

$\vec{k} \times \vec{j} = -\vec{i}$

$\vec{i} \times \vec{k} = -\vec{j}$

$$\vec{a} \times \vec{b} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix}$$

$$= \vec{i} \begin{vmatrix} a_2 & a_3 \\ b_2 & b_3 \end{vmatrix} + \vec{j} \begin{vmatrix} a_1 & a_3 \\ b_1 & b_3 \end{vmatrix} + \vec{k} \begin{vmatrix} a_1 & a_2 \\ b_1 & b_2 \end{vmatrix}$$

$\underbrace{\qquad}_{(\vec{a} \times \vec{b})_x}$ $\underbrace{\qquad}_{(\vec{a} \times \vec{b})_y}$

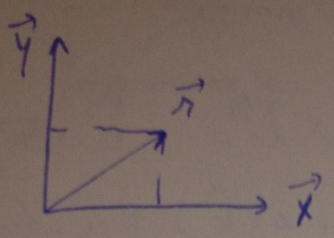
$\vec{a} \times \vec{b} = (a_2 b_3 - a_3 b_2)\vec{i} + (a_3 b_1 - a_1 b_3)\vec{j} +$

$(a_1 b_2 - a_2 b_1)\vec{k}$

$\underbrace{\qquad}_{(\vec{a} \times \vec{b})_z}$

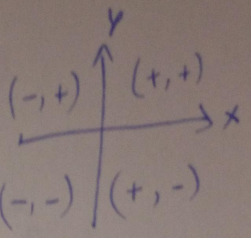
2) Coordonate polare

(parametrizare x-y)



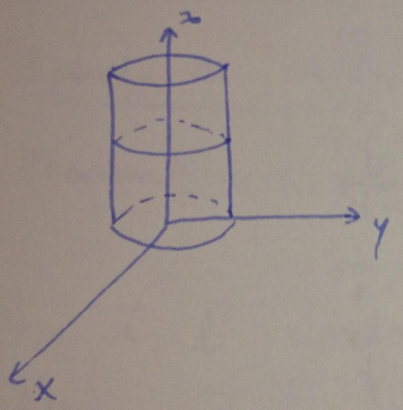
$r = |\vec{r}|$

$(x,y) \to (r, \theta$



$$\begin{cases} x = r\cos\theta \\ y = r\sin\theta \end{cases}, \quad r \in (0, \infty), \quad \theta \in [0, 2\pi)$$

Coord. cilindrice



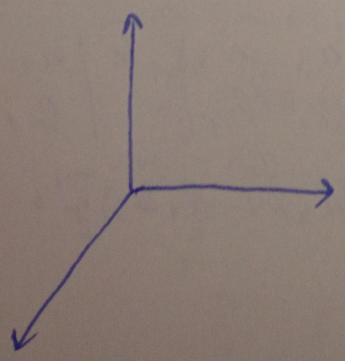
$$\begin{cases} x = r\cos\theta \\ y = r\sin\theta \\ z = z \end{cases}$$

$\vec{r} = x\vec{i} + y\vec{j} + z\vec{k}$

$(\vec{i}, \vec{j}, \vec{k}) \to (\vec{e}_r, \vec{e}_\theta, \vec{e}_z$

$\vec{r} = a_r\vec{e}_r + a_\theta\vec{e}_\theta + a_z\vec{e}_z$

Coord. sferice



$$\begin{cases} x = r\sin\theta\cos\varphi \\ y = r\sin\theta\sin\varphi \\ z = r\sin\theta \end{cases}$$

$r \in (0, \infty)$

$\theta \in [0, \pi)$

$\varphi \in [0, 2\pi)$

$\mathcal{J} = r^2\sin\theta$

Cinematica

Definiții

$\vec{v} = \lim_{\Delta t \to 0} \frac{\Delta \vec{r}}{\Delta t}$ = viteza instantanee

$(\exists)\ \vec{v} = \frac{d\vec{r}}{\Delta t} \iff \vec{r} = \vec{r}(t)$

a = accelerația

dacă $(\exists)\ \vec{v} = \vec{v}(t) \Rightarrow \vec{a} = \lim_{\Delta t \to 0} \frac{\Delta \vec{v}}{\Delta t} \neq 0$

$$\vec{a} = \frac{d\vec{v}}{dt}$$

$$\vec{a} = \frac{d}{dt}\left(\frac{d}{dt}\vec{r}\right) = \frac{d^2\vec{r}}{dt^2}$$

Notații: $\vec{v} = \frac{d\vec{r}}{dt} = \dot{\vec{r}}$

$$\vec{a} = \frac{d\vec{v}}{dt} = \dot{\vec{v}} = \ddot{\vec{r}}$$

$\vec{r} = x\vec{i} + y\vec{j} + z\vec{k}$

$\vec{v} = v_x\vec{i} + v_y\vec{j} + v_z\vec{k}$

$v_x = \frac{dx}{dt} = \dot{x}$ $\quad a_x = \ddot{x}$

$v_y = \frac{dy}{dt} = \dot{y}$ $\quad a_y = \ddot{y}$

$v_z = \frac{dz}{dt} = \dot{z}$ $\quad a_z = \ddot{z}$

def: $\vec{v} = \frac{d\vec{r}}{dt}$ $\xrightarrow[\text{speciale}]{\text{condiții}}$ $d\vec{r} = \vec{v}\,dt$

$f = f(x_1, x_2, \dots, x_n) \Rightarrow df = \frac{\partial f}{\partial x_1} + \frac{\partial f}{\partial x_2} + \frac{\partial f}{\partial x_3}$

$\int d\vec{r} = \int \vec{v}(t)\, dt$

1) $v(t) = v = ct$

$\Rightarrow \int d\vec{r} = \vec{v} \int dt = \vec{v}t + \vec{r}_0$

$r = \vec{v}t + \vec{r}_0$

$\vec{r}(t=0) = \vec{r}_0$ - poziția inițială

2) $\vec{a} = \frac{d\vec{v}}{dt} \Rightarrow d\vec{v} = \vec{a}\, dt$

$\int d\vec{v} = \int \vec{a}\, dt$, $\vec{a} = const$

$\vec{v} = \vec{a}t + \vec{v}_0$, $v(t=0) = v_0$ - viteza inițială

$\int d\vec{r} = \int \vec{v}(t)\, dt$

$\int d\vec{r} = \int (\vec{v}_0 + \vec{a}t)\, dt = \int v_0\, dt + \int \vec{a}t\, dt$

$\int d\vec{r} = \vec{v}_0 \int dt + \vec{a} \int t\, dt$

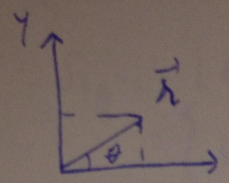
$\vec{r} = \vec{v}_0 t + \vec{a}\frac{t^2}{2} + \vec{r}_0$ - m.r.u.v ($\vec{a} = ct$)

- a) $a < 0$ $\quad u \cdot \hat{i}$
- b) $a > 0$ $\quad u \cdot a$

Mișcare: după traiectorie < rectilinie < $u \cdot v$ ($u \cdot \hat{i}$, $u \cdot a$ variate)

curbilinii < cerc, elipsă, parabolă

Aplicație:



$x = r\cos\theta$

$y = r\sin\theta$

$$\begin{cases} v_x = \dfrac{dx}{dt} = \dfrac{dr}{dt}\cos\theta - r\sin\theta\dfrac{d\theta}{dt} \\ v_y = \dfrac{dy}{dt} = \dfrac{dr}{dt}\sin\theta + r\cos\theta\dfrac{d\theta}{dt} \end{cases}$$

$\dfrac{d\theta}{dt} = \omega$ - circ.

$\dfrac{dr}{dt} = 0$

$v_x = -(r\sin\theta)\omega$

$v_y = r\cos\theta\,\omega$

$v^2 = v_x^2 + v_y^2 = r^2\omega^2\sin^2\theta + r^2\omega^2\cos^2\theta = r^2\omega^2$

$$\boxed{v = \omega r}$$

Temă $a = ?$

$a^2 = a_x^2 + a_y^2$

$a_x = \dfrac{dv_x}{dt}$

$a_y = \dfrac{dv_y}{dt}$

Operatorul ∇ (nabla)

$$\nabla \equiv \vec{i}\frac{\partial}{\partial x} + \vec{j}\frac{\partial}{\partial y} + \vec{k}\frac{\partial}{\partial z}$$

$V = V(x,y,z)$ câmp de scalari

$\vec{A} = A_x\vec{i} + A_y\vec{j} + A_z\vec{k}$ câmp vectorial

$$\nabla V(x,y,z) = \frac{\partial V(x,y,z)}{\partial x}\vec{i} + \frac{\partial V(x,y,z)}{\partial y}\vec{j} + \frac{\partial V(x,y,z)}{\partial z}\vec{k}$$

∇(câmp scalari) = gradient gradient V = vector

a) produs scalar

$$\nabla \cdot \vec{A} = \left(\vec{i}\frac{\partial}{\partial x} + \vec{j}\frac{\partial}{\partial y} + \vec{k}\frac{\partial}{\partial z}\right)\left(A_x\vec{i} + A_y\vec{j} + A_z\vec{k}\right)$$

$$= \frac{\partial A_x}{\partial x} + \frac{\partial A_y}{\partial y} + \frac{\partial A_z}{\partial z}$$

$$\nabla \cdot \vec{A} = \frac{\partial A_x}{\partial x} + \frac{\partial A_y}{\partial y} + \frac{\partial A_z}{\partial z} = \text{divergenta } \vec{A} = \text{scalar}$$

$(\vec{i}, \vec{j}, \vec{k})$ - baza standard IE^3

$$\nabla \times \vec{A} = \left(\vec{i}\frac{\partial}{\partial x} + \vec{j}\frac{\partial}{\partial y} + \vec{k}\frac{\partial}{\partial z}\right) \times \left(A_x\vec{i} + A_y\vec{j} + A_z\vec{k}\right)$$

$$\nabla \times \vec{A} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ A_x & A_y & A_z \end{vmatrix} = \vec{i}\left(\frac{\partial A_z}{\partial y} - \frac{\partial A_y}{\partial z}\right) + \dots$$

Convenția Einstein de sumare
(sumare după indici muți)

$$\delta_{ij} = \begin{cases} 1, & i = j \\ 0, & i \neq j \end{cases}$$

$$\varepsilon_{ijk} = \begin{cases} 1, & (i,j,k) \text{ bază pară} \\ -1, & (i,j,k) \text{ bază impară} \\ 0, & i-j = j-k = k-i = 0 \end{cases}$$

$$\frac{\partial}{\partial x_i} = \partial_i ; \frac{\partial}{\partial x_1} = \partial_1$$

$$(\nabla \times \vec{A})_i = \varepsilon_{ijk} \partial_j A_k$$

indici muți = indici care se repetă

$$(\nabla \times \vec{A})_i = \partial_j A_k - \partial_k A_j$$

$$(\nabla \times \vec{A})_x = \frac{\partial A_z}{\partial y} - \frac{\partial A_y}{\partial z}$$

$$\partial_i e_i = \frac{\partial e_1}{\partial x_1} + \frac{\partial e_2}{\partial x_2} + \dots + \frac{\partial e_m}{\partial x_m}, \quad i = \overline{1,m}$$

$$\Delta = \nabla \cdot \nabla = \nabla^2 \quad , \Delta = \text{laplacian}$$

$$\Delta = \left(i\frac{\partial}{\partial x} + \vec{j}\frac{\partial}{\partial y} + \vec{k}\frac{\partial}{\partial z}\right)\left(i\frac{\partial}{\partial x} + j\frac{\partial}{\partial y} + k\frac{\partial}{\partial z}\right)$$

$$\boxed{\Delta = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}}$$

$$\sqrt{\Delta} = \sqrt{\nabla^2} = \nabla$$

$V = V(x,y,z)$

$\phi = \phi(x,y,z)$

Dinamica mișcării (p, F, L, E ...)

$$p = mv = -Am\omega \sin(\omega t + \varphi)$$

$$F = m \cdot a = -Am\omega^2 \cos(\omega t + \varphi)$$

$$E_c = \frac{mv^2}{2} = \frac{m}{2} A^2 \omega^2 m^2 (\omega t + \varphi)$$

Lemă: Forța de natură elastică produce o mișcare oscilatorie!

$$\vec{F_e} = -k\vec{r} \; ; \; \vec{r} = x\vec{i} + y\vec{j} + z\vec{k}$$

$$F = -kx \Rightarrow F = F(x)$$

$$\begin{cases} F = m\ddot{x} \\ f = -kx \end{cases} \Rightarrow m\ddot{x} = -kx \;\Big|\; \frac{1}{m} \Rightarrow \boxed{\ddot{x} + \frac{k}{m} x = 0}$$

$$\frac{k}{m} = \omega^2 \Rightarrow \boxed{\ddot{x} + \omega^2 x = 0}$$ ec. dif a mișcării

$$x = A\cos(\omega t + \varphi)$$

$$\dot{x} = -A\omega \sin(\omega t + \varphi)$$

$$\ddot{x} = -A\omega^2 \cos(\omega t + \varphi)$$

$$0 = -A\omega^2 \cos(\omega t + \varphi) + \omega^2 A \cos(\omega t + \varphi) = 0$$

3D: $\vec{F} = -k\vec{r}$

1D: $F = -kx$

$$\begin{cases} E_c = \frac{kA^2}{2} \sin^2(\omega t + \varphi) \\ E_p = \frac{kA^2}{2} \cos^2(\omega t + \varphi) \end{cases}$$

CURS 2

Principiile Mecanicii newtoniene

Pr. I = Pr. inerției

Un corp își menține starea de m.r.u. sau de repaus, atâta timp cât asupra lui nu acț. alte corpuri.

(Conduce la legitimarea S.R. inerțial)

SRi = sist. care se află în repaus absolut sau se mișcă cu viteză const (m.r.u) față de un alt sistem de ref. considerat „fix."

OBS: Corolar:

Legile fizicii sunt valabile în sist. ref. in.

Nonsistem de ref. în induce S. ref. neinerțial. (se mișcă accelerat).

Pr I introduce și noțiunea de masă inerțială!

Pr. II = LEX SECUNDA

(Principiul al II-lea al lui Newton)

$$\boxed{\vec{F} = \lim_{\Delta t \to 0} \frac{\Delta \vec{p}}{\Delta t}} \quad ; \quad \vec{F} = \frac{d\vec{p}}{dt} \quad ; \quad \vec{p} = m\vec{v}$$

impuls.

$$\vec{F} = \frac{d}{dt}(m\vec{v}) = \frac{dm\vec{v}}{dt} + m\frac{d\vec{v}}{dt}$$

$$\frac{d\vec{v}}{dt} = \vec{a} \quad ; \quad \dot{m} = \frac{dm}{dt} \leftarrow \text{variația masei}$$

(masă variabilă $\Leftrightarrow m = m(t)$)

introduce ecuația racheteri!

$m = ct \Rightarrow \frac{dm}{dt} = \dot{m} = 0$

$\Rightarrow \boxed{F = m \cdot \vec{a}}$ forma comună a lui Lex Sec.

Teoretician / Experimentator : $F = \gamma \frac{m_1 m_2}{r^2}$

Legea atracției universale

$\boxed{\vec{F} = \gamma \frac{m_1 m_2}{r^3} \cdot \vec{r}}$ – Atracție

Legea Coulomb:

$F = \frac{1}{4\pi\varepsilon} \frac{q_1 q_2}{r^2}$

$F_{atracție}$: $q_1 \cdot q_2 < 0$

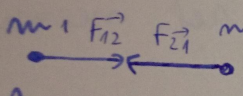
Frespingere: $q_1 \cdot q_2 > 0$

Forța elastică

$\vec{F} = -k\vec{r}$

Pr. III – Legea acțiunii și reacțiunii

– Legea egalității acțiunilor reciproce



$\vec{F}_{12} = -\vec{F}_{21}$

$|\vec{F}_{12}| = |-F_{21}| \Rightarrow F_{12} = F_{21}$

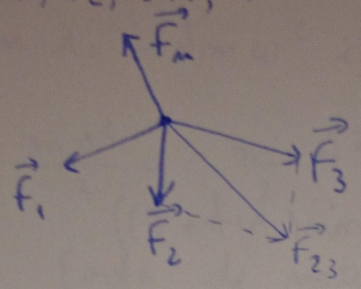
Dacă un corp acț. asupra unui corp cu o forță numită acțiune, cel de-al doilea v-a reacționa (reciprocitate) cu o forță egală în modul și de sens opus.

OBS: Aceste forțe deși nu acționează asupra aceluiași corp.

Principiul Superpoziției

Fie $\vec{F_1}, \vec{F_2}, \ldots, \vec{F_n}$

$A(n)$



$$\vec{R} = \sum_{i=1}^{n} \vec{F_i}$$

$$\vec{R} = m \cdot \vec{a} \Rightarrow \boxed{\sum_{i=1}^{n} \vec{F_i} = m\vec{a}}$$

principiul superpoziției + L.S.

Oscilații (vibrații) mecanice

Funcții matematice notabile

$x = x(t)$

$$\begin{cases} x = A\cos(\omega t + \varphi) \\ x = B\sin(\omega t + \varphi) \\ x = A\cos(\omega t + \varphi) + B\sin(\omega t + \varphi) \end{cases}$$

$\phi = \phi(t)$ faza

$x = A\cos\phi(t)$, $\phi(t) = \omega t + \varphi$

$\phi(0) = \varphi \rightarrow$ form. inițială

Fie $\begin{cases} \phi_1(t) = \omega t + \varphi_1 \\ \phi_2(t) = \omega t + \varphi_2 \end{cases}$ (−)

$$\varphi_2 - \varphi_1 = \omega t + \varphi_2 - \omega t - \varphi_1 = \Delta\varphi$$

$\Delta\phi = \Delta\varphi$ diferența de fază

$x = A\cos(\omega t + \varphi) = A\sin\left(\frac{\pi}{2} - \omega t - \varphi\right) = \cancel{A}$

$= -A\sin\left(\omega t + \varphi - \frac{\pi}{2}\right) = B\cos(\omega t + \varphi_1)$

$B = -A \; ; \quad \varphi_1 = \varphi - \frac{\pi}{2}$

$x = A\left(\cos(\omega t + \varphi) + \frac{B}{A}\sin(\omega t + \varphi)\right)$

$\operatorname{tg}\alpha$

$v = \frac{dx}{dt} = \dot{x}$

$a = \frac{dv}{dt} = \frac{d^2x}{dt^2} = \ddot{x}$

$x = A\cos(\omega t + \varphi)$

$v = -A\omega\sin(\omega t + \varphi)$

$a = -A\omega^2\cos(\omega t + \varphi)$

$\phi = \phi(t) = \omega t + \varphi$

$\frac{d\phi}{dt} = \omega$ (pulsația)

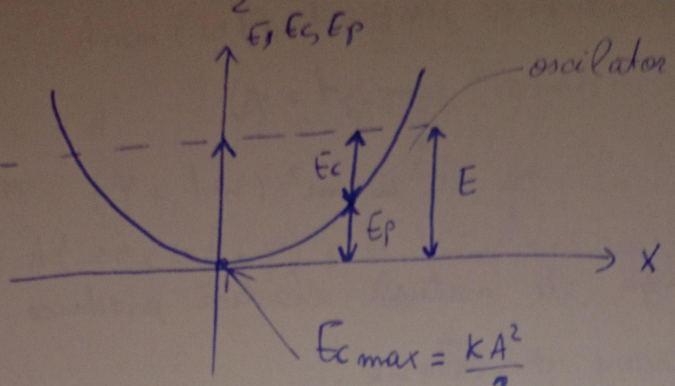
ν = frecvența

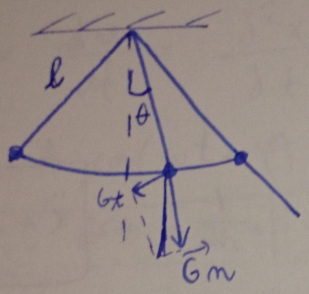
$[\nu]_{s.i.} = s^{-1}$

$[\nu]_{s.i.} = Hz$ (parazită)

Oscilator armonic liniar

$E_c + E_p = E = \frac{kA^2}{2}$





$$\begin{cases} G_t = G \sin\theta \\ G_N = G \cos\theta \end{cases}$$

$$F = -G_t = -mg\sin\theta$$

$$S = l\theta$$

$$v = \frac{ds}{dt} = l\dot{\theta}$$

$$ml\ddot{\theta} = -mg\sin\theta \;\Big|\; \frac{1}{ml} \Rightarrow \ddot{\theta} + \frac{g}{l}\sin\theta = 0$$

$$\omega^2 = \frac{g}{l}$$

$\theta < 5°$

$$\boxed{\ddot{\theta} + \omega_0^2 \sin\theta = 0}$$

$$\sqrt{\frac{g}{l}} = \frac{2\pi}{T_0} \Rightarrow \boxed{T_0 = 2\pi\sqrt{\frac{l}{g}}}$$

$$\boxed{\omega_0 = \frac{2\pi}{T_0}}$$

Dezvoltare în serie Taylor

$f = f(x)$

$$f(x) = f(x_0) + \frac{(x-x_0)}{1!}\left.\frac{df}{dx}\right|_{x_0} + \frac{(x-x_0)^2}{2!}\left.\frac{d^2f}{dx^2}\right|_{x_0} + \frac{(x-x_0)^3}{3!}\left.\frac{d^3f}{dx^3}\right|_{x_0} + \sum_{i=4}^{\infty}\frac{(x-x_0)^i}{i!}\left.\frac{d^if}{dx^i}\right|_{x_0}$$

$x_0 \to 0$

$f(x) = \sin x$

$\frac{df}{dx} = \cos x$

$\frac{d^2f}{dx^2} = -\sin x$

$\frac{d^3f}{dx^3} = -\cos x$

$$f(x) = f(0) + \frac{x}{1!}\left.\frac{df}{dx}\right|_0 + \frac{x^2}{2!}\left.\frac{d^2f}{dx^2}\right|_0 + R_{IV}$$

$R_{IV} \to 0$

$$\sin\theta \simeq \sin 0 + x\cos 0 - \frac{x^2}{2}\sin 0 - \frac{x^3}{6}\cos 0 \quad ; \quad \sin 0 = 0,\ \cos 0 = 0$$

$$\sin\theta \simeq \theta - \frac{\theta^3}{6}$$

$$\boxed{\ddot{\theta} + \omega_0^2\left(\theta - \frac{\theta^3}{6}\right) = 0} \longrightarrow T_3 = ?$$

T_3 = perioada oscilațiilor

Oscilator

$$E_p = k\frac{(x-x_0)^2}{2}$$

$E_p = E_p(x)$

termen armonic

dezv. serie Taylor

$$E_p(x) = E_p(x_0) + \frac{x-x_0}{1!}\left.\frac{dE_p}{dx}\right|_{x_0} + \frac{(x-x_0)^2}{2!}\left.\frac{d^2E_p}{dx^2}\right|_{x_0}$$

$$R_{III} = \sum_{i=3}^{\infty} \frac{(x-x_0)^i}{i!}\left.\frac{d^i f}{dx^i}\right|_{x_0} \to 0$$

$+ R_{III}$ → 0

$E_p(x_0) = 0$ (originea E_p)

termeni anarmonici

$$\left.\frac{d^2 E_p(x)}{dx^2}\right|_{x_0} = k$$

CURS 3

Medii (operații mat. de

1) Media Temporară

fie $\begin{cases} F = F(x) \\ \text{sau} \\ f = f(t) \end{cases}$

O.A.C. $\langle f(t) \rangle_t = \bar{f} = \frac{1}{T} \int_0^T f(t)\,dt$; $t \in [0, T]$

$\langle x \rangle = \langle x^2 \rangle = \not{\in}$

t = timp
T = perioadă
x = spațiul
$x \in [a, b]$

2) Media Spațială

$\langle F(x) \rangle_S = \bar{F} = \frac{1}{b-a} \int_a^b F(x)\,dx$

$F(x) = a_1 x + b_1$

$\bar{F} = \frac{1}{a-b} \int_a^b (a_1 x + b_1)\,dx = \frac{1}{b-a} \left[a_1 \int_a^b x\,dx + b_1 \int_a^b dx \right]$

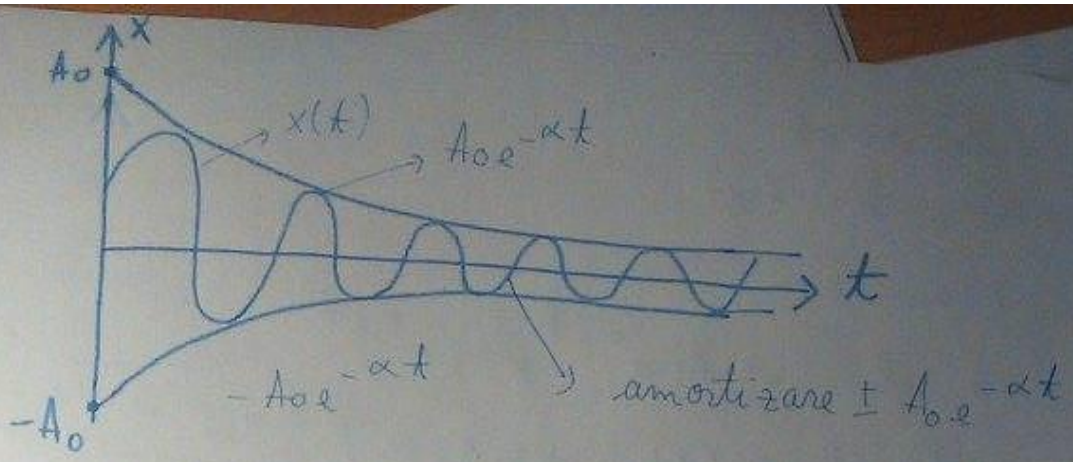
$\bar{F}_{\text{media}}$

$\bar{F} = \frac{1}{b-a} \left[\frac{a_1}{2} (b^2 - a^2) + b_1 (b-a) \right]$

$\bar{F} = \frac{a_1}{2} (b+a) + b_1$

$\bar{F} = \frac{a_1 b + a_1 a + 2 b_1}{2} = \frac{(a_1 b + b_1) + a_1 a + b_1}{2}$

$\bar{F} = \frac{F(b) + F(a)}{2}$



Caz 2: $\gamma > \omega_0$; $\Leftrightarrow \gamma^2 - \omega_0^2 > 0$

$$\alpha_{1,2} = -\gamma \pm \sqrt{\gamma^2 - \omega_0^2}$$

$$x(t) = A_1 e^{\alpha_1 t} + A_2 e^{\alpha_2 t}$$

$$x(t) = x_1(t) + x_2(t)$$

$$x_1(t) = A_1 e^{\alpha_1 t} = A_1 e^{(-\gamma + \sqrt{\gamma^2 - \omega_0^2})t}$$

$$x_2(t) = A_2 e^{\alpha_2 t} = A_2 e^{(-\gamma - \sqrt{\gamma^2 - \omega_0^2})t}$$

$$\gamma > \sqrt{\gamma^2 - \omega_0^2}$$

$$\lim_{t \to \infty} x_1(t) = \lim_{t \to \infty} A_1 e^{-(\gamma - \sqrt{\gamma^2 - \omega_0^2})t} = 0$$

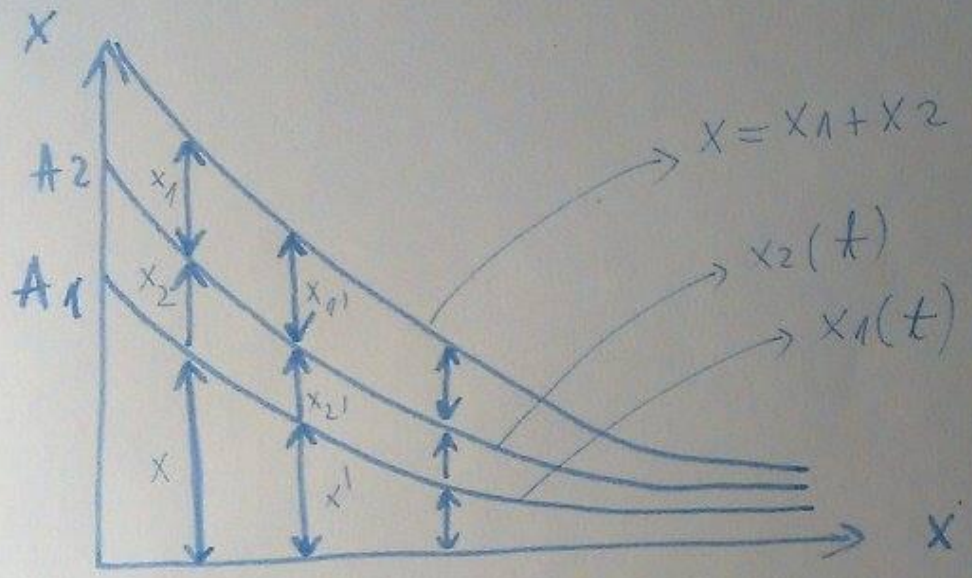
($\gamma - \sqrt{\gamma^2 - \omega_0^2} \to 0$)

$$\Rightarrow \lim_{t \to \infty} x_1(t) = 0$$

$$\lim_{t \to \infty} x_2(t) = \lim_{t \to \infty} A_2 e^{-(\gamma + \sqrt{\gamma^2 - \omega_0^2})t} = 0$$

($\gamma + \sqrt{\gamma^2 - \omega_0^2} \to 0$)

$$\Rightarrow \lim_{t \to \infty} x_2(t) = 0$$



$X_1 + X_2 = X$

$X_1' + X_2' = X'$

$F(x)$ liniară $\Rightarrow$ $\boxed{\bar{F} = \frac{F(b)+F(a)}{2}}$
$x \in [a, b]$

Compunerea oscilațiilor

Oscilații de aceeași frecvență, amplitudine $\begin{cases} \text{egală} \\ \text{inegală} \end{cases}$

Oscilații cu frecvențe diferite

ω = pulsația

$\omega = 2\pi\nu$, ν = frecvența

$T = \frac{1}{\nu}$, T = perioada

at întâlnăs? e chestie dă finețe

Funcții matematice

$x = A\cos(\omega t + \varphi)$

$x = B\sin(\omega t + \varphi)$

$x = A\cos(\omega t + \varphi_1) + B\sin(\omega t + \varphi_2)$

$\boxed{x = A\cos(\omega t + \varphi)}$

Avem $\begin{cases} x_1 = x_1(t) \\ x_2 = x_2(t) \end{cases}$ 2 oscilații cu $\omega_1 = \omega_2 = \omega$

$x_1 = x_{01}\cos(\omega t + \varphi_1)$

$x_2 = x_{02}\cos(\omega t + \varphi_2)$

$x = x(t)$ a.î. $x = x_1 + x_2$

$x(t) = x_{01}\cos(\omega t + \varphi_1) + x_{02}\cos(\omega t + \varphi_2) =$

$= x_{01}\cos\omega t\cos\varphi_1 - x_{01}\sin\omega t\sin\varphi_1 +$

$x_{02}\cos\omega t\cos\varphi_1 - x_{02}\sin\omega t\sin\varphi_2$

$x(t) = \underbrace{(x_{01}\cos\varphi_1 + x_{02}\cos\varphi_2)}_{x_0\cos\varphi}\cos\omega t + \underbrace{(x_{01}\sin\varphi_1 + x_{02}\sin\varphi_2)}_{x_0\sin\varphi}\sin\omega t$

$x(t) = x_0(\cos\varphi\cos\omega t + \sin\varphi\sin\omega t)$

$= x_0\cos(\omega t + \varphi) \Rightarrow \boxed{x(t) = x_0\cos(\omega t + \varphi)}$

$x_0\cos\varphi = x_{01}\cos\varphi_1 + x_{02}\cos\varphi_2$

$x_0\sin\varphi = x_{01}\sin\varphi_1 + x_{02}\sin\varphi_2 \Rightarrow \frac{x_{01}\sin\varphi}{x_{01}\cos\varphi} = \operatorname{tg}\varphi$

$\boxed{\varphi = \operatorname{arctg}\left\{\frac{x_{01}\sin\varphi_1 + x_{02}\sin\varphi_2}{x_{01}\cos\varphi_1 + x_{02}\cos\varphi_2}\right\}}$

$x_{01}^2\sin^2\varphi + x_{02}^2\sin^2\varphi = x_0^2$

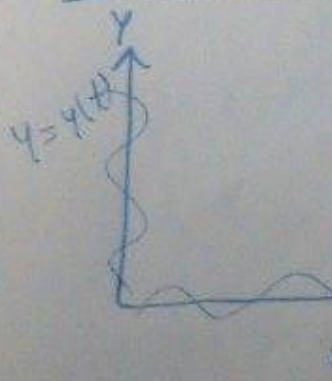
$x_0 = \sqrt{(x_{01}\cos\varphi_1 + x_{02}\cos\varphi_2)^2 + (x_{01}\sin\varphi_1 + x_{02}\sin\varphi_2)^2}$

$\boxed{x_0 = x_{01}^2 + x_{02}^2 + 2x_{01}x_{02}\cos(\varphi_1 - \varphi_2)}$

Observații oscilații pe aceeași axă (

)

Oscilații de aceeași frecvență dar pe axe diferite



$\begin{cases} x = x_0\cos\omega t \\ y = y_0\cos(\omega t + \varphi) \end{cases}$

$\frac{x}{x_0} = \cos\omega t \Rightarrow \frac{x^2}{x_0^2} = \cos^2\omega t$

$\frac{y}{y_0} = \cos\omega t\cos\varphi - \sin\omega t\sin\varphi$

$\sin\omega t\sin\varphi = \underbrace{\cos\omega t}_{\frac{x}{x_0}}\cos\varphi - \frac{y}{y_0}$

$\sin^2\omega t + \sin^2\varphi = \cos^2\omega t\cos^2\varphi + \frac{y^2}{y_0^2} - 2\frac{y}{y_0}\frac{x}{x_0}\cos\varphi$

$\sin^2 \omega t = 1 - \cos^2 \omega = 1 - \frac{x^2}{x_0^2}$

$$\left(1 - \frac{x^2}{x_0^2}\right)\sin^2\varphi = \frac{x^2}{x_0^2}\cos^2\varphi + \frac{y^2}{y_0^2} - 2\frac{x}{x_0}\frac{y}{y_0}\cos\varphi$$

$$\frac{x^2}{x_0^2}\left(\sin^2\varphi + \cos^2\varphi\right) + \frac{y^2}{y_0^2} - 2\frac{x}{x_0}\frac{y}{y_0}\cos\varphi =$$

$$= \sin^2\varphi$$

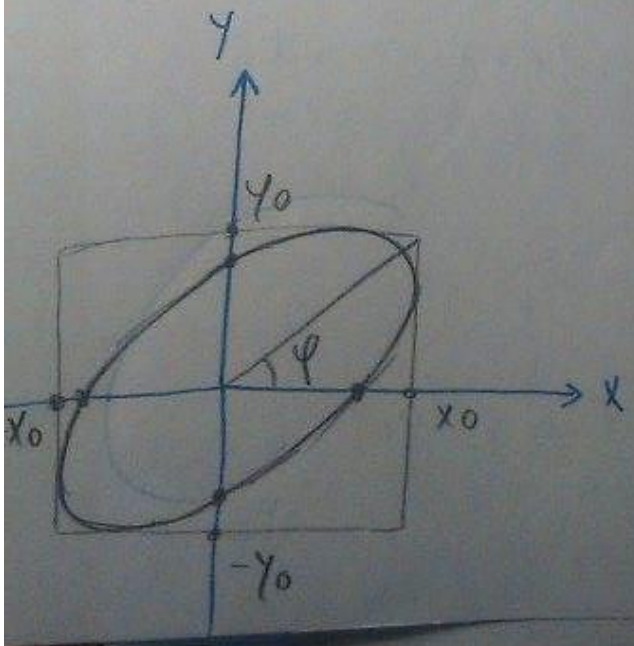
$$\frac{x^2}{x_0^2} + \frac{y^2}{y_0^2} - \left(\frac{x}{x_0}\right)\left(\frac{y}{y_0}\right)\cos\varphi - \left(\frac{y}{y_0}\right)\left(\frac{x}{x_0}\right)\cos\varphi$$

$$= \sin^2\varphi$$

2D: conică
- elipsă
- parabolă
- hiperbolă
- cerc

$$\Delta = \begin{vmatrix} \frac{1}{x_0^2} & -\frac{1}{x_0 y_0}\cos\varphi \\ -\frac{1}{x_0 y_0}\cos\varphi & \frac{1}{y_0^2} \end{vmatrix} = \frac{1}{x_0^2 y_0^2}\left(1 - \cos^2\varphi\right) =$$

$$= \frac{\sin^2\varphi}{x_0^2 y_0^2} > 0 \Rightarrow \text{elipsă}$$



$\varphi = \frac{\pi}{2}$ $\left\{ \begin{array}{l} \cos\frac{\pi}{2} = 0 \\ \sin\frac{\pi}{2} = 1 \end{array} \right.$ $\Rightarrow \frac{x^2}{x_0^2} + \frac{y^2}{y_0^2} = 1$

elipsă

$\varphi = 0$ $\left\{ \begin{array}{l} \cos 0 = 1 \\ \sin 0 = 0 \end{array} \right.$ $\Rightarrow \frac{x^2}{x_0^2} + \frac{y^2}{y_0^2} - 2\frac{x}{x_0}\cdot\frac{y}{y_0} = 0$

$\Rightarrow \left(\frac{x}{x_0} - \frac{y}{y_0}\right)^2 = 0$

$\Rightarrow \boxed{y = \pm \frac{y_0}{x_0} x}$

$\boxed{y = \pm \left(\frac{y_0}{x_0}\right) x}$

Oscilații (vibrații) cu un singur grad de libertate

1D:

$F_e = -kx$

$F_r = -\alpha v = -\alpha \dot{x}$

↑ forță de frecare prop. cu viteza

$F_f = F(t)$

$\sum_{i=1}^{3} F_i = ma$ (L.S. + P. Supr)

$F_e + F_r + F_f = ma$; $a = \frac{d^2x}{dt^2} = \ddot{x}$

$-kx - \alpha v + F(t) = m\ddot{x}$

$m\ddot{x} + \alpha\dot{x} + kx = F(t) \Rightarrow \boxed{\ddot{x} + \frac{\alpha}{m}\dot{x} + \frac{k}{m}x = \frac{1}{m}F(t)}$

Oscilația cu un singur grad de libertate:

$$\ddot{x} + 2\beta\dot{x} + \omega_0^2 x = \frac{1}{m} F(t) \quad (1)$$

$\Downarrow$ sol. $x = x(t)$

(1) ec. dif. grad II cu coef const – neomogenă!

Caz particular:

$\alpha = 0$, $F(t) = 0$

$$\ddot{x} + \omega_0^2 x = 0 \quad (2)$$

ec. dif. a oscilatorului armonic liniar

ec. dif de ord. II cu coef const - omogenă

$$x = x(t) \Rightarrow x = A e^{\alpha t}$$

$$\dot{x} = \frac{dx}{dt} = A\alpha e^{\alpha t}$$

$$\ddot{x} = \frac{d^2x}{dt^2} = A\alpha^2 e^{\alpha t}$$

$$\alpha^2 (A e^{\alpha t}) + \omega_0^2 (A e^{\alpha t}) = 0$$

$$\Rightarrow \boxed{\alpha^2 + \omega_0^2 = 0} \text{ ec. caracteristică}$$

$$\alpha_{1,2} = \pm \sqrt{-\omega_0^2} = \pm i\omega_0$$

$$\boxed{x(t) = A_1 e^{-i\omega_0 t} + A_2 e^{i\omega_0 t}}$$

$$e^{i\theta} = \cos\theta + i \sin\theta$$

$$z = e^{i\theta}$$

$$e^{in\theta} = \cos n\theta + i \sin n\theta$$

$$e^{in\theta} = (e^{i\theta})^n = (\cos\theta + i\sin\theta)^n$$

$$\cos n\theta + i \sin n\theta = (\cos\theta + i\sin\theta)^n$$

$\underbrace{\cos n\theta + i \sin n\theta}_{z_1} = \underbrace{(\cos\theta + i \sin\theta)^n}_{z_2}$

$Re(z_1) = Re(z_2)$

$Im(z_1) = Im(z_2)$

$A_1 = \frac{A_0}{2} e^{-i\varphi} \in \mathbb{C}$

$A_2 = \frac{A_0}{2} e^{i\varphi} \in \mathbb{C}$

A_1, A_2 = amplitudini complexe

$x(t) = \frac{A_0}{2} e^{-i\varphi} e^{-i\omega_0 t} + \frac{A_0}{2} e^{i\varphi} e^{i\omega_0 t} =$

$= \frac{A_0}{2} \left(e^{-i(\omega_0 t + \varphi)} + e^{i(\omega_0 t + \varphi)} \right)$

$x(t) = \frac{A_0}{2} \left(\cos(\omega_0 t + \varphi) - i \sin(\omega_0 t + \varphi) + \right.$

$\left. + \cos(\omega_0 t + \varphi) + i \sin(\omega_0 t + \varphi) \right)$

$\boxed{x(t) = A_0 \cos(\omega_0 t + \varphi)}$ - sol ec dif II

- oscilație armonică 1D

amplitudine — pulsație — fază oscilatorie

$F = -kx \Rightarrow x = A_0 \cos(\omega_0 t + \varphi)$

$m\ddot{x} + kx = 0 \Rightarrow \ddot{x} + \omega_0^2 x = 0$

Ep. în mod direct:

$\boxed{\vec{F} = -\nabla E_p}$

Teoremă: forța este - gradient din en. pot.

$\nabla E_p = \frac{\partial E_p(x,y,z)}{\partial x} \vec{i} + \frac{\partial E_p(x,y,z)}{\partial y} \vec{j} + \frac{\partial E_p(x,y,z)}{\partial z} \vec{k}$

1D: $E_p = E_p(x)$

$\vec{F} = F\vec{i}$, $\nabla E_p = \frac{dE_p(x)}{dx} \vec{i}$

$$F = -\frac{dE_p(x)}{dx} \\ F = -kx \Big\} \Rightarrow -kx = -\frac{dE_p(x)}{dx} \Rightarrow \frac{dE_p(x)}{dx} = kx$$

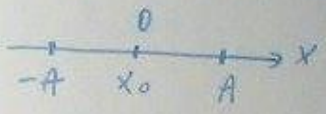
$$\Rightarrow dE_p(x) = kx\,dx \Rightarrow E_p = \int dE_p$$

$$\int E_p = k\int x\,dx$$

$$E_p(x) = \frac{kx^2}{2} + E_p(x=0)$$

$$\boxed{E_p(x) = \frac{kx^2}{2}}$$

1D:



Temă:

$$\nabla \times \nabla \times \vec{A} = -\Delta\vec{A} + \nabla(\nabla\cdot\vec{A})$$

rotor x rotor din $\vec{A}$ = - Laplacian $\vec{A}$ + gradient (div $\vec{A}$)

BANAL!

divergență

Caz particular 2

$$\ddot{x} + 2\gamma\dot{x} + \omega_0^2 x = \frac{1}{m} f(t)$$

$$F(t) = 0 \Rightarrow \boxed{\ddot{x} + 2\gamma\dot{x} + \omega_0^2 x = 0} \quad (3)$$

ec. dif grad III, coef. const, omogenă

Fie sol. de forma $x = Ae^{\alpha t}$

$$\dot{x} = A\alpha e^{\alpha t}$$

$$\ddot{x} = A\alpha^2 e^{\alpha t}$$

ec. caract. asociată ec. dif (3):

$$\boxed{\alpha^2 + 2\gamma\alpha + \omega_0^2 = 0}$$

$\alpha_{1,2} = -\gamma \pm \sqrt{\gamma^2 - \omega_0^2}$

~~$ax^2 + bx + c = 0$~~

~~$a = 1, \; b = 2\gamma$~~

~~$x_{1,2} = ?$~~

(3) are sol: $x = A_1 e^{\alpha_1 t} + A_2 e^{\alpha_2 t}$

$x = x(t)$ pt. ec. (3): $x = A_1 e^{-\gamma t + \sqrt{\gamma^2 + \omega_0^2}\, t} +$

$x = e^{-\gamma t} \left(A_1 e^{\sqrt{\gamma^2 - \omega_0^2}\, t} \right.$ $\left. A_2 e^{-\gamma t - \sqrt{\gamma^2 + \omega_0^2}\, t} \right.$
$\left. + A_2 e^{-\sqrt{\gamma^2 - \omega_0^2}\, t} \right)$

Cazul (I)

(I): $\omega_0 > \gamma \Rightarrow \sqrt{\gamma^2 - \omega_0^2} = \sqrt{-(\omega_0^2 - \gamma^2)} = i\sqrt{\omega^2 - \gamma^2}$

$x = e^{-\gamma t} \left(\frac{A_0}{2} e^{i\varphi} e^{i\sqrt{\omega_0^2 - \gamma^2}} + \frac{A_0}{2} e^{-i\varphi} e^{-i\sqrt{\omega_0^2 - \gamma^2}} \right.$

$x = e^{-\gamma t} \frac{A_0}{2} \left[e^{i(\sqrt{\omega_0^2 - \gamma^2} + \varphi)} e^{-i(\sqrt{\omega_0^2 - \gamma^2} + \varphi)} \right]$

$\omega = \sqrt{\omega_0^2 - \gamma^2}$

$x = A_0 e^{-\gamma t} \cos(\sqrt{\omega_0^2 - \gamma^2}\, t + \varphi)$

$\boxed{x = A_0 e^{-\gamma t} \cos(\omega t + \varphi)}$ POZĂ

$A_0 e^{-\gamma t}$, $\gamma > 0 \Rightarrow \lim_{t \to \infty} A_0 e^{-\gamma t} = 0$

factor de amortizare

$x = x(t)$

$\lim_{t \to \infty} x(t) = \lim_{t \to \infty} \left[A_0 e^{-\gamma t} \cos(\omega t + \varphi) \right] = 0$

$(\gamma > 0)$

$-1 \leq \cos(\omega t + \varphi) \leq 1$

Oscilație amortizată!

Easyhost
Știai că un server poate răspunde la mii de cereri într-o perioadă de câteva secunde?
Vino la proba Easyhost pentru a ne arăta ce conține...
mate... scheme ale... "Avioane" și nu ducă lipsă de idei care se pot traduce în cod? Înscrie-te acum!
Tehnoloan...

CURS 4

Mișcare de tip oscilator

1D: $\ddot{x} + 2\gamma\dot{x} + \omega_0^2 x = \frac{1}{m} F(t)$

ec. dif. ord II neomogenă, coef. const.

3D: $\ddot{\vec{r}} + 2\gamma\dot{\vec{r}} + \omega_0^2 \vec{r} = \frac{1}{m}\vec{F}(t)$ (vectorial)

$\vec{r} = x\vec{i} + y\vec{j} + z\vec{k}$

OBS: Mișcările reale sunt mișcări în spațiul (3D)

Cazul 3

(3) $F = F(t)$, $Fe = -kx$, $Ff = -\alpha\vec{v}$

1D: $\ddot{x} + 2\gamma\dot{x} + \omega_0^2 x = \frac{1}{m} F(t)$ (1) - admite soluția generală $x_g = x_g(t)$

$x_g(t) = x_g(t) = x_{\bar{o}}(t) + x_p$

$x_{\bar{o}}$ - soluția ec. omogene

x_p - soluție pentru particulară a ec. neomogenă.

Ec. omogenă:

$\ddot{x} + 2\gamma\dot{x} + \omega_0^2 x = 0$

I $\gamma < \omega_0$

II $\gamma > \omega_0$

III $\gamma = \omega_0$

/* ec. (1) repr. oscilații întreținute (se referă doar la cazul I!) */

OBS: $\gamma = \omega_0$

$\ddot{x} + 2\gamma\dot{x} + \gamma^2 x = 0$

$(\ddot{x} + \gamma\dot{x}) + \gamma(\dot{x} + \gamma x) = 0$ (2)

Notăm $X(t) = \dot{x} + \gamma x$

$\dot{X}(t) = \frac{dX(t)}{dt} = \ddot{x} + \gamma\dot{x}$

$\frac{d}{dt}$ - operator derivare liniar

$\Rightarrow$ (2) devine $\frac{dX}{dt} + \gamma X(t) = 0$

$\Rightarrow \frac{dX}{dt} = -\gamma X(t)$

$\Rightarrow \frac{dX}{X} = -\gamma\, dt \quad / \int$

$\Rightarrow \int \frac{dX}{X} = -\int \gamma\, dt \Rightarrow \ln X(t) = -\gamma t + c_0$

$(c_0 = \ln X_0)$

$\Rightarrow \ln X(t) = -\gamma t + \ln X_0$

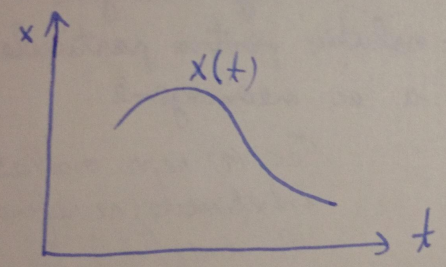
$\Rightarrow \ln \frac{X(t)}{X_0} = -\gamma t \Rightarrow X(t) = X_0 e^{-\gamma t}$

$X(t) = \dot{x} + \gamma x \Rightarrow X_0 e^{-\gamma t} = \frac{dx}{dt} + \gamma x, \quad x = x(t)$

Soluția ec. (2) este $x(t) = (c_1 t + c_2) e^{-\gamma t}$

(nu este oscilație)

$\lim_{t\to\infty} x(t) = \lim_{t\to\infty} c_1 t e^{-\gamma t} + \lim_{t\to\infty} c_2 e^{-\gamma t}$



$\lim_{t\to\infty} c_2 e^{-\gamma t} = 0$

$\lim_{t\to\infty} c_1 t e^{-\gamma t} = c_1 \lim_{t\to\infty} \frac{t}{e^{\gamma t}} \overset{l'H}{=} c_1 \lim \frac{1}{\gamma e^{\gamma t}}$

$= c_4 \lim_{t\to\infty} e^{-\gamma t} = 0, \quad c_4 = \frac{c_1}{\gamma}$

Admite sol. ec. amortizate

Rt. ec. (1) - sol. - caz I - $\gamma < \omega_0$

$x_{\bar{o}} = a_{\bar{o}} e^{-\gamma t} \cos(\omega_0 t + \varphi_0)$

$x_p = x_p(t)$

$F = F(t)$ forța

$F(t) = F_0 \cos(\omega t)$ (3)

$\omega \neq \omega_0$ $\omega = \frac{2\pi}{T}$ injectează energie în fiecare interval de timp $[0, T]$

Soluție de tip $Ae^{i\omega t}$

$F(t) \in \mathbb{C}$; $\tilde{F}(t) = F_0 e^{i\omega t}$ (*)

$\tilde{F}(t) = F_0 \cos(\omega t) + i F_0 \sin(\omega t)$

$Re(\tilde{F}(t)) = F_0 \cos(\omega t)$ (3) (tot (3))

$\ddot{x} + 2\gamma\dot{x} + \omega_0^2 x = \frac{1}{m} F_0 e^{i\omega t}$ (în $\mathbb{C}$) și

$x(t) = Ae^{i\omega t}$

$\left.\begin{aligned} \dot{x}(t) &= Ai\omega e^{i\omega t} \\ \ddot{x}(t) &= Ai^2\omega^2 e^{i\omega t} \\ i^2 &= -1 \end{aligned}\right\} \Rightarrow -\omega^2 Ae^{i\omega t} + 2\gamma i\omega A e^{i\omega t} + \omega_0^2 Ae^{i\omega t} = \frac{1}{m} F_0 e^{i\omega t}$

$\Rightarrow A\left[(\omega_0^2 - \omega^2) + i2\gamma\omega\right] = \frac{F_0}{m}$

$\Rightarrow A = \frac{F_0}{m} \cdot \frac{1}{(\omega_0^2 - \omega^2) + i2\gamma\omega}$

$$\boxed{A = \frac{F}{m} \frac{(\omega_0^2 - \omega^2) - i2\gamma\omega}{(\omega_0^2 - \omega^2)^2 + 4\gamma^2\omega^2}} \in \mathbb{C}$$

$A^* = \frac{F_0}{m} \frac{(\omega_0^2 - \omega^2) - i2\gamma^2\omega}{(\omega_0^2 - \omega^2)^2 + 4\gamma^2\omega^2} \Rightarrow AA^* = A^2$

Trecem în amplitudine complexă: $A = ae^{i\varphi}$

$A^* = ae^{-i\varphi}$

$AA^* = A^2 = a^2 e^{i\varphi} e^{-i\varphi} = a^2 \Rightarrow A^2 = a^2$, unde $a \in \mathbb{R}$

$\Rightarrow A^2 = a^2$

$A^2 = \left(\frac{F_0}{m}\right)^2 \left[\frac{(\omega_0^2 - \omega^2) - i2\gamma\omega}{(\omega_0^2 - \omega^2)^2 + 4\gamma^2\omega^2}\right]^2$

$$\boxed{A^2 = \left(\frac{F_0}{m}\right)^2 \frac{1}{(\omega_0^2 - \omega^2)^2 + 4\gamma^2\omega^2} = a^2}$$

$$a = \frac{F_0}{m} \frac{1}{\sqrt{(\omega_0^2 - \omega^2)^2 + 4\gamma^2\omega^2}}$$

$\tilde{x}_p(t) = A e^{i\omega t} = a e^{i\varphi} e^{i\omega t} = a e^{i(\omega t+\varphi)}$

$\tilde{x}_p(t) = a\cos(\omega t+\varphi) + i\, a \sin(\omega t+\varphi)$

$x_p(t) = \operatorname{Re} \tilde{x}_p(t) = a\cos(\omega t+\varphi)$

$a e^{i\varphi} = a\cos\varphi + i a \sin\varphi = \frac{F_0}{m}\left[\frac{(\omega_0^2-\omega^2)}{(\omega_0^2-\omega^2)^2+2\gamma^2\omega^2} - i\,\frac{2\gamma\omega}{(\omega_0^2-\omega)^2+2\gamma^2\omega^2}\right]$

$a\cos\varphi = \frac{F_0}{m}\,\frac{\omega_0^2-\omega^2}{(\omega_0^2-\omega^2)^2+2\gamma^2\omega^2}$

$a\sin\varphi = -\frac{F_0}{m}\,\frac{2\gamma\omega}{(\omega_0^2-\omega^2)+2\gamma^2\omega^2}$

$\frac{a\sin\varphi}{a\cos\varphi} = -\frac{2\gamma\omega}{\omega_0^2-\omega^2} \Rightarrow \operatorname{tg}\varphi = \frac{2\gamma\omega}{\omega^2-\omega_0^2}$

$\Rightarrow \boxed{\varphi = \operatorname{arctg}\frac{2\gamma\omega}{\omega^2-\omega_0^2}}$

$x_p(t) = a\cos(\omega t+\varphi)$

Sol. generală: $x_g(t) = a_0 e^{-\gamma t}\cos(\omega_0 t+\beta) + a\cos(\omega t+\varphi)$

Temă: verificare generală

Caz $\gamma^2 = 0$: $\ddot{x} + \omega_0^2 x = \frac{1}{m} F(t)$ (4) (oscilații forțate)

(4) - ec. dif. de ord. 2 neomogenă și coef. - const.

$x_g(t) = x_{\bar{0}}(t) + x_p(t)$

sau metoda variației constantelor

$\ddot{x} + \omega_0^2 x = 0$

$\ddot{x} - i\omega_0\dot{x} + i\omega_0\dot{x} + \omega_0^2 x = 0$

$(\ddot{x} - i\omega_0\dot{x}) + i\omega_0$

$(\ddot{x} + i\omega_0\dot{x}) - i\omega_0(\dot{x} + i\omega_0 x) = 0$

Notăm $\dot{x} + i\omega_0 x = X(t) \Rightarrow \dot{X}(t) = \ddot{x} + i\omega_0\dot{x}$

$\dot{X}(t) - i\omega_0 X(t) = 0 \Rightarrow \frac{dX}{dt} = i\omega_0 X$

easyhost
Știai că un server poate răspunde la mii de cereri într-o perioadă de câteva secunde?

$$\frac{dX}{X} = i\omega_0 dt \Rightarrow \int \frac{dX}{X} = i\omega_0 \int dt$$

$$\ln X = i\omega_0 t + \ln A \Rightarrow \boxed{X(t) = Ae^{i\omega_0 t}}$$

$X(t) = A(t)e^{i\omega_0 t}$ sol. ec. neomogene (4)

$$\frac{dX}{dt} = \frac{dA}{dt}e^{i\omega_0 t} + i\omega_0 A e^{i\omega_0 t}$$

$$\dot{X}(t) - i\omega_0 X(t) = \frac{1}{m}F(t)$$

$$\frac{dA}{dt}e^{i\omega_0 t} + i\omega_0 A e^{i\omega_0 t} - i\omega_0 A e^{i\omega_0 t} = \frac{1}{m}F(t)$$

$$\frac{dA}{dt}e^{i\omega_0 t} = \frac{1}{m}F(t) \Rightarrow dA = \frac{1}{m}F(t)e^{-i\omega_0 t}dt$$

$$\int dA = \frac{1}{m}\int_0^t F(t)e^{-i\omega_0 t}dt \Rightarrow \boxed{A(t) = \frac{1}{m}\int_0^t F(t)e^{-i\omega_0 t}dt}$$

$$X(t) = \frac{1}{m}\left[\int_0^t F(t)e^{-\omega_0 t}dt\right]e^{i\omega_0 t}$$

$$\int_{-\infty}^{+\infty}\ldots = \int_{-\infty}^{0}\ldots + \int_0^t\ldots + \int_t^{\infty}\ldots \quad , t \in \mathbb{R}+$$

$$\Rightarrow \int_{-\infty}^{0}\ldots = 0$$

$$\int_{-\infty}^{\infty} F(t)e^{-i\omega_0 t}dt \ (\leftarrow \text{transformata Fourier})$$

$$\Rightarrow \boxed{X(t) = \frac{1}{m}\left[\int_{-\infty}^{\infty} F(t)\cdot e^{-i\omega_0 t}dt\right]e^{i\omega_0 t}}$$

$X(t) = \dot{x}(t) + i\omega_0 x$, $X(t) \in \mathbb{C}$

$X(t) = Re(X(t)) + i\,Im(X(t))$

$Re(X(t)) = \dot{x}(t) = \frac{dx}{dt} = v$ – viteza

$Im(X(t)) = \omega_0 x \Rightarrow \boxed{x = X(t) = \frac{1}{\omega_0}Im\,X(t)}$

$\boxed{v = Re(X(t))}$

⑤ ↑ x mic

/* Mecanica newtoniană este o mecanică deterministă. Dacă se cunosc într-un punct din spațiu poziția și viteza, putem afla poziția și viteza în oricare punct. */

$\boxed{p = mv}$ - impuls

$\boxed{\omega_0^2 m = k}$

$X(t) = \dot{x}(t) + i\omega_0 x(t)$

$X^*(t) = \dot{x}(t) - i\omega_0 x(t)$

$X(t)X^*(t) = (X(t))^2 = \dot{x}^2(t) + \omega_0^2 x^2(t); \quad \dot{x} = v$

$\frac{m}{2}(X(t))^2 = \frac{mv^2}{2} + \frac{\omega_0^2 m}{2} x^2(t) \Rightarrow$

$\frac{m}{2}(X(t))^2 = \frac{mv^2}{2} + \frac{kx^2}{2}$ ($\frac{mv^2}{2} \to E_c$, $\frac{kx^2}{2} \to E_p$)

$\boxed{E = \frac{m}{2}(X(t))^2}$

Discuții:

Ex: mișcarea oscilatorie armonică

$\ddot{x} + \omega_0^2 x = 0 \quad (5)$

soluția $x(t) = A\cos(\omega_0 t + \varphi_0)$

Ec. dif (5) + condițiile inițiale = problema Cauchy

1D: (5) + c.i $\begin{cases} x(t_0 = 0) = x_0 & \text{(poziția inițială)} \\ \dot{x}(t_0 = 0) = v_0 & \text{(viteza inițială)} \end{cases}$

Fizica are rezolvarea problemei Cauchy

$x(t_0 = 0) = A\cos\varphi_0 = x_0$

$v(t) = -A\omega \sin(\omega_0 t + \varphi_0)$

$v_0 = -A\omega \sin\varphi_0$

$\cos\varphi_0 = \frac{x_0}{A}$

$\sin\varphi_0 = -\frac{v_0}{A\omega}$

$\sin^2\varphi + \cos^2\varphi = 1$

$\frac{x_0^2}{A^2} + \frac{v_0^2}{A^2\omega^2} = 1 \Rightarrow \boxed{A = \sqrt{x_0^2 + \frac{v_0^2}{\omega_0^2}}}$

$$\operatorname{tg}\varphi_0 = \frac{-\frac{v_0}{A\omega}}{\frac{x_0}{A}} = -\frac{v_0}{x_0\omega_0}$$

$$\varphi_0 = -\operatorname{arctg}\left(\frac{v_0}{x_0\omega_0}\right)$$

TERMODINAMICĂ

Procese reversibile

Temperaturi:

- Celsius (°C)
- Farenheit (°F)
- Absolută (T) ; $[T]_{S.I.} = K$, $T > 0$

t = temp. Celsius
t_0 = topirea gheții
t_f = fierberea apei

$$1°C = \frac{t_f - t_0}{100}$$

$$T(K) = 273,15 + t(°C)$$

$t = 273,16$ ← punctul triplu al apei

$T_0 = 0 + 273,15$

$T_f = 100 + 273,15$

$$\frac{T_f - T_0}{100} = \frac{100 + 273,15 - 273,15}{100}$$

$$\Rightarrow 1K = \frac{100}{100}°C \Rightarrow \boxed{1K = 1°C}$$

Transformarea politropă

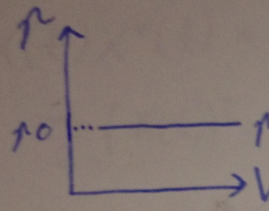
$pV^n = ct$

n = indicele politropei

p = presiune, v = volum (nivel microscopic)

1) $n = 0$

$p = ct \Rightarrow$ tr. izobară



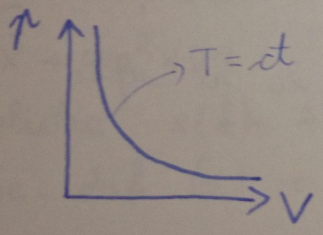
(p, V) - Ekapeyron Mendeleev

$V \not\to 0$ - $\not\exists$

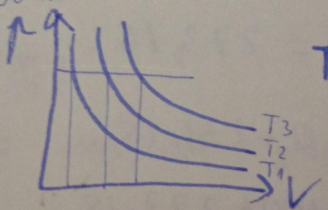
V_0 = volumul tuturor moleculelor

2) $n = 1$

$pV = ct \Rightarrow$ tr. izotermă $\Leftrightarrow t$ ct.



hiperbolă echilaterâ



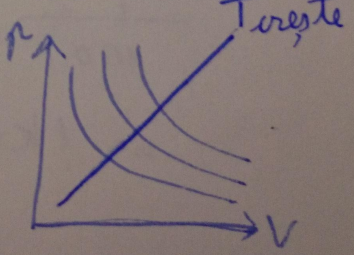
$T_1 < T_2 < T_3$

$pV = \nu RT$ - ec. de stare a gazului ideal

$\nu = \frac{m}{\mu} = \frac{m}{m_0}$

m_0 = masa unei molecule

$T = \frac{1}{\nu R} \cdot pV$



$$\left.\begin{aligned} T_2 &= \frac{1}{\nu R} p_1 V_2 \\ T_1 &= \frac{1}{\nu R} p_1 V_1 \end{aligned}\right\} \Rightarrow \frac{T_2}{T_1} = \frac{V_2}{V_1} \Rightarrow T_2 > T_1$$

$pV = \nu RT$ (ec. de stare a gazului ideal)

$\nu = \frac{m}{m_0}$ → masa moleculei

$T = \frac{1}{\nu R} pV$

CURS 5

$[V]_{SI} = m^3$; $[V] = 1\,l = 1\,dm^3 = 10^{-3} m^3$

$[p]_{SI} = \frac{N}{m^2}$ (la nivel macroscopic)

$[p] = \begin{cases} 760 \text{ mm col Hg} \\ 1 \text{ at} = 10^5 \text{ N/m}^2 \\ 1 \text{ atm (atmosferă tehnică)} \end{cases}$

$p_{hidrostatică} = \rho g h$

$f = f(p, V, T)$ → ec. de stare

$p = p(V, T)$ ↗

Presiuni reduse. Vidul

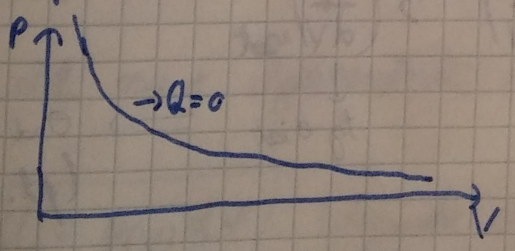
vid = grad de rarefiere maxim (0 particule)

p (tehnic) = → 10^{-3} (vid preliminar) (pompă de vidat)

→ 10^{-6} (vid bun) (pompă de difuzie)

3) $n = \gamma$ (transformare adiabatică)

$pV^\gamma = \text{const.}$ $(Q = 0)$



$\gamma = \frac{c_p}{c_V} = \frac{C_p}{C_V} = \frac{i+2}{i} = 1 + \frac{2}{i}$

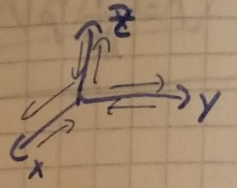
$C_p - C_V = R$ (Robert - Mayer)

C_p = căld. molară la p = const. ; $C_p = \frac{i+2}{2} R$

C_V = căld. molară la V = const. ; $C_V = \frac{i}{2} R$

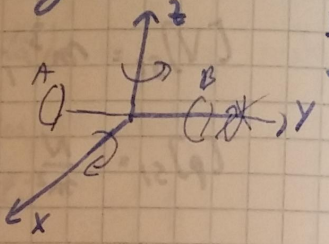
i = nr. grade de libertate

• gaz monoatomic : $i = 3$



3 translații

gaz diatomic : $i = 5$ model ganteră



3 translații
2 rotații

gaz poliatomic : $i = 6$ (3 translații + 3 rotații)

Lemă: Adiabata este mai înclinată decât izoterma

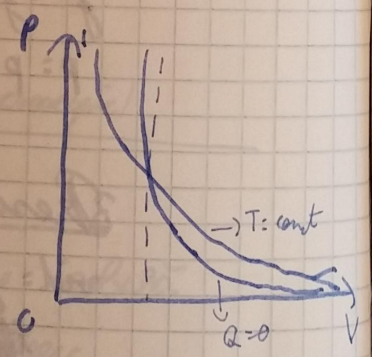
Obs: Adiabata și izoterma se întâlnesc o singură dată.

Dem.: $pV = const$

$d(pV) = 0$

$dp\,V + p\,dV = 0$

$\boxed{\left(\frac{dp}{dV}\right)_{izot} = -\frac{p}{V}}$ pt. T = const.



$pV^\gamma = const$

$d(pV^\gamma) = 0 \Rightarrow dp\,V^\gamma + \gamma p V^{\gamma-1} dV = 0 \;/\cdot V^{-\gamma}$

$dp + \gamma p V^{-1} dV = 0 \Rightarrow \frac{dp}{dV} = -\gamma p V^{-1} \Rightarrow$

$\Rightarrow \left(\frac{dp}{dV}\right)_{adiab} = \gamma\left(-\frac{p}{V}\right) > \left(\frac{dp}{dV}\right)_{izot}$

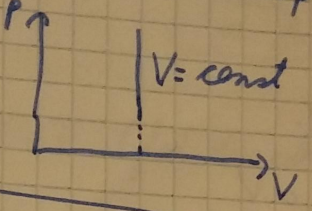
$\left(\frac{dp}{dV}\right)_{adiab} = tg\,\theta_{ad}$; $\left(\frac{dp}{dV}\right)_{izot} = tg\,\theta_{izot}$

$\Rightarrow \theta_{ad} > \theta_{izot}$

(pt. că $\theta_{izot}, \theta_{ad} \in [0, \frac{\pi}{2}]$)

4) $n \to \infty$ (izocoră)

$V = const$ ($pV^n = const \Rightarrow p^{\frac{1}{n}} V = const \overset{n \to \infty}{\Rightarrow} V = const$)



$pV = \nu RT$; R = const. gazelor ideale ; $R = K N_A$ → nr. Avogadro : $6{,}023 \times 10^{23}$ molecule/mol

ν = nr. moli ; $\nu = \frac{m}{\mu}$ → masă molară

K → const. lui Boltzmann

$pV = \frac{m}{\mu} N_A K T = \frac{N m_0}{N_A \cdot m_0} \cdot N_A K T$

$pV = NkT \Rightarrow p = \left(\frac{N}{V}\right) kT \Rightarrow \boxed{p = nkT}$ ($\frac{N}{V} \to n$)

Analiză critică

„Mărimi" speciale

U → energie internă

Q → căldură

L → lucrul mecanic

funcții de stare = U

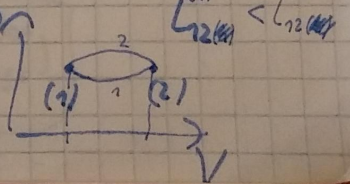
funcții de transformare (proces) = Q, L

funcție de stare $\Leftrightarrow$ admite diferențială totală exactă.

$\int_{(1)}^{(2)} dU = \int_{U_1}^{U_2} dU = U_2 - U_1 = \Delta U$

dL nu e diferențială totală exactă : $\int_{(1)}^{(2)} dL = L_{12}$

$\boxed{L = p(V)\,dV}$ (depinde de drum)



$\int_{(1)}^{(2)} dQ = Q_{12}$ (dQ nu e dif totala exacta)

Aplicatie

1) $p = const.$

$dL = p(V)\,dV$

$L_{12} = \int_{V_1}^{V_2} p(V)\,dV = p\int_{V_1}^{V_2} dV$

$\boxed{L_{12}^{izobar} = p(V_2 - V_1)}$

2) $pV = const.$

$L_{12} = \int_{V_1}^{V_2} p(V)\,dV = \int_{V_1}^{V_2} \frac{\nu RT}{V} dV = \nu RT \ln V\Big|_{V_1}^{V_2}$

$\boxed{L_{12}^{izot} = \nu RT \ln \frac{V_2}{V_1}}$

3) $V = const \Rightarrow dV = 0$

$\boxed{L_{12}^{izocor} = 0}$

4) $pV^\gamma = const$

$pV^\gamma = a \Rightarrow p = aV^{-\gamma}$

$p_1V_1^\gamma = p_2V_2^\gamma = a$

$L_{12}^{adiab} = \int_{V_1}^{V_2} p(V)\,dV = \int_{V_1}^{V_2} aV^{-\gamma}dV = a\frac{V^{1-\gamma}}{1-\gamma}\Big|_{V_1}^{V_2}$

$L_{12}^{adiab} = \frac{a}{\gamma-1}(V_1^{1-\gamma} - V_2^{1-\gamma}) = \frac{1}{\gamma-1}(aV_1^{1-\gamma} - aV_2^{1-\gamma})$

$L_{12}^{adiab} = \frac{1}{\gamma-1}(p_1V_1 - p_2V_2)$

$\boxed{L_{12}^{adiab} = \frac{p_1V_1 - p_2V_2}{\gamma-1}}$ $\boxed{L_{12}^{adiab} = \frac{\nu R(T_1 - T_2)}{\gamma-1}}$

$$\frac{1}{\gamma-1} = \frac{1}{\frac{C_p}{C_V}-1} = \frac{C_V}{R} \Rightarrow L_{12}^{adiab} = \nu C_V (T_1 - T_2)$$

$$\boxed{L_{12}^{adiab} = -\nu C_V \Delta T}$$

Def.: $U = \nu C_V T$ (fct. de stare)

$$\int_{U_1}^{U_2} dU = \int_{T_1}^{T_2} \nu C_V \, dT = \nu C_V (T_2 - T_1)$$

$$\text{Dar } \int_{U_1}^{U_2} dU = U_2 - U_1 = \Delta U$$

$$\Rightarrow \Delta U = \nu C_V \Delta T \Rightarrow \boxed{L_{12}^{ad.} = -\Delta U}$$

I. Principiul I al termodinamicii

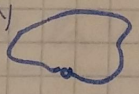
matematic: $dQ = dL + dU$

$$Q = \Delta U + L$$

$$dQ = dL + dU \Rightarrow \int_{(1)}^{(2)} dQ = \int_{(1)}^{(2)} dL + \int_{(1)}^{(2)} dU \Rightarrow \boxed{Q_{12} = L_{12} + \Delta U}$$

convenție fizică

Dt. un proces ciclic, $Q = \Delta U + L \Rightarrow Q = L$



3 cazuri

1) $Q = 0$; $L = 0$
2) $Q > 0$; $L > 0$ (mașină termică)
3) $Q < 0$; $L < 0$

Cazuri particulare

1) $p = \text{const.}$

$$Q_p = p(V_2 - V_1) + \nu C_V (T_2 - T_1) = \nu R (T_2 - T_1) + \nu C_V (T_2 - T_1)$$

$$\boxed{Q_p = \nu C_p (T_2 - T_1)}$$

Temă: transformare liniară $\boxed{p = aV + b}$; $L = ?$; $Q = ?$

Def. mat. a p. I

Principiul I al termodinamicii e principiul care introduce energia internă ($U \rightarrow$ fct. de stare).

II. Principiul al II-lea al termodinamicii

1) Def 1 → introduce entropia ($S \rightarrow$ fct. de stare)

$$\int_{(1)}^{(2)} dS = \int_{S_1}^{S_2} dS = S_2 - S_1 = \Delta S$$

entropie = măsură a gradului de dezordine dintr-un sistem (în mod natural, entropia crește)

Discuție: dS = formă Pfaff Lonemă

dQ nu e dif. tot. exactă

$dQ \rightarrow \frac{1}{T}$ admite factor integrant

$\frac{1}{T} dQ = dS$

2) Def. 2 (Caratheodory)

În vecinătatea oricărei stări, există stări care nu pot fi atinse în mod adiabatic.

3) Def. 3

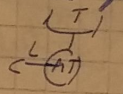
O adiabată și o izotermă nu se pot întâlni de 2 ori astfel încât să formeze un ciclu.

4) Thompson (Lord Kelvin)

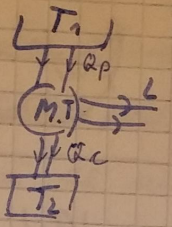
Căldura nu poate să fie luată de pe un corp (răcindu-l) și transferată pe alt corp (încălzindu-l)

fără a se produce schimbări în alte corpuri (adică lucru mecanic).

Obs.: O mașină termică monotermă nu poate să realizeze singură transferul căldurii



5) Ostwald: imposibilitatea construirii unui perpetuum mobile de speța a II-a



$T_1 > T_2$

Calculare entropiei

$$dS = \frac{dQ}{T}$$

$$dQ = pdV + \nu C_V dT$$

$$p = \frac{\nu RT}{V} \Rightarrow dS = \nu R \frac{dV}{V} + \nu C_V \frac{dT}{T}$$

$$\int dS = \nu R \int \frac{dV}{V} + \nu C_V \int \frac{dT}{T}$$

$$\boxed{S = S_0 + \nu R \ln V + \nu C_V \ln T}$$

$$\int_{S_1}^{S_2} dS = \nu R \int_{V_1}^{V_2} \frac{dV}{V} + \nu C_V \int_{T_1}^{T_2} \frac{dT}{T}$$

$$\boxed{\Delta S = \nu R \ln \frac{V_2}{V_1} + \nu C_V \ln \frac{T_2}{T_1}}$$

Ieșă: $S = f(p,T)$
$S = f(p,V)$

Formula lui Boltzmann

$S = -k \ln \Omega$;

Ω = prob. termică a macrostării
Ω = nr. de microstări al macrostării
k = const. Boltzmann

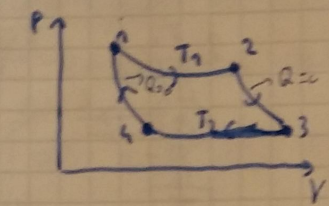
→ formula face legătura dintre termodinamică și fizica statistică („absolut exact")

7.05.2015

CURS 6

Postulatele Kelvin-Planck

Ciclul Carnot (2 ad + 2 izot.)

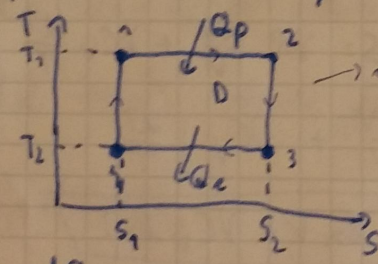


$T_1 > T_2 \Rightarrow \eta_c = 1 - \frac{T_2}{T_1}$

$\eta = 1 - \frac{|Q_c|}{Q_p} = \frac{Q_p - |Q_c|}{Q_p} = \frac{L_{ef}}{Q_p}$ → L_{ef} efectuat (lucru mec. pe ciclu)

$\begin{cases} Q_c < 0 \\ Q_p > 0 \end{cases}$ $\eta < 1$

(S, T) → coordonate speciale



→ ciclu Carnot

$dS = \frac{dQ}{T} \Rightarrow dQ = TdS \Rightarrow \int dQ = \int TdS = \text{Aria } D$

Th. un proces ciclic: $\oint$ ⇒ integrală pe contur închis

$\oint dQ = \oint TdS$

$dQ = dU + pdV \Rightarrow \oint (dU + pdV) = \oint TdS$

stare init = stare finală $\Rightarrow \oint dU = 0$ $\Rightarrow$

$\Rightarrow \oint pdV = \oint TdS \Rightarrow \oint dL = \oint TdS \Rightarrow \oint TdS = L_{ef}$

$Q_p = Q_{12} = T_1 \int_{S_1}^{S_2} dS = T_1 (S_2 - S_1)$

$L_{ef} = \oint TdS = \int_{S_1}^{S_2} T_1 dS + \int_{S_2}^{S_1} T_2 dS = (T_1 - T_2)(S_2 - S_1)$

$L_{ef} = (T_1 - T_2)(S_2 - S_1)$; $\eta_c = \frac{L_{ef}}{Q_p} = \frac{(T_1 - T_2)(S_2 - S_1)}{T_1 (S_2 - S_1)} = 1 - \frac{T_2}{T_1}$

Postulate

1) Randamentul ciclului Carnot nu depinde de natura gazului de lucru. Randamentul ciclului Carnot depinde doar de T_1 (temp. rez. maxim) și T_2 (temp. rez. minim)

2) Randamentul ciclului Carnot e cel mai mare rand al oricărui ciclu între aceleași temp T_1 și T_2.

Corolar: Mașina termică ce util. ciclul Carnot e cea mai eficientă.

Funcții din termodinamică. Potențiale termodinamice

Tot. term. ar trebuie să fie fct. de stare (și admită toate derivatele)

1) U (energia internă)

$U = \nu C_V T$ (nu e suficient)

Pi: $dQ = dU + dL \Rightarrow dU = dQ - dL \Rightarrow dU = TdS - pdV$

6 să ne ocupăm cu $f = f(p, V, T, S..)$

$$df = \frac{\partial f}{\partial p} dp + \frac{\partial f}{\partial V} dV + \frac{\partial f}{\partial T} dT + \frac{\partial f}{\partial S} dS$$

Ex.: $f = f(p, V, T)$ - fct. de stare

Fie A_1, A_2 variabile termodinamice; $f = f(A_1, A_2) \Rightarrow$ potențial termodinamic (fct. de 2 var)

$$\Rightarrow df = \left(\frac{\partial f}{\partial A_1}\right)_{A_2} dA_1 + \left(\frac{\partial f}{\partial A_2}\right)_{A_1} dA_2 \quad ; \quad \left(\frac{\partial f}{\partial p}\right)_T \text{ sau } \left(\frac{\partial f}{\partial p}\right)_V$$

(A_2 = var. constantă)

6

$dU = TdS - pdV \Rightarrow U = U(S, V) \Rightarrow dU = \left(\frac{\partial U}{\partial S}\right)_V dS + \left(\frac{\partial U}{\partial V}\right)_S dV$

Deci, $\begin{cases} \left(\frac{\partial U}{\partial S}\right)_V = T & (K) \\ \left(\frac{\partial U}{\partial V}\right)_S = -p & (P) \end{cases}$

Lema lui Schwarz: $\frac{\partial^2 f}{\partial x_1 \partial x_2} = \frac{\partial^2 f}{\partial x_2 \partial x_1}$ (orice dif. tot. exactă permite lema lui Schwarz)

$$\frac{\partial^2 U}{\partial S \partial V} = \frac{\partial^2 U}{\partial V \partial S}$$

$$\left.\begin{aligned} \frac{\partial^2 U}{\partial S \partial V} &= \frac{\partial}{\partial S}\left(\frac{\partial U}{\partial V}\right) = \frac{\partial}{\partial S}(-p) = -\left(\frac{\partial p}{\partial S}\right)_V \\ \frac{\partial^2 U}{\partial V \partial S} &= \frac{\partial}{\partial V}\left(\frac{\partial U}{\partial S}\right) = \frac{\partial T}{\partial V} = \left(\frac{\partial T}{\partial V}\right)_S \end{aligned}\right\} \Rightarrow$$

$$\Rightarrow \boxed{\left(\frac{\partial T}{\partial V}\right)_S = -\left(\frac{\partial p}{\partial S}\right)_V} \text{ (ecuația I a lui Maxwell)}$$

Obținem $\delta = \delta(p, V, T)$ eliminând dU, dS din $(\alpha), (\beta)$.

$$dQ = dU + pdV$$

$$\boxed{dQ = dU - \left(\frac{\partial U}{\partial V}\right)_S dV} \rightarrow U \rightarrow \text{candidat perfect pt. potențial termodinamic}$$

---

2) Entalpia

$$H = U + pV$$

$$\left.\begin{aligned} dH &= dU + pdV + Vdp \\ dU &= dQ - pdV \end{aligned}\right\} \Rightarrow dH = dQ + Vdp$$

$$dH = TdS + Vdp \Rightarrow H = H(S, p)$$

$$dH = \left(\frac{\partial H}{\partial S}\right)_p dS + \left(\frac{\partial H}{\partial p}\right)_S dp$$

$$\Rightarrow \begin{cases} \left(\frac{\partial H}{\partial S}\right)_p = T \\ \left(\frac{\partial H}{\partial p}\right)_S = V \end{cases}$$

$$\left.\begin{aligned} \frac{\partial^2 H}{\partial S \partial p} &= \left(\frac{\partial V}{\partial S}\right)_p \\ \frac{\partial^2 H}{\partial p \partial S} &= \left(\frac{\partial T}{\partial p}\right)_S \end{aligned}\right\} \overset{\text{Schwarz}}{\Rightarrow}$$

$$\Rightarrow \boxed{\left(\frac{\partial V}{\partial S}\right)_p = \left(\frac{\partial T}{\partial p}\right)_S} \quad \text{(Maxwell 2)}$$

$H = U + pV \Rightarrow U = H - pV \Rightarrow \boxed{U = H - \left(\frac{\partial H}{\partial p}\right)_S p}$ → fct. stare; → Q, L (am dem cu ce din U se poate calcula)

---

3) Energia libera (F)

Def: $F = U - TS$

$dF = dU - TdS - SdT$

$dF = TdS - pdV - TdS - SdT$

$dF = -pdV - SdT \Rightarrow F = F_{(V,T)}$

$$dF = \left(\frac{\partial F}{\partial V}\right)_T dV + \left(\frac{\partial F}{\partial T}\right)_V dT \Bigg\} \Rightarrow$$

$$\Rightarrow \begin{cases} \left(\frac{\partial F}{\partial V}\right)_T = -p \\ \left(\frac{\partial F}{\partial T}\right)_V = -S \end{cases}$$

$$\left(\frac{\partial^2 F}{\partial T \partial V}\right) = -\left(\frac{\partial p}{\partial T}\right)_V \Bigg\} \Rightarrow \boxed{\left(\frac{\partial p}{\partial T}\right)_V = \left(\frac{\partial S}{\partial V}\right)_T}$$

$$\left(\frac{\partial^2 F}{\partial V \partial T}\right) = -\left(\frac{\partial S}{\partial V}\right)_T$$

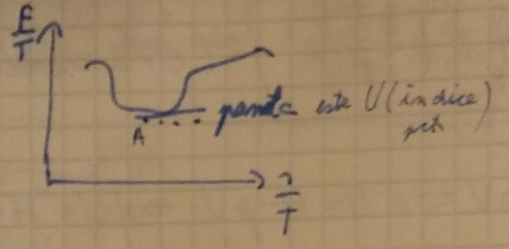
(Maxwell 3)

$U = F + TS \Rightarrow \boxed{U = F - T \cdot \left(\frac{\partial F}{\partial T}\right)_V} \Rightarrow F$ pot. term.

$$U = F - \left(\frac{\partial F}{\partial T}\right)_V T$$

$$\frac{\partial}{\partial T}\left(\frac{F}{T}\right) = \frac{1}{T}\left(\frac{\partial F}{\partial T}\right)_V - \frac{1}{T^2}F = -\frac{1}{T^2}\left[F - T\left(\frac{\partial F}{\partial T}\right)_V\right] = -\frac{1}{T^2}U \Rightarrow$$

$$\Rightarrow U = \frac{\frac{\partial}{\partial T}\left(\frac{F}{T}\right)}{-\frac{1}{T^2}} \Rightarrow \boxed{U = \frac{\frac{\partial}{\partial T}\left(\frac{F}{T}\right)}{\frac{\partial}{\partial T}\left(\frac{1}{T}\right)}}$$



$$tg_A\,\alpha = \frac{\partial\left(\frac{E}{T}\right)}{\partial\left(\frac{1}{T}\right)}$$

---

4) Entalpia liberă (Potential Gibbs) (G)

$$G = H - TS \;;\quad H = U + pV$$

$$dG = dH - TdS - SdT$$

$$dH = dU + dpV + pdV = TdS - pdV + Vdp + pdV = TdS + Vdp$$

$$\Rightarrow \boxed{dG = Vdp - SdT} \;;\quad G = G(p,T)$$

$$dG = \left(\frac{\partial G}{\partial p}\right)_T dp + \left(\frac{\partial G}{\partial T}\right)_p dT$$

Deci,
$$\begin{cases} \left(\frac{\partial G}{\partial p}\right)_T = V \\ \left(\frac{\partial G}{\partial T}\right)_p = -S \end{cases}$$

$$\frac{\partial^2 G}{\partial p \partial T} = \frac{\partial^2 G}{\partial T \partial p} \Rightarrow \boxed{-\left(\frac{\partial S}{\partial p}\right)_T = \left(\frac{\partial V}{\partial T}\right)_p}$$

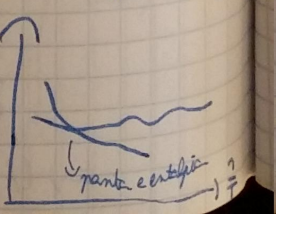
(Maxwell 4)

$$H = G + TS = G - T\cdot\left(\frac{\partial G}{\partial T}\right)_p$$

$$\boxed{H = G - \left(\frac{\partial G}{\partial T}\right)_p \cdot T} \rightarrow \text{G pot. termodinamic}$$

$$\frac{\partial}{\partial T}\left(\frac{G}{T}\right) = \frac{1}{T}\frac{\partial G}{\partial T} - \frac{1}{T^2}G = -\frac{1}{T^2}\left(G - T\left(\frac{\partial G}{\partial T}\right)_p\right) \Rightarrow$$

$$\Rightarrow H = \frac{\frac{\partial}{\partial T}\left(\frac{G}{T}\right)}{-\frac{1}{T^2}} \Rightarrow H = \frac{\frac{\partial}{\partial T}\left(\frac{G}{T}\right)}{\frac{\partial}{\partial T}\left(\frac{1}{T}\right)}$$



Principiul al III-lea al termodinamicii
(Nernst și Planck)

Preambul

F energie liberă

$F = U - TS$

$dF = dU - TdS - SdT$

$dF = TdS - pdV - TdS - SdT$

$dF = -pdV - SdT$

Iz. $T = const \Rightarrow dT = 0 \Rightarrow dF = -pdV \Rightarrow \int dF = -\int pdV = -\int dL$

$\int_{(1)}^{(2)} dF = -\int_{(1)}^{(2)} dL \Rightarrow F_2 - F_1 = -L_{12}$

Iz. proces reversibil: $F_1 = F_2 = 0 \Rightarrow F_1 = F_2 = 0$

$L_{12} = 0 \Leftrightarrow V = ct.$

Deci, $\begin{matrix} V=ct \\ T=ct \end{matrix} \Rightarrow F_1 = F_2$ (proc. reversibil)

Iz. proces ireversibil:

$F_1 - F_2 \geq 0 \Rightarrow F_1 \geq F_2$

$(\Sigma_1) \rightarrow (\Sigma_2) \rightarrow \dots \rightarrow (\Sigma_n)$ stări ale procesului

$F_1 \geq F_2 \geq \dots \geq F_n$

$dF = \left(\frac{\partial F}{\partial V}\right)_T dV + \left(\frac{\partial F}{\partial T}\right)_V dT$

$F = U + T\left(\frac{\partial F}{\partial T}\right)_V$. Aplicăm operatorul Δ. $\rightarrow \Delta F = F_2 - F_1$

$$\boxed{\Delta F = \Delta U = \Delta\left[T\left(\frac{\partial F}{\partial T}\right)_V\right]}$$

Boltzmann
$S = -k \ln \Omega$

$S = -k \ln P(E)$
↳ probabilitatea ...

8.09.2015

CURS 7

Principiul al III-lea. Consecințe

Q, L: Planck, Nernst etc.

ΔU, ΔF etc. (experimental)

$\lim_{T \to 0} S_0 = 0$ (adiabată de 0 coincide cu izot de 0)

Când $T \to 0$, S nu mai e f. de stare ci o const: $\lim_{T \to 0} \Delta S = 0$

→ P III stabilește entropia de 0 ($S_0 = 0$ la 0K!)

① Imposibilitatea atingerii temperaturii 0 absolut ($T_0 \neq 0K$)

filosofi: moartea termică a universului

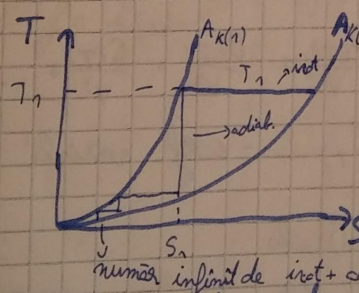
S crește! (S → măsură a gradului de dezordine)

Dar $dS = \frac{dQ}{T} \to$ scade ; $S = \int \frac{dQ}{T} \to$ scade ⇒ moartea univ.

S, T ⊕ A_k → parametri intensivi

$S = S(T, A_k)$; $k = \overline{1,n}$

$dS = \sum_{k=1}^{n} \frac{\partial S}{\partial A_k} dA_k + \frac{\partial S}{\partial T} dT$



(1) și (2) stări

Așadar, grafic se demonstrează că nu putem ajunge la 0K cu un nr. finit de pași.

Ce învățăm: termodinamica proceselor reversibile

procese ideale (gaz ideal) ≠ procese term. reale (gaz real) (despre astea am vorbit)
↳ ecuația van der Waals

Ecuația van der Waals

Ideal: $pV = \nu RT$

$p = \frac{\nu RT}{V}$; $V \to 0$ (V_0 b lim = vol. mol.)

$p \sim \frac{\nu RT}{V - Nb}$ (p e mai mare) $N \to$ nr. mol.

$[b]_{SI} = m^3$
↳ covolum

$p = \frac{\nu RT}{V - Nb} - a\frac{N^2}{V^2}$

$n = \frac{N}{V}$ → nr. de ciocniri $= n(n-1) \sim n^2 = \frac{N^2}{V^2}$

Deci, $\boxed{\left(p + a \cdot \frac{N^2}{V^2}\right)(V - Nb) = \nu RT}$ $a, b \to$ constante

$p = NkT$ (ideal)

Teoria virialului (coeficienți viriali)

$\left(p + a\frac{N^2}{V^2}\right)(V - Nb) = NkT$

$\nu = \frac{N}{N_A}$; $R = kN_A \Rightarrow \nu R = Nk$

$pV = NkT$; $p = \frac{NkT}{V}$ (dezv. în serie)

$p = \frac{NkT}{V}\left(1 + \frac{A(T)}{V} + \frac{B(T)}{V^2} + \frac{C(T)}{V^3} + \dots\right)$ ① dezv. în serie cos T, V

$A(T), B(T), C(T) \dots$ - coef. viriali

Cf. van der Waals

$p = \frac{NkT}{V - Nb} - a\frac{N^2}{V^2} \Rightarrow p = \frac{NkT}{V\left(1 - \frac{Nb}{V}\right)} - \frac{aN^2}{V^2}$

$$p = \frac{NkT}{V}\left[\left(1-\frac{Nb}{V}\right)^{-1} - \frac{aN}{VkT}\right]$$

Desv. serie Taylor:

$$f(x) = f(x_0) + \frac{x-x_0}{1}\frac{df}{dx}(x_0) + \dots \qquad \frac{1}{1-x} = 1+x+x^2+\dots$$

$$\left(1-\frac{Nb}{V}\right)^{-1} = \frac{1}{1-\frac{Nb}{V}} = 1 + \frac{Nb}{V} + \left(\frac{Nb}{V}\right)^2 + \dots + \cdot$$

Deci, $p = \frac{NkT}{V}\left(1 + \frac{Nb}{V} + \left(\frac{Nb}{V}\right)^2 + \dots - \frac{aN}{VkT}\right)$

$$p = \frac{NkT}{V}\left[1 + \frac{N}{V}\left(b - \frac{a}{kT}\right) + \frac{N^2b^2}{V^2} + \frac{N^3b^3}{V^3} + \dots\right]$$

Din (1) $\Rightarrow$
$$\begin{cases} A(T) = \left(b - \frac{a}{kT}\right)\cdot N \\ B(T) = N^2b^2 \\ C(T) = N^3b^3 \\ \vdots \end{cases}$$

Fizica statistică

Statistica matematică

→ lucrează cu funcții de distribuție (repartiție), densitatea fct. de distr.,

$F = F(x)$ fct. de repart.

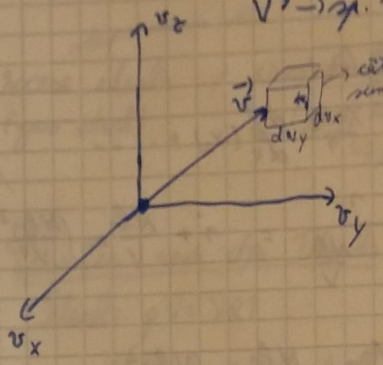
$f = f(x)$ dens. fct. de repart.

Teorie:

Fie $f = f(x)$.
$$\begin{cases} 1)\ f = f(x) \geq 0 \quad \forall x \in \mathbb{R} \\ 2)\ F(x) = \int_{-\infty}^{x} f(x)\,dx \\ 3)\ \int_{-\infty}^{\infty} f(x)\,dx = 1 \\ 4)\ \int_a^b f(x)\,dx = P(a \leq x \leq b); \end{cases}$$

Repartiția Maxwell (repartiția molec după viteze)

$V^3 \to$ sp. vitezelor



$\vec{v} = v_x \vec{i} + v_y \vec{j} + v_z \vec{k}$

$v^2 = v_x^2 + v_y^2 + v_z^2$

Cu această rep, Maxwell vrea să găs. nr. de molec. care au vit. cuprinsă în intervalul $\vec{v} \in [\vec{v}, \vec{v} + \Delta\vec{v}] \Leftrightarrow$

$$\Leftrightarrow \begin{cases} v_x \in [v_x, v_x + \Delta v_x] \\ v_y \in [v_y, v_y + \Delta v_y] \\ v_z \in [v_z, v_z + \Delta v_z] \end{cases}$$

$d\vec{v} = dv_x\, dv_y\, dv_z$

Matematic: câte molec au vârful vitezei în cubul $d\vec{v} = dv_x \cdot dv_y \cdot dv_z$

Fie acest număr $dn(v_x, v_y, v_z)$

$dn = dn(v_x, v_y, v_z) \sim n_0$ → concentrația

$dn \sim e^{-\frac{m_0 v^2}{2kT}}$ $\quad k \to$ const. Boltzmann

$dn \sim d\vec{v} = dv_x \cdot dv_y \cdot dv_z$

Rep. Maxwell: $dn(v_x, v_y, v_z) = n_0 C_M e^{-\frac{m_0 v^2}{2kT}} dv_x dv_y dv_z$ → coef. lui Maxwell

sau $dn(v_x, v_y, v_z) = C_M n_0 e^{-\frac{m_0 v_x^2}{2kT}} \cdot e^{-\frac{m_0 v_y^2}{2kT}} \cdot e^{-\frac{m_0 v_z^2}{2kT}} dv_x dv_y dv_z$

Fie $g(x)$ fct. Atunci $\bar{g}(x) = \langle g(x) \rangle = \int_{-\infty}^{\infty} g(x) f(x)\, dx$

$\frac{dn}{n_0} = f(v_x, v_y, v_z)\, dv_x dv_y dv_z$

↳ densă fct. de rep. Maxwell

$$f(v_x, v_y, v_z) = C_M e^{-\frac{m_0 v_x^2}{2kT}} e^{-\frac{m_0 v_y^2}{2kT}} e^{-\frac{m_0 v_z^2}{2kT}}$$

$$F = F(v_x, v_y, v_z)$$

$$\bar{F} = \langle F \rangle = \iiint_D F \cdot f(v_x, v_y, v_z)\, dv_x\, dv_y\, dv_z$$

$$\bar{F} = \langle F(v_x, v_y, v_z) \rangle = C_M \int_{-\infty}^{\infty}\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} F(v_x, v_y, v_z)\, e^{-\frac{m_0 v_x^2}{2kT}} e^{-\frac{m_0 v_y^2}{2kT}} e^{-\frac{m_0 v_z^2}{2kT}} dv_x dv_y dv_z$$

$$\iiint_D f(v_x, v_y, v_z)\, dv_x\, dv_y\, dv_z = 1$$

$$\int_{-\infty}^{\infty}\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} C_M e^{-\frac{m_0 v_x^2}{2kT}} e^{-\frac{m_0 v_y^2}{2kT}} e^{-\frac{m_0 v_z^2}{2kT}} dv_x dv_y dv_z = 1 \Rightarrow$$

$$\Rightarrow C_M = \frac{1}{\left(\int_{-\infty}^{\infty} e^{-\frac{m_0 v_x^2}{2kT}} dv_x\right) \cdot \left(\int_{-\infty}^{\infty} e^{-\frac{m_0 v_y^2}{2kT}} dv_y\right) \cdot \left(\int_{-\infty}^{\infty} e^{-\frac{m_0 v_z^2}{2kT}} dv_z\right)}$$

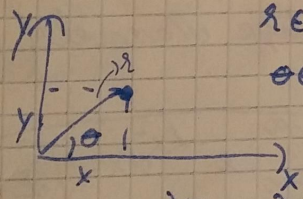
$$\int_{-\infty}^{\infty} e^{-\alpha x^2} dx = \sqrt{\frac{\pi}{2}}$$

$$\int_{-\infty}^{\infty} e^{-\alpha x^2} dx = 2\int_0^{\infty} e^{-\alpha x^2} dx = 2\,J(\alpha)$$

$$\int_0^{\infty} e^{-\alpha y^2} dy = 2\,J(\alpha)$$

$$J^2(\alpha) = J(\alpha) \cdot J(\alpha) = \left(\int_0^{\infty} e^{-\alpha x^2} dx\right) \cdot \left(\int_0^{\infty} e^{-\alpha y^2} dy\right) \Rightarrow$$

$$\Rightarrow J^2(\alpha) = \int_0^{\infty}\int_0^{\frac{\pi}{2}} e^{-\alpha r^2} r\, dr\, d\theta \;;\; dx\,dy = J(r,\theta)\, dr\, d\theta$$



$r \in (0, \infty)$

$\theta \in (0, \frac{\pi}{2})$

$x = r\cos\theta$

$y = r\sin\theta$

$$J(r,\theta) = \begin{vmatrix} \frac{\partial x}{\partial r} & \frac{\partial x}{\partial \theta} \\ \frac{\partial y}{\partial r} & \frac{\partial y}{\partial \theta} \end{vmatrix} = r$$

$$J^2(\alpha) = \left(\int_0^{\frac{\pi}{2}} d\theta\right) \cdot \left(\int_0^{\infty} e^{-\alpha r^2} \cdot r\, dr\right)$$

$$J^2(\alpha) = \frac{\pi}{2} \int_0^{\infty} e^t \left(-\frac{1}{2t}\right) dt \;;\; J^2(\alpha) = \frac{\pi}{2} \cdot \frac{1}{4} \;?$$

15.05.2019

CURS 8)

Repartiția Maxwell (a molec. după viteză)

$dn(v_x, v_y, v_z) = n_0 C_M e^{-\frac{m_0 v_x^2}{2kT}} e^{-\frac{m_0 v_y^2}{2kT}} e^{-\frac{m_0 v_z^2}{2kT}} dv_x dv_y dv_z$

dn ↓ nr. de molec. cu $\vec{v} \in [\vec{v}, \vec{v}+d\vec{v}]$; $v = v_x \vec{i} + v_y \vec{j} + v_z \vec{k}$

$\int_{(\mathcal{D})} dn(v_x, v_y, v_z) = n_0$ sau $\frac{1}{n_0} \int_{(\mathcal{D})} dn(v_x, v_y, v_z) = 1$

($\mathcal{D}$) ↓ întreg domeniul

$\frac{dn(v_x, v_y, v_z)}{n_0} = f(v_x, v_y, v_z) dv_x dv_y dv_z$ ← dens. de repartiție Maxwell $\Rightarrow$

$\Rightarrow f(v_x, v_y, v_z) = C_M e^{-\frac{m_0 v_x^2}{2kT}} e^{-\frac{m_0 v_y^2}{2kT}} e^{-\frac{m_0 v_z^2}{2kT}}$

$\int_{(\mathcal{D})} f(v_x, v_y, v_z) dv_x dv_y dv_z = 1$ → condiție de normare la unitate

Deci, $\iiint_{(\mathcal{D})} C_M e^{-\frac{m_0 v_x^2}{2kT}} e^{-\frac{m_0 v_y^2}{2kT}} e^{-\frac{m_0 v_z^2}{2kT}} dv_x dv_y dv_z = 1$ (1)

În acest caz, $(\mathcal{D}) \equiv \mathbb{R} \times \mathbb{R} \times \mathbb{R}$; $\iiint \to (\int)(\int)(\int)$

$(1) \Rightarrow C_M \left(\int_{-\infty}^{\infty} e^{-\frac{m_0 v_x^2}{2kT}} dv_x \right) \left(\int_{-\infty}^{\infty} e^{-\frac{m_0 v_y^2}{2kT}} dv_y \right) \cdot \left(\int_{-\infty}^{\infty} e^{-\frac{m_0 v_z^2}{2kT}} dv_z \right) = 1$

C_M se află din condiția de normare la unitate pt. densitatea fct. de repartiție Maxwell ($C_M = \frac{1}{(\int_{-\infty}^{\infty} \dots) \cdot (\int_{-\infty}^{\infty} \dots) \cdot (\int_{-\infty}^{\infty} \dots)}$)

Integrala Poisson: $\boxed{I(\alpha) = \int_{-\infty}^{\infty} e^{-\alpha x^2} dx = \sqrt{\frac{\pi}{\alpha}}}$

$e^{-\alpha x^2}$ → fct. pară $\Rightarrow I(\alpha) = 2\int_0^{\infty} e^{-\alpha x^2} dx \Rightarrow \boxed{\int_0^{\infty} e^{-\alpha x^2} dx = \frac{1}{2}\sqrt{\frac{\pi}{\alpha}}}$

$\int_{-\infty}^{\infty} e^{-\frac{m_0 v_x^2}{2kT}} dv_x = \sqrt{\frac{\pi \cdot 2kT}{m_0}}$ $\left(\alpha = \frac{m_0}{2kT}\right)$

$\int_{-\infty}^{\infty} e^{-\frac{m_0 v_x^2}{2kT}} dv_x = \int_{-\infty}^{\infty} e^{-\frac{m_0 v_y^2}{2kT}} dv_y = \int_{-\infty}^{\infty} e^{-\frac{m_0 v_z^2}{2kT}} dv_z = \left(\frac{m_0}{2kT\pi}\right)^{-\frac{1}{2}}$

$$C_M = \left(\frac{m_0}{2\pi kT}\right)^{\frac{3}{2}}$$

$$f(v_x, v_y, v_z) = \left(\frac{m_0}{2\pi kT}\right)^{\frac{3}{2}} e^{-\frac{m_0 v_x^2}{2kT}} \cdot e^{-\frac{m_0 v_y^2}{2kT}} \cdot e^{-\frac{m_0 v_z^2}{2kT}}$$ → densitatea de probabilitate Maxwell

---

Fie $\mathcal{F} = \mathcal{F}(v_x, v_y, v_z)$. ~~Atunci $\langle \mathcal{F} \rangle = \bar{\mathcal{F}} = \int_{(\Omega)} \mathcal{F}(v_x, v_y, v_z) \cdot f(v_x, v_y, v_z)$~~

Atunci $\langle \mathcal{F} \rangle = \bar{\mathcal{F}} = \int_{(\Omega)} \mathcal{F}(v_x, v_y, v_z) \cdot f(v_x, v_y, v_z) dv_x dv_y dv_z$

↳ media fct.

~~$\bar{v}_x = \langle \bar{v}_x \rangle = C_M \int_{-\infty}^{\infty}\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} v_x e^{-\frac{m_0 v_x^2}{2kT}} \cdot e^{-\frac{m_0 v_y^2}{2kT}} e^{-\frac{m_0 v_z^2}{2kT}} dv_x dv_y dv_z$~~

$\langle \bar{v}_x \rangle = C_M \left(\int_{-\infty}^{\infty} v_x e^{-\frac{m_0 v_x^2}{2kT}} dv_x \right) \cdot \underbrace{\left(\int_{-\infty}^{\infty} e^{-\frac{m_0 v_y^2}{2kT}} dv_y \right)}_{\sqrt{\frac{\pi 2kT}{m_0}}} \cdot \underbrace{\left(\int_{-\infty}^{\infty} e^{-\frac{m_0 v_z^2}{2kT}} dv_z \right)}_{\sqrt{\frac{\pi 2kT}{m_0}}}$

$\langle v_x^2 \rangle = \left(\int_{-\infty}^{\infty} v_x^2 e^{-\frac{m_0 v_x^2}{2kT}} dv_x \right) \cdot \frac{2\pi kT}{m_0} \cdot C_M$

$E_{cx} = \frac{m_0 v_x^2}{2} \Rightarrow \langle E_{cx} \rangle = \left\langle \frac{m_0 v_x^2}{2} \right\rangle = \frac{m_0}{2} \langle v_x^2 \rangle = \frac{m_0}{2} \bar{v}_x^2$

$\langle\ \rangle$ → operator de mediere (este liniar) $\left(\int_{-\infty}^{\infty} \left(\sum c_i x_i \right) f(x) dx = \right.$

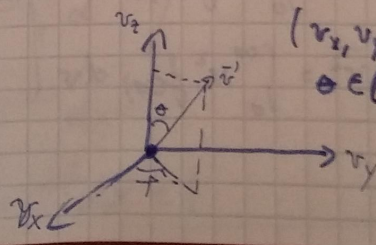
$= \sum \left(c_i \int_{-\infty}^{\infty} x_i f(x) dx \right) = \sum c_i \bar{x}_i$

Ne interesează nr. de molecule „orientate" după modulul vitezei.

$v^2 = v_x^2 + v_y^2 + v_z^2 \quad (v = +\sqrt{v^2})$

Fie $f = f(v)$ dens. fct. de rep. a moleculei după modulul vitezei.

Spațiul vitezelor:



$(v_x, v_y, v_z) \rightarrow (v, \theta, \varphi)$

$\theta \in [0, \pi)$; $\varphi \in [0, 2\pi)$

$$\begin{cases} v_x = v\sin\theta\cos\varphi \\ v_y = v\sin\theta\sin\varphi \\ v_z = v\cos\theta \end{cases} \qquad dv_x\,dv_y\,dv_z = \mathcal{J}(v,\theta,\varphi)\,d\theta\,d\varphi\,dv$$

$$\mathcal{J}(v,\theta,\varphi) = v^2\sin\theta$$

$v = +\sqrt{v^2}$ (toate moler. aflate pe sfera de rază v sunt indiscernabile (au aceeași viteză în modul!)

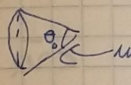
f se mai numește și fct. „radială" a lui Maxwell.

θ, φ unghiuri; $\mathcal{J}\,dv\,d\theta\,d\varphi = v^2 dv\,d\Omega$ $\rightarrow d\Omega = \frac{\mathcal{J}}{v^2}\,d\theta\,d\varphi = \sin\theta\,d\theta\,d\varphi$

$$\Omega_1 = \int d\Omega_1 = \int_0^{\pi}\int_0^{2\pi} \sin\theta\,d\theta\,d\varphi = \int_0^{\pi}\sin\theta\,d\theta \cdot \int_0^{2\pi} d\varphi =$$

$= 4\pi$ steradiani (măsură unghiului (spațial))

↳ unghiul total în jurul unui punct



unghi solid

$$dv_x\,dv_y\,dv_z = \mathcal{J}(v,\theta,\varphi)\,dv\,d\theta\,d\varphi = v^2\underbrace{(\sin\theta\,d\theta\,d\varphi)}_{d\Omega}\,dv$$

$$f(v_x, v_y, v_z)\,dv_x\,dv_y\,dv_z = f(v)\,\mathcal{J}\,dv\,d\theta\,d\varphi$$

$\Omega = 4\pi$ steradieni ; $f(v) = C_M\, e^{-\frac{m_0 v^2}{2kT}} \cdot \underbrace{4\pi v^2}_{\sim 2.}$

Deci, $\boxed{f(v) = 4\pi\left(\frac{m_0}{2\pi kT}\right)^{\frac{3}{2}} v^2 e^{-\frac{m_0 v^2}{2kT}}}$ → dens. fct. de repartiție după modulul vitezei

$|v| = v = +\sqrt{v^2}$; $v \geq 0$

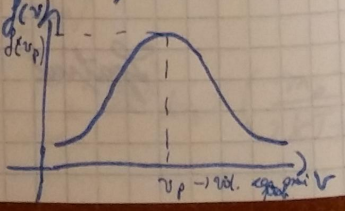
$v \in \mathbb{R}_+$; $\boxed{\int_0^{\infty} f(v)\,dv = 1}$ (de calculat)

Fie $F = F(v) \Rightarrow \bar{F}(v) = \langle F(v)\rangle = \int_0^{\infty} F(v)\,f(v)\,dv$

$$\bar{v} = \int_0^{\infty} v\,f(v)\,dv$$

$$\bar{v^2} = \int_0^{\infty} v^2 f(v)\,dv$$

(2) → ne conduce la viteza cea mai probabilă



$v = v_p \Leftrightarrow \left.\frac{df}{dv}\right|_{v_p} = 0$

$$\frac{df}{dv} = 4\pi \left(\frac{m_0}{2\pi kT}\right)^{\frac{3}{2}} \left(2v - \frac{2m_0 v}{2kT} \cdot v^2\right) e^{-\frac{m_0 v^2}{2kT}}$$

$$\frac{df}{dv} = 0 \Rightarrow \left(1 - \frac{m_0}{2kT} v^2\right) = 0 \Rightarrow v^2 = \frac{2kT}{m_0} \Rightarrow \boxed{v_p = \sqrt{\frac{2kT}{m_0}}}$$

$$f_m = f(v_p) = 4\pi \cdot \left(\frac{m_0}{2\pi kT}\right)^{\frac{3}{2}} \cdot \frac{2kT}{m_0} \cdot e^{-1} = 4\pi \cdot \pi^{-\frac{3}{2}} \left(\frac{2kT}{m_0}\right)^{1-\frac{3}{2}} e^{-1}$$

$$\boxed{f_m = 4\pi \cdot \left(\frac{m_0}{2kT\pi}\right)^{\frac{1}{2}} \cdot \frac{1}{e}}$$

$$\bar{v} = \langle v \rangle = \int_0^\infty v f(v)\, dv = \int_0^\infty v \cdot 4\pi \left(\frac{m_0}{2\pi kT}\right)^{\frac{3}{2}} v^2 e^{-\frac{m_0 v^2}{2kT}} dv$$

$$\bar{v} = \langle v \rangle = 4\pi \left(\frac{m_0}{2\pi kT}\right)^{\frac{3}{2}} \int_0^\infty v^3 e^{-\frac{m_0 v^2}{2kT}} dv$$

$$J(\alpha) = \int_0^\infty e^{-\alpha x^2} dx = \frac{1}{2}\sqrt{\frac{\pi}{\alpha}} = \frac{1}{2}\sqrt{\pi}\,\alpha^{-\frac{1}{2}}$$

$$\frac{d}{d\alpha} J(\alpha) = \frac{d}{d\alpha}\left(\frac{1}{2}\sqrt{\pi}\,\alpha^{-\frac{1}{2}}\right) \Rightarrow$$

$$\Rightarrow \frac{d}{d\alpha}\left(\int_0^\infty e^{-\alpha x^2} dx\right) = \int_0^\infty \frac{d}{d\alpha}\left(e^{-\alpha x^2}\right) dx = -\int_0^\infty x^2 e^{-\alpha x^2} dx \Rightarrow$$

Dar $\frac{dJ(\alpha)}{d\alpha} = \frac{1}{2}\sqrt{\pi}\left(-\frac{1}{2}\right) \cdot \alpha^{-\frac{3}{2}} = -\frac{\sqrt{\pi}}{4}\alpha^{-\frac{3}{2}}$

$$\Rightarrow \int_0^\infty x^2 e^{-\alpha x^2} = \frac{\sqrt{\pi}}{4} \cdot \alpha^{-\frac{3}{2}}$$

$$J_1(\alpha) = \int_0^\infty x^2 e^{-\alpha x^2} dx = \frac{\sqrt{\pi}}{4} \cdot \alpha^{-\frac{3}{2}}$$

$$\frac{dJ_1(\alpha)}{d\alpha} = \int_0^\infty x^2 (-x^2) e^{-\alpha x^2} dx$$

$$\frac{dJ_1(\alpha)}{d\alpha} = \ldots = \frac{\sqrt{\pi}}{4} \cdot \left(-\frac{3}{2}\right) \alpha^{-\frac{5}{2}} \Rightarrow$$

$$\Rightarrow \boxed{\int_0^\infty x^4 e^{-\alpha x^2} dx = \frac{3\sqrt{\pi}}{8}\alpha^{-\frac{5}{2}}}$$

Temă: $\int_0^\infty x^{2p} e^{-\alpha x^2} dx = ?$

$\int_0^\infty x^{2p+1} e^{-\alpha x^2} dx = ?$

$J_2(\alpha) = \int_0^\infty x e^{-\alpha x^2} dx = ?$

$-\alpha x^2 = t; \quad -2\alpha x dx = dt$

$\int_0^\infty x e^{-\alpha x^2} = -\frac{1}{2\alpha} \int_0^{-\infty} e^t dt = -\frac{1}{2\alpha} \cdot e^t \Big|_0^{-\infty} =$

$= -\frac{1}{2\alpha}(0 - 1) = \frac{1}{2\alpha}$

$\frac{d}{d\alpha} J_2(\alpha) = \int_0^\infty x \cdot \frac{d}{d\alpha}(e^{-\alpha x^2}) dx = \int_0^\infty -x^3 e^{-\alpha x^2} dx$ $\Rightarrow$

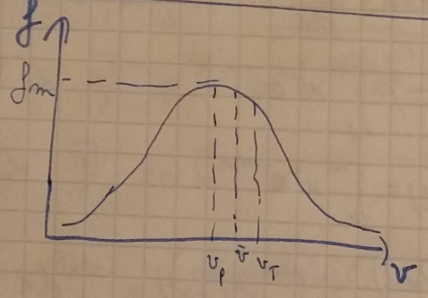
Deci $\frac{d}{d\alpha} J_2(\alpha) = \frac{1}{2} \cdot (-\frac{1}{\alpha^2})$

$\Rightarrow \boxed{\int_0^\infty x^3 e^{-\alpha x^2} dx = \frac{1}{2\alpha^2}}$

Astfel $\vec{v} = \langle v \rangle = 4\pi \cdot \pi^{-\frac{3}{2}} \left(\frac{m_0}{2kT}\right)^{\frac{3}{2}} \int_0^\infty v^3 e^{-\frac{m_0 v^2}{2kT}} dv$

$\bar{v} = 4\pi^{-\frac{1}{2}} \left(\frac{m_0}{2kT}\right)^{\frac{3}{2}} \cdot \frac{1}{2 \cdot \left(\frac{m_0}{2kT}\right)^2} = 2\pi^{-\frac{1}{2}} \left(\frac{m_0}{2kT}\right)^{-\frac{1}{2}} \Rightarrow$

$\Rightarrow \boxed{\bar{v} = \sqrt{\frac{8kT}{\pi m_0}}}$; $\vec{v} = \sqrt{\frac{4}{\pi} \cdot \frac{2kT}{m_0}} = \sqrt{\frac{4}{\pi}} v_p > v_p$



$\overline{v^2} = \langle v^2 \rangle = \int_0^\infty v^2 f(v) dv = \int_0^\infty v^2 \cdot 4\pi \cdot \left(\frac{m_0}{2\pi kT}\right)^{\frac{3}{2}} v^2 e^{-\frac{m_0 v^2}{2kT}} dv$

$\overline{v^2} = 4\pi \left(\frac{m_0}{2\pi kT}\right)^{\frac{3}{2}} \int_0^\infty v^4 e^{-\frac{m_0 v^2}{2kT}} dv$

$\int_0^\infty x^4 e^{-\alpha x^2} dx = \frac{3\sqrt{\pi}}{8} \alpha^{-\frac{5}{2}}$ $\Rightarrow$

$$\Rightarrow \overline{v^2} = 4\pi \cdot \pi^{-\frac{3}{2}} \left(\frac{m_0}{2kT}\right)^{\frac{3}{2}} \Big/ \frac{3\sqrt{\pi}}{8} \cdot \left(\frac{m_0}{2kT}\right)^{-\frac{5}{2}}$$

$$\overline{v^2} = \frac{3}{2} \cdot \frac{2kT}{m_0} \Rightarrow \boxed{\overline{v^2} = \frac{3kT}{m_0}} \Rightarrow \sqrt{\overline{v^2}} = v_T$$ ← viteză termică

$$\boxed{v_T = \sqrt{\overline{v^2}} = \sqrt{\frac{3kT}{m_0}}}$$

$$v_T = \sqrt{\frac{3}{2}} \cdot \sqrt{\frac{2kT}{m_0}} = \sqrt{\frac{3}{2}}\, v_p > v_p$$

$$\bar{v} = \sqrt{\frac{8}{3\pi}}\, v_p$$

$$v_p = \sqrt{\frac{2kT}{m_0}} = \sqrt{\frac{2kN_A T}{m_0 N_A}} = \sqrt{\frac{2RT}{\mu}}$$

$$\bar{v} = \sqrt{\frac{8RT}{3\pi\mu}} \; ; \; v_T = \sqrt{\frac{3RT}{\mu}}$$

$$\boxed{v_p < \bar{v} < v_T}$$

Legea echipartiției energiei cinetice pe gradele de libertate

$$E_c = \frac{m_0 v^2}{2} = \frac{m_0 (v_x^2 + v_y^2 + v_z^2)}{2} = \frac{m_0 v_x^2}{2} + \frac{m_0 v_y^2}{2} + \frac{m_0 v_z^2}{2}$$

$$E_c = E_{cx} + E_{cy} + E_{cz}$$

$$\bar{E}_c = \langle E_c \rangle = \int_0^\infty E_c f(v)\, dv = \int_0^\infty \frac{m_0 v^2}{2} f(v)\, dv =$$

$$= \frac{m_0}{2} \int_0^\infty v^2 f(v)\, dv = \frac{m_0 \overline{v^2}}{2} = \frac{m_0}{2} \cdot \frac{3kT}{m_0} \Rightarrow$$

$$\Rightarrow \boxed{\bar{E}_c = \frac{3}{2} kT}$$ → $i = 3$ (3 grade de libertate: translație pe Ox, Oy, Oz)

Izotropia spațiului: $\bar{E}_{cx} = \bar{E}_{cy} = \bar{E}_{cz} = \frac{1}{2} kT$ ← energia cinetică pe grad de libertate

$$\langle E_{cx} + E_{cy} + E_{cz} \rangle = 3 \langle E_{cx} \rangle = 3 \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \frac{m_0 v_x^2}{2} \cdot$$

$$\cdot f(v_x, v_y, v_z)\, dv_x\, dv_y\, dv_z = \frac{3m_0}{2} \left(\int_{-\infty}^{\infty} v_x^2 e^{-\frac{m_0 v_x^2}{2kT}} dv_x \right) \cdot \left(\int_{-\infty}^{\infty} e^{-\frac{m_0 v_y^2}{2kT}} dv_y \right) \cdot$$

$$\cdot \left(\int_{-\infty}^{\infty} e^{-\frac{m_0 v_z^2}{2kT}} dv_z \right) = \frac{3m_0}{2} \cdot \ldots \cdot \frac{2\pi kT}{m_0}$$

$$E_{cx} = \frac{m_0}{2} \cdot \ldots \cdot \frac{2\pi kT}{m_0}$$

$$\bar{E}_{cy} = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} e^{-\frac{m_0 v_x^2}{2kT}} \cdots = E_{cx} = \bar{E}_{cy}$$ ← din calcul

29 aprilie 2015

Fizică
- cursul 10 -

Repartiția Maxwell

$$dn = n_0 \left(\frac{m_0}{2\pi kT}\right)^{\frac{3}{2}} e^{-\frac{m_0 v^2}{2kT}} dv_x dv_y dv_z \quad ; \quad v^2 = v_x^2 + v_y^2 + v_z^2$$

Densitatea funcției de repartiție

$$f(v_x, v_y, v_z) = \left(\frac{m_0}{2\pi kT}\right)^{\frac{3}{2}} e^{-\frac{m_0 v_x^2}{2kT}} e^{-\frac{m_0 v_y^2}{2kT}} e^{-\frac{m_0 v_z^2}{2kT}}$$

$v_x \in [v_x, v_x + \Delta v_x]$

$v_y \in [v_y, v_y + \Delta v_y]$

$v_z \in [v_z, v_z + \Delta v_z]$

(Repartiția moleculelor după viteză)
= Maxwell - Boltzmann

Repartiția moleculelor după viteze și poziții

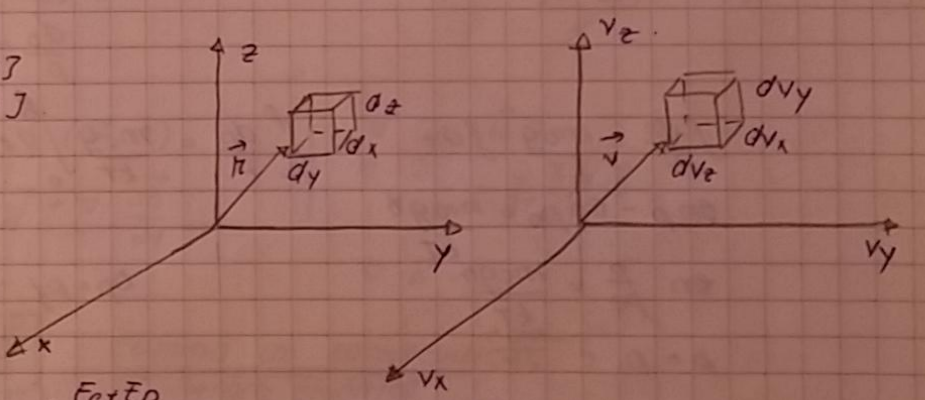
Căutăm moleculele care au $\vec{v} \in [\vec{v}, \vec{v} + \Delta\vec{v}]$ și $\vec{r} \in [\vec{r}, \vec{r} + \Delta\vec{r}]$

sau $v_x \in [v_x, v_x + \Delta v_x]$; $v_y \in [v_y, v_y + \Delta v_y]$, $v_z \in [v_z, v_z + \Delta v_z]$

și $\begin{cases} x \in [x, x + \Delta x] \\ y \in [y, y + \Delta y] \\ z \in [z, z + \Delta z] \end{cases}$

$d\vec{v} = dv_x dv_y dv_z$

$d\vec{r} = dx\,dy\,dz$



$$dn = n_0 \left(\frac{m_0}{2\pi kT}\right)^{\frac{3}{2}} e^{-\frac{E_c + E_p}{kT}} dx\,dy\,dz\,dv_x\,dv_y\,dv_z$$

$$f(v_x, v_y, v_z, x, y, z) = \left(\frac{m_0}{2\pi kT}\right)^{\frac{3}{2}} e^{-\frac{m_0 v_x^2}{2kT}} \cdot e^{-\frac{m_0 v_y^2}{2kT}} \cdot e^{-\frac{m_0 v_z^2}{2kT}} \cdot e^{-\frac{E_p}{kT}}$$

Câmp gravitațional terestru

$E_p = mgz$; $E_p = mgh$

$E_p = E_p(x, y, z)$

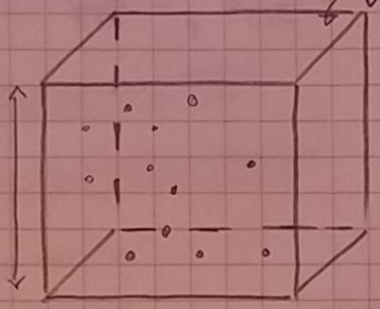
$$f(v_x, v_y, v_z, x, y, z) = \left(\frac{m_0}{2\pi kT}\right)^{\frac{3}{2}} e^{\frac{-m_0 v_x^2}{2kT}} e^{\frac{-m_0 v_y^2}{2kT}} \cdot e^{\frac{-m_0 v_z^2}{2kT}} \cdot e^{\frac{-mgh}{kT}}$$

Fie $\mathcal{F} = \mathcal{F}(v_x, v_y, v_z, x, y, z) \Rightarrow$

$$\Rightarrow \tilde{\mathcal{F}} = \langle \mathcal{F} \rangle = \iiint\limits_{\mathbb{R}^6} F(v_x, v_y, v_z, x, y, z) f(v_x, v_y, v_z, x, y, z)\, dv_x\, dv_y\, dv_z\, dx\, dy\, dz$$

$$(\mathcal{D}) = \mathbb{R} \times \mathbb{R} \times \mathbb{R} \times \mathbb{R} \times \mathbb{R} \times \mathbb{R} = \mathbb{R}^6$$

Legea „barometrică"



$p = \rho g z$ (1)

(gaz la altitudinea z)

(1) $dp = \rho g\, dz$

gaz → ecuațiile gazelor ideale)

$$pV = \nu RT, \quad \nu = \frac{m}{\mu}$$

$p = \frac{m}{\mu} \cdot \frac{1}{V} RT$ k = constanta lui Boltzman

T = temperatura absolută

$\mu = N_A m_0$; $R = N_A \cdot k$; $\frac{m}{V} = \rho$

$p = \rho \frac{N_A kT}{N_A m_0} \Rightarrow \rho = p \cdot \frac{m_0}{kT}$

$\Rightarrow$ (2) $dp = p \frac{m_0}{kT} g\, dz \Rightarrow$

$\Rightarrow$ ~~dz~~

$$\frac{dp}{p} = \frac{m_0 g}{kT} dz$$

$$\int \frac{dp}{p} = \frac{m_0 g}{kT} \int dz \Rightarrow \int_{p_0}^{p} \frac{dp}{p} = \frac{m_0 g}{kT} \int_0^h dz$$

$$\ln p - \ln p_0 = \frac{m_0 g h}{kT}$$

$$\ln \frac{p}{p_0} = \frac{m_0 g h}{kT} \qquad \Delta p = p_f - p_i < 0$$

$$p = p_0 e^{\frac{-mgh}{kT}}$$

$\frac{dp}{p} < 0$; $\frac{dp}{p} = -\frac{m_0 g}{kT} dz$

$$\Rightarrow \boxed{p = p_0 \cdot e^{\frac{-mgh}{kT}}}$$

(foarte bună în atmosferă)

(TEMĂ Calculați $x^2 y z^3 v_x v_z^2 = ?$

$x^2 v_x^2 = ?$)

Teoria câmpului electromagnetic

Ecuațiile lui Maxwell

Glosar
- $\vec{E}$ - intensitatea cp electric
- $\vec{H}$ - intensitatea cp magnetic
- $\vec{D}$ - inducția cp electric
- $\vec{B}$ - inducția cp magnetic
- $\vec{j}$ - densitatea de conducție
- ρ → densitatea de sarcini conținute în volum
  → $\rho = \frac{dq}{dv}$

Ecuațiile lui Maxwell

I

$$\begin{cases} \nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t} \\ \nabla \times \vec{H} = \vec{j} + \frac{\partial \vec{D}}{\partial t} \\ \nabla \cdot \vec{B} = 0 \\ \nabla \cdot \vec{D} = \rho \end{cases}$$

$\boxed{\rho = \frac{dq}{dv}}$, q = sarcina electrică

∇ = nabla; (1) $\nabla = \vec{i}\frac{\partial}{\partial x} + \vec{j}\frac{\partial}{\partial y} + \vec{k}\frac{\partial}{\partial z}$ } gradient (1)

(2) $\nabla \cdot \vec{B} = \frac{\partial B_x}{\partial x} + \frac{\partial B_y}{\partial y} + \frac{\partial B_z}{\partial z}$ } divergență (2)
rotor (3)

(3) $\nabla \times \vec{E} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ E_x & E_y & E_z \end{vmatrix}$

$\vec{B} = B_x\vec{i} + B_y\vec{j} + B_z\vec{k}$

$\vec{E} = E_x\vec{i} + E_y\vec{j} + E_z\vec{k}$

$\bullet\ (\nabla \times \vec{E}) = \varepsilon_{ijk}\partial_j E_k$ (*)

densități

$\vec{j}$ = curent de conducție
$[j]_{dim} = \left[\frac{\partial \vec{D}}{\partial t}\right]_{dim}$

$\frac{\partial \vec{D}}{\partial t}$ = densitate de curent de deplasare

(*) $\varepsilon_{ijk} = \begin{cases} 1, & (ijk) \text{ par} \\ -1, & (ijk) \text{ impar} \\ 0, & i=j \text{ sau } j=k \text{ sau } k=i \end{cases}$

(Einstein sumare după indici muți)

$\vec{I}$ = sistem de 4 ecuații diferențiale (12 ecuații scalare)
cu

$\vec{E}, \vec{H}, \vec{B}, \vec{D}, \vec{j}, \rho$ — 1 = 16 necunoscute scalare.
3 scalare 3sc 3sc 3sc 3sc

Există legături între
$\vec{E}$ și $\vec{D}$; $\vec{B}$ și $\vec{H}$ etc (legi de material:

$\vec{D} = \varepsilon \vec{E}$
$\vec{B} = \mu \vec{H}$)

ε = permitivitate electrică
μ = permeabilitate magnetică.

μ, ε = mărimi absolute

ε_0 = permitivitate în vid
μ_0 = permeabilitatea vidului.

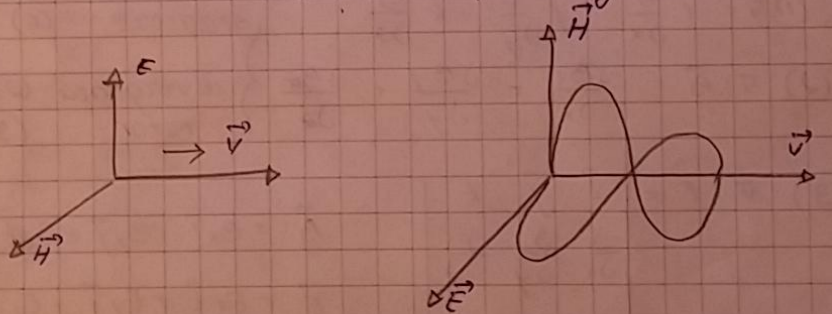
$\varepsilon = \varepsilon_r \varepsilon_0$
$\mu = \mu_r \mu_0$.

$$\varepsilon = \begin{pmatrix} \varepsilon_{11} & \varepsilon_{12} & \varepsilon_{13} \\ \varepsilon_{21} & \varepsilon_{22} & \varepsilon_{23} \\ \varepsilon_{31} & \varepsilon_{32} & \varepsilon_{33} \end{pmatrix} \rightarrow \text{forma diagonală } \begin{pmatrix} E_x & 0 & 0 \\ 0 & E_y & 0 \\ 0 & 0 & E_z \end{pmatrix}$$

$E_x = E_y = E_z$ material izotrop.

Completare. Maxwell introduce pt prima dată densitatea de deplasare; el face astfel o „unificare" a curenți de câmpului electromagnetic.

o variație a cp. electric produce o variație a cp magnetic și viceversa.



Cp. electromagnetic este un triedru trirectangular drept; are ca axe $\vec{E}, \vec{H}$ și $\vec{v}$ (viteza de propagare).

$\nabla \cdot \vec{B} = 0$
$\nabla \cdot \vec{D} = \rho$ ⇒ există sarcini electrice → viteza se dezmembrează

$\bar{e} = -1.6 \cdot 10^{-19} C$
$[q]_{SI} = C$ (coulomb)

Lemă Nu există sarcini magnetice elementare

$\vec{B} = \mu \vec{H}$, $\vec{D} = \varepsilon \vec{E}$

$$\begin{cases} \nabla \times \vec{E} = -\mu \frac{\partial \vec{H}}{\partial t} \\ \nabla \times \vec{H} = \vec{j} + \varepsilon \frac{\partial \vec{E}}{\partial t} \\ \nabla \cdot \vec{H} = 0 \\ \nabla \cdot \vec{E} = \frac{\rho}{\varepsilon} \end{cases}$$

Comentariu

$\nabla \cdot \vec{B} = 0 \Leftrightarrow \vec{B} = \nabla \times \vec{A}$

$(\exists)\ \vec{A} \neq 0$

Tema $\nabla \cdot (\nabla \times \vec{A}) = 0$.

- Daca $\exists$ rotor diferit de 0 $\Rightarrow$ avem un camp vectorial rotational.
- Daca nu $\exists$ rotori diferiti de 0 $\Rightarrow$ avem un ep. vect. ~~rotora~~ nerotational.

Analog $\vec{E}$ $\quad V = V(x,y,z)$ potentialul

$\vec{E} = -\nabla V$

$\nabla \cdot \vec{E} = \frac{\rho}{\varepsilon} \Rightarrow \nabla(\nabla V) = -\frac{\rho}{\varepsilon}$

$\nabla \cdot \nabla = (\vec{i}\frac{\partial}{\partial x} + \vec{j}\frac{\partial}{\partial y} + \vec{k}\frac{\partial}{\partial z})(\vec{i}\frac{\partial}{\partial x} + \vec{j}\frac{\partial}{\partial y} + \vec{k}\frac{\partial}{\partial z})$

$\nabla \cdot \nabla = \Delta = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}$

$$\boxed{\Delta V = -\frac{\rho}{\varepsilon}}$$ V = potential scalar

Aplicatie

$\nabla \times \vec{H} = \vec{j} + \frac{\partial \vec{D}}{\partial t} \quad | (\nabla \cdot)$

$\nabla \cdot (\nabla \times \vec{H}) = \nabla \cdot \vec{j} + \nabla \cdot \frac{\partial \vec{D}}{\partial t}$

$\nabla \cdot (\nabla \times \vec{H}) = 0$

$\nabla \cdot \frac{\partial \vec{D}}{\partial t} = \frac{\partial}{\partial t}(\nabla \cdot \vec{D})$

$\nabla \cdot \vec{D} = \rho$

$$\Rightarrow \boxed{0 = \nabla \cdot \vec{j} + \frac{\partial \rho}{\partial t}}$$ Ecuatia de conservare a energiei electrice.

$$\begin{cases} (1)\ \nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t} \\ (2)\ \nabla \times \vec{H} = \vec{j} + \frac{\partial \vec{D}}{\partial t} \\ (3)\ \nabla \cdot \vec{B} = 0 \end{cases}$$

Forma locala (diferentiala) (microscopica) a ecuatiilor Maxwell

$\nabla \cdot \vec{D} = \rho$.

$\rho = \frac{dq}{dV}$

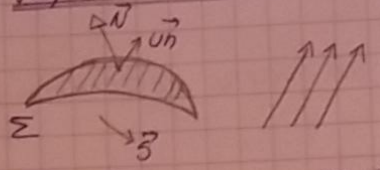
3 scalare 3sc 3sc 3sc 3sc

Forma generala (integrala / în tot spatiul)

trecerea de la forma locala la forma generală

Flux - divergenta
circulatie - rotor

Rapel matematic



$\vec{u_n}$ = versorul lui $\vec{N}$
$|\vec{u_n}| = 1$

suprafata Σ orientată!

$\vec{S} = \vec{u_n} \cdot S$

cp de vectori

ϕ = flux

$\phi = \vec{B} \cdot \vec{S} = B \vec{u_n} S$

$\phi = B S \cos\alpha$

$\alpha = \sphericalangle(\vec{u_n}, \vec{B})$

$\int_\Sigma \vec{B} \vec{u_n} \, dS$ = flux integral

Flux - divergentă

$\vec{A}$ = camp vectorial

$\nabla \cdot \vec{A}$, $\vec{A} \vec{u_n} dS$

$$\int_V \nabla \cdot \vec{A} \, dV = \int_\Sigma \vec{A} \vec{u_n} \, dS$$

V = volum
Σ delimitează pe V
⇒ Σ = bordul lui V
(multime → bord
volum → suprafață
3D → 2D)

Circulatie - rotor

$\vec{A} \cdot d\vec{l}$ $\nabla \times \vec{A}$

$$\oint_\Gamma \vec{A} \cdot d\vec{l} = \int_{\Sigma_\Gamma} \nabla \times \vec{A} \, \vec{u_n} \, dS$$

1D → 2D

Revenind la sistemul Maxwell (forma locală),

(1) $\int (\nabla \times \vec{E}) \vec{u_n} \, dS = \int (-\frac{\partial \vec{B}}{\partial t}) \vec{u_n} \, dS$

$\int_{\Sigma_\Gamma} (\nabla \times \vec{E}) \vec{u_n} \, dS = \oint_\Gamma \vec{E} \, d\vec{l}$

$$\oint_\Gamma \vec{E} \cdot d\vec{l} = -\frac{\partial}{\partial t} \left(\int_{\Sigma_\Gamma} \vec{B} \vec{u_n} \, dS \right)$$

Legea lui Faraday

$\oint_\Gamma \vec{E} \cdot d\vec{l} = e$

$\Rightarrow$ $e = -\frac{d\phi}{dt}$ forma comună

e = tensiunea electromagnetică

„Eu, curentul cel indus, toata viata m-am opus cauzei ce m-a produs"
(⇒ explicatia semnului „-" din $e = -\frac{d\Phi}{dt}$

(2) $\nabla \times \vec{H} = \vec{j} + \frac{\partial \vec{D}}{\partial t} \Rightarrow \int_{\Sigma_n} (\nabla \times \vec{H})\, \vec{u_n}\, dS =$

$= \int_{\Sigma_n} \vec{j}\, \vec{u_n}\, dS + \int_{\Sigma_n} \frac{\partial \vec{D}}{\partial t}\, \vec{u_n}\, dS$

$\int_{\Sigma_n} (\nabla \times \vec{H})\, \vec{u_n}\, dS = \int_n \vec{H} \cdot d\vec{l} \Rightarrow$

$\Rightarrow \boxed{\int_n \vec{H} \cdot d\vec{l} = \int_{\Sigma_n} \vec{j} \cdot \vec{u_n}\, dS + \frac{\partial}{\partial t} \left(\int_{\Sigma_n} \vec{D} \cdot \vec{u_n}\, dS \right)}$

Legea Ampère (legea circuitului magnetic)

$\boxed{\int_n \vec{H} \cdot d\vec{l} = I_c + I_d}$ → legea circuitului total

$\begin{cases} I_c = \int_{\Sigma_n} \vec{j} \cdot \vec{u_n}\, dS \\ I_d = \frac{\partial}{\partial t} \int_{\Sigma_n} \vec{D} \cdot \vec{u_n}\, dS \end{cases}$

(3) $\nabla \cdot \vec{B} = 0$ $\int_{V_\Sigma} \nabla \cdot \vec{B}\, dV = \int_\Sigma \vec{B} \cdot \vec{u_n}\, dS$; dar

$\int \nabla \cdot \vec{B}\, dV = 0 \Rightarrow \boxed{\int_\Sigma \vec{B} \cdot \vec{u_n}\, dS = 0}$

Legea fluxului (inductiei) magnetice (=) legea lui Gauss a ind magnetice (Gauss propriu-zisă)

Fluxul inductiei magnetice printr-o suprafata inchisa este 0

CURS 10

Ecuații Maxwell

$$\begin{cases} \nabla \times \bar{E} = -\dfrac{\partial \bar{B}}{\partial t} \\ \nabla \times \bar{H} = \bar{J} + \dfrac{\partial \bar{D}}{\partial t} \\ \nabla \cdot \bar{B} = 0 \\ \nabla \cdot \vec{D} = \rho \end{cases}$$

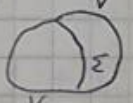
J - densitate de curent (de conducție)

$$\rho = \frac{dq}{dV}$$

$\nabla \cdot \vec{D} = \rho$ Flux = divergență

$\bar{A} = A_x \bar{i} + A_y \bar{j} + A_z \bar{k}$ = câmp vectorial

$\int \bar{A}\, \bar{u}_n\, ds$ = flux integral



$$\int_{V_\Sigma} \nabla \cdot \bar{A} = \int_\Sigma \bar{A}\, u_n\, dS$$

$$\int_{V_\Sigma} \nabla D = \int_{V_\Sigma} \rho\, dV$$

$$\int_{V_\Sigma} \nabla \vec{D}\, dv = \int_\Sigma \vec{D}\, \vec{u}_n\, ds \;/\Rightarrow$$

$$\boxed{\int_\Sigma \vec{D}\, \vec{u}_n\, ds = \int_{V_\Sigma} \rho\, dV}$$

↑ Legea fluxului inducției electrice
sau
Legea lui Gauss pt. inducția electrică

$\vec{D}$ = ind. electrică

$\vec{D} = \Sigma \vec{E}$; $\vec{E}$ = intensitatea câmpului electric

Σ = permitivitatea (absolută)

$$\Rightarrow \int_\Sigma \vec{E}\, \vec{u}_n\, ds = \frac{1}{\Sigma} \int_{V_\Sigma} \rho\, dV$$

$$\rho = \frac{dq}{dV} \Rightarrow \rho\, dV = dq$$

$$\int_\Sigma \vec{E} \cdot \vec{u}_n\, ds = \frac{1}{\Sigma} \int_{V_\Sigma} dq = \int dq = q \text{ sarcini conținute în vol}$$

$$\boxed{\int_\Sigma \vec{E}\, \vec{u}_n\, ds = \frac{q}{\Sigma}}$$

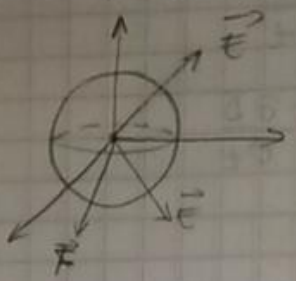
Lema: Fluxul integral al intensității câmp. el = sarcina / permitivitate

Aplicații:

$$\int_\Sigma \vec{E}\cdot\vec{u}_n\,dS = \frac{q}{\varepsilon}$$

$$\vec{E} = ct \Rightarrow \vec{E}\cdot\int_\Sigma \vec{u}_n\,dS = \frac{q}{\varepsilon}$$

$$\int_\Sigma \vec{u}_n\,dS = \Sigma$$



Cazul sferic

$$dV = dx\,dy\,dz$$

$$dV = r^2 \sin\theta\,dr\,d\theta\,d\varphi$$

$$S = r^2\iint \sin\theta\,d\theta\,d\varphi$$

$$S = \left(\int_0^{\pi}\sin\theta\,d\theta\right)\left(\int_0^{2\pi} d\varphi\right)r^2 = 4\pi r^2$$

$$\Rightarrow E\cdot 4\pi r^2 = \frac{q}{\varepsilon} \Rightarrow E = \frac{1}{4\pi\varepsilon}\cdot\frac{q}{r^2}$$

$$\vec{F} = q\vec{F}$$

(forța, sarcina)

$$\boxed{F = \frac{1}{4\pi\varepsilon}\cdot\frac{2q}{r^2}}$$

$\begin{cases} F \text{ atracție} \Leftrightarrow 2\cdot q < 0 \\ F \text{ de respingere} \Leftrightarrow 2\cdot q > 0 \end{cases}$

$$\nabla\cdot\vec{D} = \rho$$

$$D = \varepsilon\vec{E}$$

$$\boxed{\nabla\cdot\vec{E} = \frac{\rho}{\varepsilon}}$$

Potențial

V = potențial electric – scalar

A = pot. – vector datorat $(\nabla\cdot\vec{B} = 0) \Rightarrow \exists A$ a.î. $\vec{B} = \nabla\times\vec{A}$ (rotor)

$$\vec{E} = -\nabla\cdot V;\quad V = V(x, y, z)$$

$$\nabla\cdot\vec{E} = \nabla\cdot(\nabla V) = \Delta V$$

$$\Delta V = \frac{\rho}{\varepsilon}$$

$\rho = 0 (\nexists q) \Rightarrow \nabla V = 0$

Ecuațiile diferențiale Laplace și Poisson. Rezolvate $\Rightarrow$

$$V = V(x, y, z). \rightarrow \vec{E} = -\nabla V \Rightarrow \vec{D} = \vec{E}\cdot\Sigma$$

$V = \frac{4\pi R^3}{3}$

$\frac{dV}{dR} = 4R\pi^2 = S \; ; \; S = \frac{dV}{dR}$

$B = \nabla \times \vec{A}$ etc.

Alte 'rute'

$\nabla \times \vec{H} = \vec{j} + \frac{\partial \vec{D}}{\partial t} \quad | \; (\nabla \times)$

$\nabla \times \nabla \times \vec{H} = \nabla \times \vec{j} + \nabla \times \left(\frac{\partial D}{\partial t}\right)$

Știm:

$\nabla \times \nabla \times \vec{A} = -\Delta \vec{A} + \nabla \underbrace{(\nabla \cdot \vec{A})}$

$\vec{A}$ câmp vectorial — gradient (divergență A)

$\nabla \times \nabla \times \vec{H} = \nabla \times \vec{j} + \nabla \times \left(\frac{\partial \vec{D}}{\partial t}\right)$;

$\nabla \times \frac{\partial}{\partial t} \Rightarrow \frac{\partial}{\partial t} \cdot \nabla \times$

$\Rightarrow -\Delta \vec{H} + \nabla \underbrace{(\nabla \cdot H)}_{\| \; 0} = \nabla \times j + \frac{\partial}{\partial t} (\nabla \times \vec{D}) \begin{cases} \nabla \cdot \vec{B} = 0 \\ \vec{B} = \mu \vec{H} \\ D = \varepsilon \vec{E} \end{cases} \Rightarrow \nabla \cdot \vec{H} = 0$

$\nabla \times \vec{E} = -\frac{\partial B}{\partial t} \; | \Rightarrow \; \nabla \times \vec{D} = -\varepsilon \frac{\partial \vec{B}}{\partial t}$

$-\varepsilon \Delta \vec{H} = \nabla \times \vec{j} - \varepsilon \mu \left(\frac{\partial H}{\partial t}\right) \Rightarrow$

$$\boxed{\Delta H + \nabla \times \vec{j} + \varepsilon \mu \frac{\partial^2 \vec{H}}{\partial t^2} = 0} \quad (I)$$

$\Delta = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}$

↳ ec. dif. coef. d. omogenă (ord. 2)

Dacă $\nabla \times \vec{j} = 0 \Rightarrow \boxed{\Delta H - \varepsilon \mu \frac{\partial^2 H}{\partial t^2} = 0} \Rightarrow H.$

OPTICĂ ELECTROMAGNETICĂ

Lumina este undă electromagnetică. Lumina este duală (undă corpuscul):

1. Undă electromagnetică
2. Fotoni

Lumina e alcătuită din fotoni (efect corpuscular).

Interacțiuni:

Lumina vs lumina

lumina vs substanță

Fenomene optice:

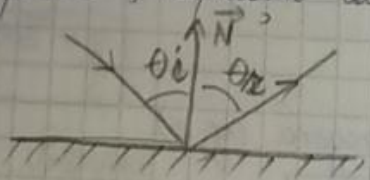
- reflexie
- refracție
- dispersie
- absorbție

} lumina vs substanță

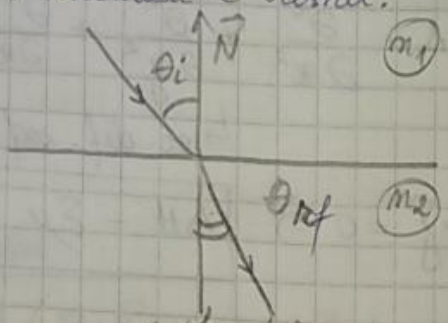
- interferență
- difracție

} lumină vs lumină

Reflexia și refracția la început studiată de Fresnel.



reflexia



refracția

Raza de lungime incid., N și raza reflectată sunt în același plan. $\theta_i = \theta_r$

n_1, n_2 - indici de refracție

~~Unda incid, N~~ $n_1 \sin\theta_i = n_2 \sin\theta_{rf}$

Optica =>

Vizibilul = ROGVAIV

Optica ≡ [infraroșu (IR), ultraviolet (UV), ROGVAIV]

ROGVAIV ∈ [4000 - 8000 Å]

$1 Å = 10^{-10} m$

ROGVAIV ∈ [400 - 800 nm] $1 nm = 10^{-9} m$

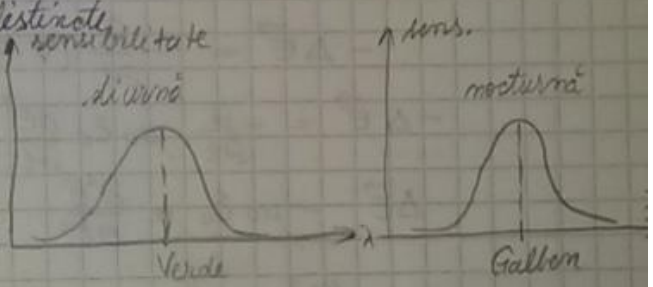
ROGVAIV 7 culori ale spectrului, datorat lui Newton

lumina "albă"

Spectrofotometric, 6 culori distincte

ROGVAIV

↙ lumina nocturnă ↘ lumina diurnă



$\Psi = \Psi(\vec{r}, t)$

$\vec{r} = x\vec{i} + y\vec{j} + z\vec{k}$

$\Psi = \Psi(x, y, z, t)$

↓ $\vec{\Psi} = \vec{\Psi}(x, y, z, t)$

↓ să respecte la ecuația de undă

$$\Delta\vec{\Psi}(x, y, z, t) - \frac{1}{c^2} \cdot \frac{\partial^2 \Psi(x, y, z, t)}{\partial t^2} = 0$$

↓ lumina să fie undă EM

$\Psi(x, y, z, t) \in [\vec{E}, \vec{H}, \vec{D}, \vec{B}]$

c = viteza de propagare a undei

Ecuațiile câmp. EM (Maxwell)

$\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$ $\quad \rho = 0 ; j = 0$

$\nabla \times \vec{H} = \vec{j} + \frac{\partial \vec{D}}{\partial t}$ vid $\quad \varepsilon_0, \mu_0$ ale vidului

$\nabla \times \vec{B} = 0$

$\nabla \times D = \rho$

→

Ecuațiile în vid

$$\nabla \times \vec{E} = -\frac{\partial B}{\partial t}$$

$$\nabla \times H = \frac{\partial \vec{D}}{\partial t}$$

$$\nabla \times D = 0$$

$$\nabla \times B = 0 \; ; \quad D = \varepsilon_0 \vec{E}$$

$$B = \mu_0 \vec{H}$$

1) $\nabla \times \nabla \times \vec{E} = \nabla \times \left(-\frac{\partial B}{\partial t}\right) \oplus \nabla \times \nabla E = -\Delta \vec{E} + \nabla(\nabla \cdot \vec{E})$

$\nabla \cdot D = 0 \Rightarrow \nabla \vec{E} = 0 \Rightarrow \nabla E = 0 \Rightarrow$

$$\nabla \times \nabla \times \vec{E} = -\Delta \vec{E}$$

$$-\Delta \vec{E} = -\frac{\partial}{\partial t}(\nabla B) \; ; \quad \nabla \times \vec{B} = \mu_0 \nabla H = \mu_0 \cdot \frac{\partial D}{\partial t} = \mu_0 \varepsilon_0 \frac{\partial \vec{E}}{\partial t}$$

$$-\Delta \vec{E} = -\frac{\partial}{\partial t} \mu_0 \varepsilon_0 \frac{\partial \vec{E}}{\partial t} = -\mu_0 \varepsilon_0 \frac{\partial^2 \vec{E}}{\partial t^2} \Rightarrow$$

$$\Delta \vec{E} - \mu_0 \varepsilon_0 = \frac{\partial^2 \vec{E}}{\partial t^2}$$

$$\mu_0 \varepsilon_0 = \frac{1}{c^2}$$

$$\Rightarrow \boxed{\Delta \vec{E} - \frac{1}{c^2} \frac{\partial^2 E}{\partial t^2} = 0} \rightarrow \text{calcul de undă pt } \vec{E}$$

MECANICĂ

q_i = poziția (coord.)

$\dot{q}_i = \frac{dq_i}{dt}$ viteza

$(q_i \; \dot{q}_i) \; ; \; i = \overline{1,f}$

$(q_1 \dots q_f, \dot{q}_1, \dots)$

Spațiul = 2f coordonate

$(q_i, \dot{q}_i) \in S^{1t}$

MECANICĂ TEORETICĂ

q_i = poziții (coord.) generalizate

$\dot{q}_i = \frac{dq_i}{dt}$ viteze generalizate

$(q_i \; \dot{q}_i)$; $i = \overline{1, f}$; f = nr grade de libertate

$(q_1 \dots q_f, \dot{q}_1, \dots \dot{q}_f)$ $\in f$ coordonate

Spațiul = $2f$ coordonate

$(q_i, \dot{q}_i) \in S^{2f}$ = spațiul configurațiilor

13.05.2015

Phoenix

CURS 11 - FIZICĂ

Formalismul Lagrange

q_i = coord generalizate

$\dot{q}_i$ = viteze generalizate

$\dot{q}_i = \frac{dq}{dt}$; $i = \overline{1,f}$; f = nr de grade de libertate

$(q_i, \dot{q}_i) = \overbrace{q_1 q_2 \dots q_f}^{2f}$

Spațiul configurațiilor

$L = L(q_i, \dot{q}_i, t)$ → funcția Lagrange (lagrangeana)

$L = L(\underbrace{q_1, q_2 \dots \dot{q}_1, \dot{q}_2 \dots \dot{q}_f}_{2f+1}, t)$

S = acțiunea

$S = \int_{t_1}^{t_2} L(q_i, \dot{q}_i, t)\, dt$; $i = \overline{1,f}$

$[L]_{SI} = J$ (joule) ⇒ dim L = dim Energie

Ec. dimensională

$[\,]^{\alpha} m^{\beta} T^{\gamma}$; dim $E_c = L^2 M T^{-2}$; $\alpha = 2$; $\beta = 1$; $\gamma = -2$

lungime, masă, timp ; $E_c = \frac{m v^2}{2}$

$[S] = Js$ ⇒ dim $S = L^2 M T^{-1}$

(joule · secundă)

Ecuația Einstein - Planck

$W_2 - W_1 = R$ J

$\dim E = [\dim R]\,[\dim \nu]$

$[\dim R] = \frac{\dim E}{\dim \nu} = L^2 M T^{-1} \Rightarrow R$ - constanta lui Planck

Lema: Dacă acțiunea într-o pb de fizică este apropiată de valoarea constantei lui Planck, atunci pb este rezolvată corect doar în mecanica cuantică.

$$\left.\begin{array}{l} W_2 - W_1 = \frac{R}{2\pi} 2\pi \text{ J} \\ \omega = 2\pi \text{ J} \\ \frac{R}{2\pi} = \hbar \end{array}\right\} \Rightarrow \boxed{W_2 - W_1 = \Delta W = \hbar \omega}$$

Suprem funcția S unei probe variaționale. Problema variațională a lui S este de extrem, respective minim.

Fie δ o variație mică (ca și Δ)

δS variație mică a lui S

δq_i = variație mică a lui S, q_i

$\delta \dot{q}_i$ = variație mică a lui $\dot{q}_i$

$\delta t = 0$ - t

δ_{q_i} → variație mică în raport cu q_i

$\boxed{\delta_{q_i} S = 0}$ principiul variațional al lui S

$\delta_{q_i} S = S'(q_i, \dot{q}_i, t) - S(q_i, \dot{q}_i, t)$

$S'(q_i, \dot{q}_i, t) \to S(\dot{q}_i, \ddot{q}_i, t)$ $\quad i = \overline{1,f}$ - schimbare de coordonate

$q_i \to \dot{q}_i = q_i + \delta q_i$

$\dot{q}_i \to \ddot{q}_i = \dot{q}_i + \delta \dot{q}_i$

$t \to t' = t + \delta t$

$$\begin{Bmatrix} q_i \\ \dot{q}_i \\ t \end{Bmatrix} \longrightarrow \begin{Bmatrix} q_i' \\ q_i'' \\ t' \end{Bmatrix} = \begin{Bmatrix} q_i + \delta q_i \\ \dot{q}_i + \delta \dot{q}_i \\ t + \delta t \end{Bmatrix}$$

$$\delta_{q_i} S = \int_{t_1+\delta t_1}^{t_2+\delta t_2} L(q_i + \delta q_i, \dot{q}_i + \delta \dot{q}_i, t+\delta t)\,dt - \int_{t_1}^{t_2} L(q_i, \dot{q}_i, t)\,dt$$

↓ se dezvoltă în serie Taylor

în condițiile: $\delta q_i(0) = \delta q_i(t) = 0$, $\delta t(0) = 0$ $\Rightarrow$ Ecuațiile Lagrange

$$\frac{\partial L}{\partial q_i} - \frac{d}{dt}\left(\frac{\partial L}{\partial \dot{q}_i}\right) = 0; \quad i = \overline{1,f}$$

$$\begin{cases} \frac{\partial L}{\partial q_1} - \frac{d}{dt}\left(\frac{\partial L}{\partial \dot{q}_1}\right) = 0 \\ \frac{\partial L}{\partial q_2} - \frac{d}{dt}\left(\frac{\partial L}{\partial \dot{q}_2}\right) = 0 \\ \cdots \\ \frac{\partial L}{\partial q_f} - \frac{d}{dt}\left(\frac{\partial L}{\partial \dot{q}_f}\right) = 0 \end{cases}$$

f ecuații diferențiale — Pb. Cauchy

Condiții inițiale: $\begin{cases} q_i(0) = q_i^0 \\ \dot{q}_i(0) = \dot{q}_i^0 \end{cases}$ $\Rightarrow$ $q_i, \dot{q}_i$

$$\begin{cases} q_1(0) = q_1^0, \ldots q_f(0) = q_f^0 & f \text{ const } c_1, c_2 \ldots c_f \\ \dot{q}_1(0) = \dot{q}_1^0 \ldots \dot{q}_f(0) = \dot{q}_f^0 & f \text{ const } c_{f+1} \ldots c_{2f} \end{cases}$$

$$\Rightarrow \begin{cases} q_i = q_i(c_1 \ldots c_f, c_{f+1} \ldots c_{2f}, t) \Rightarrow q_i = q_1(t) \\ \dot{q}_i = \dot{q}_i(c_1 \ldots c_f, c_{f+1} \ldots c_{2f}, t) \Rightarrow \dot{q}_i = \dot{q}_i(t) \end{cases} \quad i = \overline{1,f}$$

Cerința mecanicii clasice (definită din $\vec{\ }$)

Schema formalismului Lagrange

Starea inițială

$$\begin{cases} q_i(0) = q_i^0 \\ \dot{q}_i(0) = \dot{q}_i^0 \\ L = L(q_i, \dot{q}_i, t) \\ T = T(q_i(t), \dot{q}_i(t), t) \text{ funcție în } S^2 \end{cases}$$

Evoluția în timp a sistemului

$$\frac{\partial L}{\partial q_i} - \frac{d}{dt}\left(\frac{\partial L}{\partial \dot{q}_i}\right) = 0, \quad \overline{i = 1, f}$$

Starea finală

$$\begin{cases} q_i = q_i(t) \\ \dot{q}_i = \dot{q}_i(t) \\ T = T(q_i(t), \dot{q}_i(t), t) \end{cases}$$

$T = T(q_i(t), \dot{q}_i(t), t)$ - funcție în S^2

Semnificația fizică a funcției Lagrange:

În sisteme conservative de puncte materiale:

$L = E_c - E_p$

E_c - en. cinetică și E_p - en. potențială

Sistem conservativ: $E = E_c + E_p = ct$

Demonstrație:

$q_i \longrightarrow x_i$, $\dot{q}_i \longrightarrow \dot{x}_i$; $i = \overline{1, f}$

$E_{ci} = \frac{m_i \dot{x}_i^2}{2}$; $E_{ci} = E_{ci}(\dot{x}_i)$ $\exists$ $\frac{\partial E_{ci}(x_i)}{\partial \dot{x}_i} \neq 0$ dar $\not\exists$

$\frac{\partial E_{ci}(x_i)}{\partial x_i} = 0$ (1)

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$E_{pi} = E_p(x_i)$; $\exists \frac{\partial E_p x_i}{\partial x_i} \neq 0$, dar $\nexists \frac{\partial E_p(x_i)}{\partial x_i} = 0$

$\vec{F} = \nabla E_p$; $\Rightarrow F_i = \frac{dE_p(x_i)}{dx_i}$

$F_i = m\ddot{x}$; $m\dot{x} = \frac{d}{dt}(p_i) = p_i$

$F_i = m\ddot{x}_i$; $i = \overline{1,f}$

Dm (1) $\Rightarrow \overline{F_i} = \frac{dE_p(x_i)}{dx_i} \neq 0 = \frac{-dE_p}{dx_i} + \frac{dE_c}{dx_i} = \frac{d}{dx_i}(E_c - E_p)$ (2)

$E_c = \frac{1}{2}\sum_{i=1}^{f} m_i \dot{x}_i^2$; $\frac{\partial E_c}{\partial \dot{x}_i} = \frac{1}{2} m_i \cdot 2\dot{x}_i = m_i \dot{x}_i = p_i$

$\overline{F_i} = \frac{\partial E_c}{\partial x_i} - 0 = \frac{\partial E_c}{\partial x_i} - \frac{\partial E_p}{\partial x_i} = \frac{d}{dx_i}(E_c - E_p)$ (1)

$F_i = \frac{dp_i}{dt} \Rightarrow \frac{d}{dx_i}(E_c - E_p) = \frac{d}{dt} p_i = \frac{d}{dt}\left(\frac{d}{d\dot{x}_i}(E_c - E_p)\right)$

(1) $\wedge$ (2) $\Rightarrow$ $\boxed{E_c = E_p \cdot L}$ $\Rightarrow \frac{\partial L}{\partial x_i} = \frac{d}{dt}\left(\frac{\partial L}{\partial \dot{x}_i}\right) = 0$

Ec. Lagrange

APLICAȚII : Oscilatorul armonic liniar

$F = -kx$; $E_p = \frac{kx^2}{2}$

$E_c = \frac{m\dot{x}^2}{2}$

$L = E_c - E_p = \frac{m\dot{x}^2}{2} - \frac{kx^2}{2}$

$\frac{\partial L}{\partial x} - \frac{d}{dt}\left(\frac{\partial L}{\partial \dot{x}}\right) = 0$

$\frac{\partial L}{\partial x} = -kx$

$\frac{\partial L}{\partial \dot{x}} = m\dot{x}$

$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{x}}\right) = \frac{d}{dt}(m\dot{x}) = m\ddot{x}$

$\Rightarrow -kx - m\ddot{x} = 0$

$\Rightarrow \ddot{x} + \frac{k}{m} x = 0$

$\boxed{\ddot{x} + \omega_0^2 x = 0}$; $\boxed{\omega_0^2 = \frac{k}{m}}$

↓ ec. diferențială

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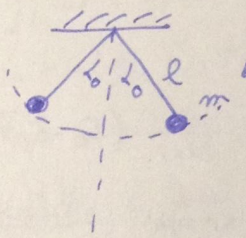
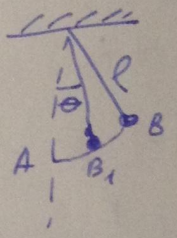
① $q \to x$; $\dot{q} \to \dot{x}$

2) Pendulul fizic : (m, ℓ)





$\overline{AB_1}$: $S = \ell\theta$

$v = \frac{dS}{dt} = \ell \frac{d\theta}{dt} = \ell\dot{\theta}$; $\boxed{\dot{\theta} = \frac{d\theta}{dt}}$ $\Rightarrow$ $\boxed{v = \ell\dot{\theta}}$

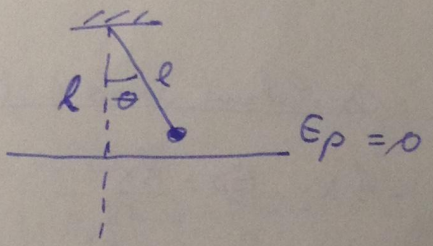
$q \to \theta$; $\dot{q} \to \dot{\theta}$ → coord generalizată; → viteză generalizată

$E_c = \frac{mv^2}{2} = \frac{m}{2}\ell^2\dot{\theta}^2$

$E_p = mgh$

$h = \ell - \ell\cos\theta$

$h = \ell(1-\cos\theta)$



$L = E_c - E_p = \frac{m\ell^2\dot{\theta}^2}{2} - mgh$

$L = \frac{m\ell^2}{2}\dot{\theta}^2 - mg\ell(1-\cos\theta)$; $L = L(\theta, \dot{\theta}, t)$

Ec. Lagrange : $\boxed{\frac{\partial L}{\partial \theta} - \frac{d}{dt}\left(\frac{\partial L}{\partial \dot{\theta}}\right) = 0}$ Obținerea ec. dif. de mișcare

$\frac{\partial L}{\partial \theta} = -mg\ell \sin\theta$

$\frac{\partial L}{\partial \dot{\theta}} = \frac{d}{dt}\left(\frac{m\ell^2}{2}\dot{\theta}^2\right) = m\ell^2\dot{\theta}$

$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot\theta}\right) = \frac{d}{dt}\left(ml^2\dot\theta\right) = ml^2\ddot\theta$$

$$\frac{d}{dt}(\dot\theta) = \frac{d}{dt}\left(\frac{d}{dt}\right)\theta = \frac{d^2\theta}{dt^2} = \ddot\theta$$

Ec. Lagrange pt $L = L(\theta, \dot\theta, t)$

$$-mgl\sin\theta - ml^2\ddot\theta = 0 \quad \left(\frac{-1}{ml^2}\right)$$

? $\boxed{\ddot\theta + g_e \sin\theta = 0}$; $g_e = \omega_0^2$

Luăm aprox. ordi

$$\ddot\theta + \omega_0^2\theta = 0$$

$$\theta = \theta_0 \cos(\omega_0 t + \varphi)$$

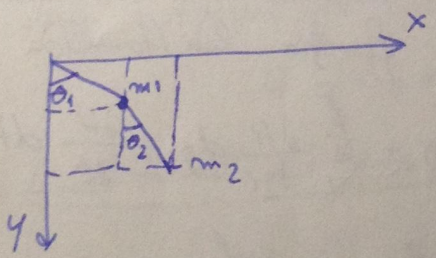
$$\omega_0^2 = g_e \text{ ; } \omega_0 = \frac{2\pi}{T_0}$$

$$\boxed{T_0 = 2\pi\sqrt{\frac{l}{g}}}$$

Aprox de ordii : la examen!

Pendulul dublu

masă punctuală



$$\begin{cases} x_1 = l_1 \sin\theta_1 \\ y_1 = l_1 \cos\theta_1 \\ x_2 = l_1 \sin\theta + l_2 \sin\theta_2 \\ y_2 = l_1 \cos\theta + l_2 \cos\theta_2 \end{cases}$$

$$V_1^2 = \dot x_1^2 + \dot y_1^2 \Rightarrow \begin{cases} E_{c1} = \frac{m_1 \dot x_1^2}{2} + \frac{m_1 \dot y_1^2}{2} \\ E_{c2} = \frac{m_2 \dot x_2^2}{2} + \frac{m_2 \dot y_2^2}{2} \end{cases}$$

$V_1^2 = \dot x_1^2 + \dot y_1^2$
$E_{p1} = m_1 g y_1$
$E_{p2} = m_2 g y_2$

$$\Rightarrow \begin{aligned} \dot x_1 &= l_1 \cos(\theta_1)\cdot\dot\theta_1 \\ \dot y_1 &= l_1(-\sin\theta_1)\cdot\dot\theta_1 \end{aligned}$$

$$\Rightarrow L = \frac{m_1 \dot{x}_1^2}{2} + \frac{m_1 \dot{y}_1^2}{2} + \frac{m_2 \dot{x}_2^2}{2} + \frac{m_2 \dot{y}_2^2}{2} - m_1 g (l_1 \cos\theta_1 + l_2 \cos\theta$$

$$L = L(\theta_1, \theta_2, \dot{\theta}_1, \dot{\theta}_2, t)$$

Caz particular: $m_1 = m_2 = m$, $l_1 = l_2 = l$

$$\Rightarrow \begin{cases} \frac{\partial L}{\partial \theta_1} - \frac{d}{dt}\left(\frac{\partial L}{\partial \dot{\theta}_1}\right) = 0 \\ \frac{\partial L}{\partial \theta_2} - \frac{d}{dt}\left(\frac{\partial L}{\partial \dot{\theta}_2}\right) = 0 \end{cases}$$

Formalismul Hamilton

$L = L(q_i, \dot{q}_i, t)$; $(q_i, \dot{q}_i)$ $\forall$ $i = \overline{1, f}$

$H = H(p_i, q_i, t)$ (p_i, q_i) p_i, q_i = variabile canonic conjugate; ele respectă mecanica clasică

p_i – impuls generalizat; q_i – coord. generalizată

$(p_i, q_i) \rightarrow (p_1 \dots p_f, q_1 \dots q_f) \in S^f \rightarrow$ spațiul fazelor

$$\boxed{p_i = \frac{\partial L}{\partial \dot{q}_i} \text{ (Definiție)}}$$

$$H = \sum_{i=1}^{f} p_i \dot{q}_i - L(q_i, \dot{q}_i, t)$$; H – hamiltonianul sistemului sau funcția lui Hamilton

$$H = H(p_i, q_i, t) \Rightarrow dH = \sum_{i=1}^{f} \frac{\partial H}{\partial p_i} dp_i + \sum_{i=1}^{f} \frac{\partial H}{\partial q_i} dq_i + \frac{\partial H}{\partial t} dt \quad (\alpha)$$

$$H = \sum_{i=1}^{f} p_i \dot{q}_i - L(q_i, \dot{q}_i, t) \Rightarrow dH = \sum_{i=1}^{f} p_i d\dot{q}_i + \sum_{i=1}^{f} \dot{q}_i dp_i - dL$$

$$\Rightarrow dL = \sum_{i=1}^{f} \frac{\partial L}{\partial q_i} dq_i + \sum_{i=1}^{f} \frac{\partial L}{\partial \dot{q}_i} d\dot{q}_i + \frac{\partial L}{\partial t} dt$$

$$dH = \sum_{i=1}^{f} p_i d\dot{q}_i + \sum_{i=1}^{f} \dot{q}_i dp_i - \sum_{i=1}^{f} \frac{\partial L}{\partial q_i} dq_i - \sum_{i=1}^{f} \frac{\partial L}{\partial \dot{q}_i} d\dot{q}_i - \frac{\partial L}{\partial t} dt \quad (\beta)$$

$\Rightarrow \sum p_i d\dot{q}_i$

$\equiv$ (B) $\Rightarrow$ $\boxed{\frac{dH}{dt} = -\frac{\partial L}{\partial t}}$; $\dot{q}_i = \frac{\partial H}{\partial p_i}$

$\sum_{i=1}^{f} p_i dq_i$

Ec. Lagrange : $\frac{\partial L}{\partial q_i} - \frac{d}{dt}\left(\frac{\partial L}{\partial \dot{q}_i}\right) = 0 \Rightarrow \frac{\partial L}{\partial \dot{q}_i} - \frac{\partial p_i}{\partial t} = \dot{p}_i$

$\sum_{i=1}^{f} \frac{\partial H}{\partial q_i} dq_i = -\sum \dot{p}_i dq_i \Rightarrow p_i = -\frac{\partial H}{\partial q_i}$

$\Rightarrow$ Ecuațiile Hamilton : $\begin{cases} \dot{q}_i = \frac{\partial H}{\partial p_i} \\ \dot{p}_i = -\frac{\partial H}{\partial q_i} \\ \frac{\partial H}{\partial t} = -\frac{\partial L}{\partial t} \end{cases}$, $i = \overline{1,f}$ $\quad \begin{cases} \dot{q}_i = \frac{dq_i}{dt} \\ \dot{p}_i = \frac{dp_i}{dt} \end{cases}$

$\Rightarrow$ Valoarea superioară a formalismului lui Hamilton față de formalismul Lagrange constă în ec. diferențiale de ord I.

$\Rightarrow$ Alte avantaje : găsim direct (p_i, q_i)

3) $H = E_c + E_p = E_{totală}$

Lema : Pt un sist. de pct materiale, sist conservă

$$H = E_c + E_p = E$$

Hamiltoniana are semnif. fizică (H = en. totală a sistemului)

4) Există și un alt formalism : Hamilton-Jacobi

Schema formalismului Hamilton :

- Stare inițială

$\begin{cases} p_i(0) = p_i^0 \\ q_i(0) = q_i \\ H = H(p_i, q_i, t) \end{cases}$

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• Evoluția în timp a sistemului:

$$\begin{cases} \dot{q}_i = \dfrac{\partial H}{\partial p_i} \\ \dot{p}_i = -\dfrac{\partial H}{\partial q_i} \\ \dfrac{\partial H}{\partial t} = -\dfrac{\partial L}{\partial t} \end{cases}$$

• Stare finală

$$\begin{cases} q_i = q_i(t) \\ p_i = p_i(t) \\ F = F(p_i(t), q_i(t), t) \\ F \in S^2 f \rightarrow \text{spațiul fazelor} \end{cases}$$

Integrarea sistemului → Mărimi asupra sistemului

⇒ paranteza Poisson:

$$F = F(p_i, q_i, t)$$

$$dF = \sum \frac{\partial F}{\partial p_i} dp_i + \sum \frac{\partial F}{\partial q_i} dq_i + \frac{\partial F}{\partial t} dt$$

$$\frac{dF}{dt} = \sum_{i=1}^{f} \frac{\partial F}{\partial p_i} \frac{dp_i}{dt} + \sum_{i=1}^{f} \frac{\partial ?}{\partial q_i} \frac{dq_i}{dt} + \frac{\partial ?}{\partial t}$$

$$\frac{\partial F_i}{dt} = p_i = \frac{\partial H}{\partial p_i} \; ; \; \frac{dq_i}{\partial t} = \dot{q}_i = -\frac{\partial H}{\partial p_i}$$

$$\frac{\partial H}{\partial t} = \sum_{i=1}^{f} \frac{\partial F}{\partial p_i} - \sum_{i=1}^{f} \frac{\partial F}{\partial q_i} \cdot \frac{\partial H}{\partial p_i} + \frac{\partial F}{\partial t}$$

și notăm $\{H, F\} = \underbrace{\sum_{i=1}^{f} \frac{\partial F}{\partial p_i} \cdot \frac{\partial H}{\partial q_i} - \sum_{i=1}^{f} \frac{\partial F}{\partial q_i} \cdot \frac{\partial H}{\partial p_i}}_{\text{paranteza Poisson}}$

$$\frac{dF}{dt} = \{H, F\} + \frac{\partial F}{\partial t}$$

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Cursul 12

MECANICĂ CUANTICĂ

1) MN (mecanică newtoniană)
↳ deterministă / clasică

2) MC (mecanica cuantică)
≡ fizica " "
↳ probabilistică

$\varphi, \psi, \dots \theta, \xi$: funcții din MC

$\varphi = \varphi(q)$

1. mărginită : $\lim_{q \to \infty} \varphi(q) = ct$

2. continuă și cu derivate continue (C^2 bine)

3. $\in$ spațiu Hilbert , $\varphi \in \mathcal{H}$

de modul pătrat sumabilă

$$\int_{-\infty}^{+\infty} |\varphi(q)|^2 dq < \infty$$

↓ ~~im~~ convergentă

Produs scalar (superior, complex etc) generalizat

$\mathcal{H}$ înzestrat cu produs scalar

Fie $\varphi, \psi \in \mathcal{H}$, atunci $(\varphi(q_2), \psi(q_1)) = (\varphi, \psi) =$

$$- \int_{-\infty}^{\infty} \varphi^*(q) \psi(q) dq$$ produs scalar

φ^* complex conjugat

Proprietăți :

↳ de către

Operator

Fie A operator ; $\varphi \in S$ atunci $A\varphi \in S$ (op de închidere)

Proprietăți

liniar $A(\alpha_1 \varphi_1 + \dots + \alpha_m \varphi_m) = \alpha_1 A \varphi_1 + \dots + \alpha_m A \varphi_m$

$\varphi_1, \dots \varphi_m \in S \Rightarrow \varphi_i \ i = \overline{1,m}$

$\psi \in \mathcal{H}$; $A\psi \in \mathcal{H} \Rightarrow \exists A^+$ operator adjunct

operator A, A^+ ↖ adjunctul lui

$(A\psi, \varphi) = (\varphi, A^+\varphi)$ dacă și numai dacă (?!)

Fie A, A^+ cu doi operatori asociați $\Rightarrow A = A^+$ atunci

A^+ s.n. autoadjunct (hermitic)

$\Rightarrow (A\psi, \varphi) = (\varphi, A\psi)$

Lem: $(A\psi, \varphi) = (\varphi, A^+\varphi) = (\varphi, A, \varphi)$.

Lemă: Toți op. din M.C. sunt hermitici.

Operatorul comutator

Fie A, B op. a.î. $AB \neq BA \Rightarrow \exists [A; B] = AB - BA$

caz trivial $[A, B] = 0 \Rightarrow AB = BA \Rightarrow$ comutator $0 \Rightarrow$

A comută cu B

MC - operatori hermitici

Ecuația cu funcții și valori proprii

Fie A - op:

$A\psi = a\psi$ cc cu fct și val proprii

ψ: fctii proprie; a = val proprie asociată ψ prin A.

Lemă

Val proprii asociate operatorilor hermitici din MC - reale

$A = A^+$ (hei) $A\psi = a\psi \Rightarrow$ ~~$\psi \in (\psi, A\psi)$~~ $\psi \in \mathcal{H}$

$\Rightarrow a \in \mathbb{R}$

Dem: $(A\psi, \psi) = (\psi, A^+\psi)$

$(A\psi, \psi) = (a\psi, \psi) = a^*(\psi, \psi)$ $\Rightarrow$

$(\psi, A^+\psi) = (\psi, A\psi) = (\psi, a\psi) = a(\psi, \psi)$

$\Rightarrow a^*(\psi, \psi) = a(\psi, \psi) \Rightarrow (a^* - a)(\psi, \psi) = 0 \Rightarrow$

$a^* - a = 0 \Rightarrow a^* = a \Rightarrow a \in \mathbb{R}$

$z = x_1 + iy_1 \in \mathbb{C}$;

$z = x_2 + iy_2$ $\quad x_1, x_2 \in \mathbb{R}$

$z_1 = z_2 \Leftrightarrow \begin{cases} x_1 = x_2 \\ y_1 = y_2 \end{cases}$

$(\varphi, \varphi) = \int_{-\infty}^{\infty} \varphi^*(q), \varphi(q)\, dq = \int_{-\infty}^{\infty} |\varphi(q)|^2 dq$

$\sqrt{(\varphi, \varphi)} = \|\varphi\|$

$\varphi^*(q)\, \varphi(q) = |\varphi(q)|^2$;

$(\varphi, \psi) = 0 \Rightarrow \varphi$ și ψ ortogonali

$\varphi_1, \varphi_2, \ldots \varphi_m$ → ortogonale

$(\varphi_i, \varphi_j) = 0$ $\quad (\varphi_i, \varphi_i) = 1$

$i = \overline{1,m}$

$j = \overline{1,m}$ $\quad = \int_{-\infty}^{\infty} |\varphi(q)|^2 dq = 1$

condiția de normare la unitate

$(\varphi_i, \varphi_j) = \delta_{ij}$; $\quad \delta_{ij} \begin{cases} 1, i = j \\ 0, i \neq j \end{cases}$

ortonormale, - fc.

Principiile și postulatele MC

3 principii + 3 postulate

Principiile

1. Principiul superpoziției

Dacă un sistem cuantic se află în stările $\varphi_1, \varphi_2, \ldots \varphi_m$, descrise de φ, se poate afla și în cea descrisă de $\varphi = \sum_{i=1}^{m} c_i \varphi_i$

Termeni / cuvinte în MC

- Observație $\mathcal{A}$
- Observabilă cuantică $\hat{A}$ $\quad (\hat{A} = \hat{A}^+)$

2 Principiul de corespondență $\quad \hat{q}$ - operator în MC

- Observabile în MC

variabilele dinamice din MN (ex q_i, p_i, E)

Fiecăreia îi corespunde una cuantică, din MC

Relațiile dintre observabilele cuantice ↓ conservă relațiile dintre var. cuantice din MN.

$q_i \longrightarrow \hat{q}_i$

$p_i \longrightarrow \hat{p}_i$

3. Principiul relațiilor de comutare

Între oscar cuantici poziții, impuls - energie, Teorm. relații de comutare. (Conf 2) =>

$$\hat{q}_i \to q_i$$
$$\hat{p}_i \to -i\hbar \frac{\partial}{\partial q_i}$$
$$\hat{E} \to i\hbar \frac{\partial}{\partial t}$$

Einstein Planck

$$W_1 - W_2 = \varepsilon_{12} = h\nu$$
$$\frac{h}{2\pi} \cdot 2\pi\nu$$
$$\frac{h}{2\pi} = \hbar \Rightarrow E = \hbar\omega$$

$\hbar = \frac{h}{2\pi}$ constanta Planck (barată)

$[q_i, q_j] = 0$, $[p_i, p_j] = 0$; $[q_i, p_j] = i\hbar\delta_{ij}$

3D:

x, y, z $\quad P = p_x \bar{i} + p_y \bar{j} + p_z \bar{k}$

p_x, p_y, p_z $\quad \bar{r} = x\bar{i} + y\bar{j} + z\bar{k}$

$[q_i\ q_j] = q_i q_j - q_j q_i \quad : q_i, q_j \in R$

$i = \overline{1,n}$; $j = \overline{1,n}$ (comutativă în R)

$$p_i = -i\hbar \frac{\partial}{\partial q_i} \qquad \frac{\partial}{\partial q_i}\left(\frac{\partial}{\partial q_j}\right) = \frac{\partial}{\partial q_j}\left(\frac{\partial}{\partial q_i}\right) \to 0$$
$$p_j = -i\hbar \frac{\partial}{\partial q_j}$$

- lema lui Schwartz

1D: x, p_x ; $p_x = -i\hbar \frac{\partial}{\partial x}$

$[x, p_x] = i\hbar$ $\quad (i = j \Rightarrow x, p_x, \delta_{ij} = 1)$

$i = \sqrt{-1}$

$[x, p_x] = x p_x - p_x x$

fie o fct ψ $\psi = \psi(x)$

$$\Rightarrow (x p_x - p_x x)\psi(x) = x p_x \psi(x) - p_x x \psi(x) =$$
$$= x\left(-i\hbar \frac{\partial \psi(x)}{\partial x}\right) - \left[-i\hbar \frac{\partial}{\partial x}(x\psi(x))\right]$$
$$= -i\hbar x \frac{\partial \psi}{\partial x} + i\hbar\psi(x) + i\hbar x \frac{\partial \psi}{\partial x} = i\hbar\psi(x).$$

$(xp_x - p_x x)\psi(x) = i\hbar\psi(x)$

$(xp_x - p_x x) = i\hbar \Rightarrow [x, p_x] = i\hbar$ ged

$p_y = -i\hbar\frac{\partial}{\partial y}$; $p_z = -i\hbar\frac{\partial}{\partial z}$;

Postulatele MC

1) Dacă facem obs cu Ob de obs. cuant A asupra unui sistem aflat în stările cuantice repr. de $\psi_1, \psi_2, \ldots \psi_m$ ⇒ eg. ia fi $a_1, a_2, \ldots a_m$, respectiv val. proprii asociate stărilor $\psi_1, \psi_2, \ldots \psi_m$ corresp. din ecuația $A\psi_i = a_i\psi_i$; $i = \overline{1,m}$

2) Dacă facem Ob de A asupra unui sist. cuantic aflat în mai multe stări cuantice, val medie a lui A este $\langle A \rangle = \frac{\int_{-\infty}^{\infty} \psi^*(q) A \psi(q)\, dq}{\int_{-\infty}^{\infty} \psi^*(q)\psi(q)\, dq}$

Dacă $\psi \in \mathcal{H}$ este normată la unitate:

$\int_{-\infty}^{\infty} \psi^*(q)\psi(q)\, dq = \int_{-\infty}^{\infty} |\psi(q)|^2\, dq = 1$ este

$\langle A \rangle = \int_{-\infty}^{\infty} \psi^*(q)\, A\, \psi(q)\, dq$

3) Dacă se face Ob de A asupra unui sist. aflat în starea caract. de ψ_i vom ști cu certitudine val. a_i, respectiv val proprie a ~~stării~~ asociată lui ψ_i, din ecuația $A\psi_i = a_i\psi_i$

Ecuația lui Schrödinger

H, hamiltoniana sistemului ; $H = W + E_p$

W = Energia cinetică = $\frac{mv^2}{2} = \frac{m^2v^2}{2m} = \frac{p^2}{2m}$; $p = mv$

$E_p = V(\bar{r}, t)$

$\psi = \psi(\bar{r}, t) \in \mathcal{H}$; $r = xi + yj + zk$ |⇒

$\psi = \psi(x, y, z, t)$; $V = V(x, y, z, t)$

$H = \frac{p^2}{2m} + V(\bar{r}, t)$

→ ec cu fctii și val. proprii a lui H

$\boxed{H\psi(r, t) = E\psi(\bar{r}, t)}$

$p^2 = p_x^2 + p_y^2 + p_z^2$

$p_x = -i\hbar \frac{\partial}{\partial x}$

$p_x^2 = p_x(p_x) = -i\hbar \frac{\partial}{\partial x}\left(-i\hbar \frac{\partial}{\partial x}\right)$

$p_x^2 = -\hbar^2 \frac{\partial^2}{\partial x^2}$

$p_y^2 = -\hbar \frac{\partial^2}{\partial y^2}$;

$p_z^2 = -\hbar^2 \frac{\partial^2}{\partial z^2}$

$p^2 = p_x^2 + p_y^2 + p_z^2 = -\hbar^2\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}\right) =$

$= -\hbar^2 \Delta$;

$p^2 = -\hbar^2 \Delta$

$H = -\frac{\hbar^2}{2m}\Delta + V(\bar{r}, t) \Rightarrow \left[-\frac{\hbar^2}{2m}\Delta + V(\bar{r}, t)\right]\psi(\bar{r}, t) =$

$= E\,\psi(\bar{r}, t)$

Sau $E \to i\hbar \frac{\partial}{\partial t}$

$\left[-\frac{\hbar^2}{2m}\Delta + V(\bar{r}, t)\right]\psi(\bar{r}, t) = i\hbar \frac{\partial}{\partial t}\psi(\bar{r}, t)$

↳ ec. lui Sch. temporală

= ec dif ord 2 în coord

-"- , în timp

Sol: $\psi = \psi(\bar{r}, t)$

$\psi(\bar{r}, t) = \psi(\bar{r})\, \theta(t)$

caz particular: $\psi(\bar{r}, t) = \psi(\bar{r})\, e^{-\frac{i}{\hbar}Et}$

$[E \cdot t] = J \cdot s$ caimensiunea actiunii

$V(\bar{r}, t) \to V(\bar{r})$

$\left[-\frac{\hbar^2}{2m}\Delta\psi(\bar{r}) + V(\bar{r})\psi(\bar{r})\right] e^{-\frac{i}{\hbar}Et} = i\hbar \frac{\partial}{\partial t} e^{-\frac{i}{\hbar}Et}\psi(\bar{r})$

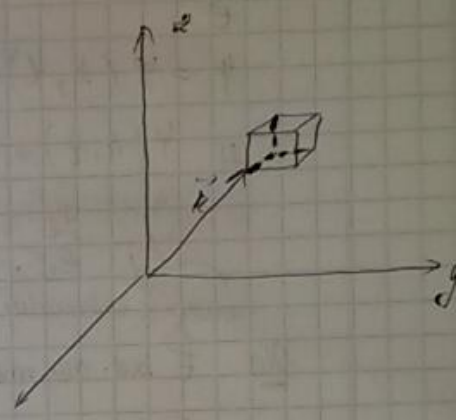
$\left[-\frac{\hbar^2}{2m}\Delta + V(\bar{r})\right]\psi(\bar{r})\, e^{-\frac{i}{\hbar}Et} = E \cdot e^{-\frac{i}{\hbar}Et}$

$-\frac{\hbar^2}{2m}\Delta\psi(\bar{r}) + V(\bar{r})\psi(\bar{r}) = E\psi(\bar{r}) \Rightarrow \psi(\bar{r}) = \psi_E(\bar{r})$

$$-\frac{\hbar^2}{2m}\Delta\psi(\bar{r}) + V(\bar{r})\psi(\bar{r}) = E\psi(\bar{r})$$

$$\Delta\psi(\bar{r}) + \frac{2m}{\hbar^2}\left[E - V(\bar{r})\right]\psi(\bar{r}) = 0$$

↑ ec. Schr. atemporală



E = energia sistemului

$\psi\Psi(\bar{r},t) = \psi_E(\bar{r})\, e^{-\frac{i}{\hbar}Et}$

$|\Psi(\bar{r},t)|^2 d\bar{r}$; $d\bar{r} = dxdydz$

$|\psi(x,y,z,t)|^2 dxdydz$ →

Probab. de a găsi particula cuantică în cubul de $V = dxdydz$. (deci de energie E)

$$|\Psi(\bar{r},t)|^2 = \left|\psi_E(\bar{r})\, e^{-\frac{i}{\hbar}Et}\right|^2 = |\psi_E(\bar{r})|^2 \underbrace{\left|e^{-\frac{i}{\hbar}Et}\right|^2}_{\left(e^{\frac{i}{\hbar}Et}\right)^2 = 1}$$

$|\psi_E(\bar{r})|^2 dxdydz$

$\psi_{E_i}(\bar{r})$, $i = \overline{1,n}$

$\hat{H}\psi_1 = E_1\psi_1$ ... $\hat{H}\psi_n = E_n\psi_n$

$\psi_1 \Rightarrow E_1$ → $\psi_n \Rightarrow E_n$

$$(\psi_{E_i}, \psi_{E_j}) = \begin{cases} \delta_{ij} & \text{, discret} \\ \delta(E_i - E_j) & \text{, continuu.} \end{cases}$$

$\psi_E(q)$; $\psi_{E'}(q)$ ⇒ $\int \psi_E \psi_{E'} = \delta(E, E')$

Cursul 13

Schrodinger

E.

$\Psi = \Psi(r,t)$ funcția de undă

$\hat{H}\ \Psi(r,t) = E\Psi(\bar{r},t)$ ... f și val. proprii

$\Psi_1\ \ \Psi_2$ ... f. proprii

$E_1\ \ E_2$ ... val proprii

energiile sistemului

Obs E sist. nu mai e o f. continuă

E sist. va avea val discrete $E_1, .. E_n$, val. proprii corresp. $\Psi_1, .. \Psi_n$

$\hat{H}\ \Psi_{Ei} = E_i\ \Psi_{Ei} \quad i = \overline{1,n}$

$\hat{p} = -i\hbar \frac{\partial}{\partial q_i} \quad (p_i, q_i)$ canonic conjugate

$p_x = -i\hbar \frac{\partial}{\partial x} \quad (p_x, x)$

$p_y = -i\hbar \frac{\partial}{\partial y} \quad (p_y, y)$

$p_z = \ldots$

$H \rightarrow$ hamiltoniana sistemului

$H = W + E_p$

$E_p \rightarrow V = V(\bar{r},t)$

$W = \frac{mv^2}{2} = \frac{p^2}{2m}$

$p = mv$ impuls

$p = p_x^2 + p_y^2 + p_z^2$

$p_x^2 = -i\hbar \frac{\partial}{\partial x}\left(-i\hbar \frac{\partial}{\partial x}\right) = -\hbar^2 \frac{\partial^2}{\partial x^2}$

idem y, z

$p^2 = p_x^2 + p_y^2 + p_z^2 = -\hbar^2 \Delta$

$p^2 = -\hbar^2 \Delta$

$\Delta = \frac{\partial^2}{\partial x^2} + .. y + .. z$

$H = -\frac{\hbar^2}{2m} \Delta + V(\bar{r},t)$

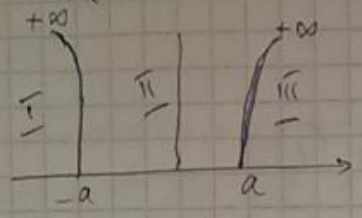
$|\Psi(r,t)|^2 d\bar{r}$

$|\Psi(x,y,z,t)|^2 dx\,dy\,dz$

Particula cuantică în groapa de potențial cu pereți infiniți

$V = V(\vec{r}, t)$; $V(\vec{r}) \longrightarrow V\vec{r} = V(x)$

1D



3D $dV = dx\,dy\,dz$

1D dx

fizic $|\psi(x)|^2 dx$

$$V = V(x) = \begin{cases} \infty, & x \in (-\infty; -a) \\ 0; & x \in [-a, a] \\ \infty; & x \in (a, \infty) \end{cases}$$

$$V(x) = \begin{cases} 0, & |x| \leq a \\ \infty, & |x| > a \end{cases}$$

problemă de cuantică

- funcțiile proprii ψ_E și val. proprii ~~ale~~ → Energiile E :

$$H\psi_E = E\psi_E$$

Obs :

$x \in (-\infty, -a)$ $\begin{cases} V(x) \to \infty \\ \psi = \psi(x) = 0 \end{cases}$

$x \in [-a, a]$ $\begin{cases} V(x) = 0 \\ \psi = \psi(x) \neq 0 \end{cases}$

$x \in (a, \infty)$ $\begin{cases} V(x) \to \infty \\ \psi = \psi(x) = 0 \end{cases}$

3 regiuni :

I) $\psi_I(x) = 0$

II $\psi_{II}(x) \neq 0$, $|\psi(x)|^2 dx$

III $\psi_{III}(x) = 0$

Rezolvăm problema în regiunea II

$H\psi_{II} = E\psi_{II}$ rezolva ec. Sch.

$H = W + V(x)$; $W = \frac{p_x^2}{2m} = -\frac{\hbar^2}{2m}\frac{d^2}{dx^2}$

$$H = -\frac{\hbar^2}{2m}\frac{d^2}{dx^2} + V(x)$$

$$H_{\underline{II}} = -\frac{\hbar^2}{2m}\cdot\frac{d^2}{dx^2}$$

$$-\frac{\hbar^2}{2m}\frac{d^2\psi(x)}{dx^2} = E\psi(x)$$

$$\frac{d^2\psi(x)}{dx^2} + \frac{2mE}{\hbar^2}\psi(x) = 0 \quad (1)$$

În reg $\underline{II}$, ec dif. de ord $\underline{II}$, omogenă

$$\frac{2mE}{\hbar^2} = k^2 \Rightarrow \frac{d^2\psi(x)}{dx^2} + k^2\psi(x) = 0 \Rightarrow$$

$$\Rightarrow \psi(x) = A\cos kx + B\sin kx$$

ec. (1) + (cond de limită (Cauchy) = CL).

CL:

$$\begin{cases} \psi(x) \underset{x\to -a}{=} 0 \\ \psi(x) \underset{x\to a}{=} 0 \end{cases} \qquad \begin{cases} \psi(-a) = 0 \\ \psi(a) = 0 \end{cases} \rightarrow \text{calculăm limita și } B$$

$$k = \sqrt{\frac{2mE}{\hbar^2}} \; ; \; \hbar = \frac{h}{2\pi}$$

$x \in \mathbb{R}$; $(-\infty; +\infty)$

~~sin (-a)~~

$$\psi(-a) = A\cos ka - B\sin ka = 0$$

$$\psi(a) = A\cos ka + B\sin ka = 0$$

sin - impară
cos - pară
$f(-x) = f(x)$
$f(-x) = -f(x)$

$$\begin{cases} A\cos ka - B\sin ka = 0 \\ + \quad + \quad = 0 \end{cases}$$

(+) $A = 0$

(-) $B = 0$

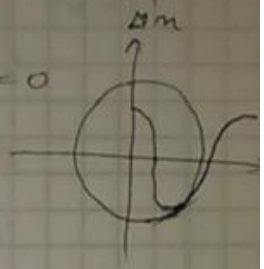
avem două posibilități ψ $\begin{cases} \text{① } A \neq 0, \; B = 0 \\ \quad \psi = \text{pară} \; ; \; \psi(x) = A\cos kx \\ \text{② } A = 0 \; ; \; B \neq 0 \\ \quad \psi - \text{impară} \quad \psi(x) = B\sin kx \end{cases}$

$\hat{H}\psi = E\psi$ $\quad \psi^{Par}(x) = A\cos kx$

$\psi^{Impar}(x) = B\sin kx$

~~$\sin a = 0$~~

$\psi^{P}(a) = 0 \Rightarrow A\cos ka = 0$



$\cos ka = 0$

$ka = (2\Lambda+1)\frac{\pi}{2}$

$m = 2\Lambda+1$ impar

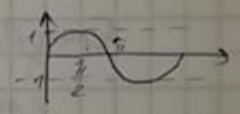
$$\boxed{\psi^{P}_{m}(x) = A\cos\frac{m\pi}{2a}x}$$

~~$k = \frac{2\pi}{2a}$~~

$\psi^{i}(x) = B\sin kx$

c.l. $\psi^{i}(a) = 0 \Rightarrow B\sin ka = 0 \Rightarrow \sin ka = 0$

$\sin x = a \;;\; x = (-1)^{k}\arcsin a + k\pi$



~~$ka = \frac{2\Lambda\pi}{2}$~~ ; $ka = \frac{m\pi}{2}$; $m = 2\Lambda$

$$\boxed{\psi^{i}_{m}(x) = B\sin\frac{m\pi}{2a}x \quad m = 2\Lambda}$$

- A, B se găsesc din cond de normare la unitate

$$\int_{-a}^{a}|\psi(x)|^2\,dx = 1$$

$$\int_{-a}^{a}|\psi^{P}_{m}(x)|^2\,dx = 1 \quad \text{cond. } (x)$$

$$\int_{-a}^{a}|\psi^{i}_{m}(x)|^2\,dx = 1 \quad \text{cond } (xx)$$

① $\int_{-a}^{a} A^2\cos^2\frac{m\pi}{2a}x\,dx = 1 \;;\; A^2\int\ldots = 1$

$\cos 2\alpha = \cos^2 - \sin^2$

$\cos^2\alpha = \frac{1+\cos 2\alpha}{2} = \frac{1+1-2\sin^2\alpha}{2}$ $\quad \sin^2\alpha = \frac{1-\cos 2\alpha}{2}$

$$\int_{a}^{a}\cos\ldots = \int_{a}^{a}\frac{1+\cos 2\alpha}{2}dx = \underbrace{\frac{1}{2}\int_{a}^{a}dx}_{a} + \frac{1}{2}\int_{-a}^{a}\cos\frac{2m\pi}{a}$$

$= \frac{a}{m\pi}\sin t = 0. \Rightarrow$

$A^2 \cdot a = 1 \Rightarrow A = \sqrt{\frac{a}{a}}$

$\psi^p_n(x) = \frac{\sqrt{a}}{a} \cos \frac{n\pi}{2a} x$

$(**) \int_{-a}^{a} B^2 \sin^2 \frac{n\pi}{2a} x = 1$

$\Rightarrow B^2 \int \dots = 1$

$\sin^2 \frac{n\pi}{a} = \frac{1 - \cos \frac{n\pi}{a} x}{2} \Rightarrow \int_{-a}^{a} \sin^2 \frac{n\pi}{2a} x \, dx =$

$= \frac{a}{2} \int_{-a}^{a} dx - \int \cos^2 \frac{2\pi}{n} x \, dx$

$\frac{2a}{2} = a$ $\qquad$ 0

$B^2 \cdot a = 1 \Rightarrow B = \frac{\sqrt{a}}{a}$

$$\boxed{\psi^i_{n\,(n=2s)}(x) = \frac{\sqrt{a}}{a} \sin \frac{n\pi}{2a} \cdot x}$$

Energiile

$\frac{2mE}{\hbar^2} = k^2$; $E = \frac{k^2\hbar^2}{2m}$; $k^2 = \frac{n^2\pi^2}{4a^2}$

$E = \frac{n^2\pi^2}{4a^2} \Big/ 2m \cdot \frac{\hbar^2}{4\pi^2} = \frac{n^2\hbar^2}{32a^2m} = \boxed{E_n = \frac{n^2\hbar^2}{32a^2m}, \; n \in \mathbb{N}}$

$E_1 = \frac{\hbar^2}{32a^2m}$; $E_2 = \frac{4\hbar^2}{32a^2m} = \frac{\hbar^2}{8a^2m} = 4E_1$

$E_3 = 9E_1$

E spectru discret $\in \{E_1, E_2, \dots\}$

funcții energii

ψ_1^p $\qquad$ E_1

ψ_2^i $\qquad$ E_2

ψ_3^p $\qquad$ E_3

$H_1 \psi_1^p = E_1 \psi_1^p$

$\psi_1^p = A \cos \frac{\pi}{2a} x$

$A = \frac{\sqrt{a}}{a}$

$H \psi_2^i = E_2 \psi_2^i$

$\psi_2^i = B \frac{\sqrt{a}}{a} \sin \frac{2\pi}{2a} x$ etc.